

Structural Cryptanalysis of QR-UOV: Projective Direct Forgery, Local-Separator Descent, and Equivalent-Key Recovery

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31 July 2026

Abstract

QR-UOV compresses Unbalanced Oil and Vinegar (UOV) by representing variables in a degree-three quotient ring. We give two complementary structural attacks and a local-separator refinement that, under an explicit symbolic-homotopy gate model, yields a candidate Level-I break.

The direct attack sends a chosen target to $(0, \dots, 0, 1)$, constructs an $(m - 3)$ -dimensional common isotropic space for three zero-target quadrics, and quotients the remaining homogeneous system by scaling. The resulting square projective cores have dimensions 50, 74, and 101. The equivalent-key attack recovers the hidden oil graph from one regular directional root.

The main new refinement splits the homotopy before lifting and replaces a global primitive-element requirement by local separation of one eligible simple target branch. A generic characteristic polynomial and its coefficient polars remain valid even when the chosen linear form collides on other branches. On a simple target factor, a single Frobenius sketch is exact. In the inherited Boolean-gate model, the resulting costs are 140.383995, 191.027133, and 247.125439 bits, respectively 2.616, 15.973, and 24.875 bits below the NIST category targets. Using the parent paper’s weaker root-probability bound still gives 140.531461 bits at Level I.

We also implement the complete local-separator pipeline on reduced QR-UOV parameters. From only the public key, message, and salt, the program constructs the residual system, lifts and recombines all branches, performs rational specialization and exact Frobenius descent, reconstructs one point, and outputs a forgery accepted by the unchanged public verifier. Five deterministic instances and a supplementary larger-field instance all succeed and replay from public transcripts. This closes the end-to-end composability objection at toy scale, but it does not validate the full-parameter time or memory extrapolation. The Level-I statement remains conditional on the stated characteristic-polynomial degree transfer, the fast online-product schedule, the public-random expansion model, and astronomical symbolic memory.

1 Introduction

Background and attack goals

A multivariate signature scheme publishes a quadratic map

$$P : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q^m$$

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and signs a target y by finding x with $P(x) = y$. UOV makes this inversion easy with a secret split between *vinegar* and *oil* variables. No central equation contains an oil–oil product, so fixing the vinegar variables leaves a linear system in the oil variables [2].

An attacker does not need the submitters’ seed. Two endpoints are enough:

- a *direct forgery* finds one preimage of the target derived from a message and salt; and
- an *equivalent-key attack* finds any hidden decomposition that can invert every target accepted by the unchanged verifier.

This paper gives one attack of each kind. They work in different coordinate systems and expose different parts of QR-UOV’s structure. Their combination therefore says more about parameter repair than either attack alone.

QR-UOV uses a quotient ring of degree $\ell = 3$. Its recommended $q = 127$ profiles have base-field dimensions

$$(n, m) = (210, 54), \quad (306, 78), \quad (411, 105),$$

and extension-field dimensions

$$(V, O, m) = (52, 18, 54), \quad (76, 26, 78), \quad (102, 35, 105)$$

over $K = \mathbb{F}_{127^3}$. NIST advanced QR-UOV to the third round of its Additional Digital Signatures process in May 2026; third-round teams may submit updated specifications and implementations [3, 4].

Attack I: remove three equations, then quotient by scaling

Let $y \neq 0$ be a verification target. A public output transformation sends it to

$$(0, \dots, 0, 1).$$

Call the first three transformed quadrics Q_1, Q_2, Q_3 . Choose a random seven-dimensional seed subspace S_0 . Chevalley–Warning guarantees a nonzero common zero $w \in S_0$, found by enumerating $\mathbb{P}^6(\mathbb{F}_q)$. For every fixed hidden oil space, S_0 intersects it nontrivially with probability at most

$$\frac{q^m - 1}{q - 1} \frac{q^7 - 1}{q^n - 1},$$

so w is off oil with overwhelming probability over public attack coins. This fixed-key statement avoids converting an average density calculation into an unjustified per-key rejection-sampling claim.

Starting from w , the rest of the subspace construction is deterministic. If W is a j -dimensional subspace on which all three quadrics vanish, intersect their three polar-orthogonal spaces. Modulo W , the result has dimension at least $n - 4j$. Whenever it has dimension at least seven, restrict to any seven-dimensional subspace and use Chevalley–Warning again to add one vector. The construction reaches dimension r whenever

$$n \geq 4r + 3.$$

For $r = m - 3$, the official sets have slacks 3, 3, 0.

Full target normalization makes all transformed target coordinates except the last one zero. On the constructed r -space, the next $r - 1 = m - 4$ homogeneous quadrics therefore define a square system in $\mathbb{P}^{r-1}$. A random affine chart turns it into $r - 1$ quadrics in $r - 1$ variables. The last transformed quadric is not included in the solver core: it only tests whether the projective

point can be scaled to the nonzero target, and then supplies that scale. Thus the actual geometric solver dimensions are

$$50, \quad 74, \quad 101.$$

The affine dimensions 51, 75, 102 are the $\alpha = 3$ Hashimoto line already used in QR-UOV's direct-attack analysis [29, 1]; the one-variable projective quotient is an additional exact reduction.

For a point outside the oil space, one public-random pull-back form has uniform evaluation and a full affine family of first derivatives. Evaluation at two off-oil points is independent unless the points are base-field scalar multiples. A second-moment calculation proves that an eligible projective root with the required nonsingularity exists with probability at least

$$0.326336998767509 \dots$$

A random nonzero chart contains any fixed projective root with probability strictly greater than $1 - 1/q$. The root-probability theorem uses an idealized model in which the secret graph seed is independent of the public central-form seed. The official construction has this property in the ideal-XOF and ideal-cipher models except on a seed collision; Section 24 charges that event explicitly. Every reconstructed vector is evaluated under the original public key, so failure can omit a signature but cannot create a false one.

Attack II: one directional root gives an equivalent key

In normalized extension-field coordinates, the hidden oil space is the graph

$$\mathcal{O} = \{(-Gc, c) : c \in K^O\}.$$

Fixing a public direction $c \in K^O$ yields m public quadrics in only V variables, with planted solution Gc . If the planted derivative has rank V , one recovered root determines every column of G through public linear systems. The graph is checked against every public identity, pulled back to the official base-field coordinates, and accompanied by an invertible signing matrix. An accepted output is therefore an exact universal-forgery key.

The solver adapts the nonsingular-solutions homotopy of Safey El Din and Schost to characteristic 127 [26]. An explicit Vandermonde product start and a full random separator replace the two short integer-indexed families responsible for the published characteristic restriction. A flattened coefficient algebra and sparse irreducible working fields improve the finite-size ledger. The audit also corrects a previous undercount: rational reconstruction requires precision $2C_0$, not C_0 .

The fixed-output security scissors

The attacks react oppositely to the vinegar count V . For quotient degree ℓ , the base-field dimension is $n = \ell V + m$. Target-orthogonal slicing of s equations is possible whenever

$$\ell V \geq s(m - s).$$

After projectivization, that reduction has square solver core $m - s - 1$ rather than $m - s$. Taking $s = \ell$ gives a direct core $m - \ell - 1$ whenever $V \geq m - \ell$. If $V \leq m - \ell - 1$, the directional core is already at most $V \leq m - \ell - 1$. Since there is no integer between these two cases,

$$r_{\text{attack}}(V) \leq m - \ell - 1$$

for every vinegar count and every nonzero target. For QR-UOV, the ceiling is $m - 4$. This is an algebraic-core statement, not a claim that time depends only on dimension. Its design implication is exact: increasing V while holding m fixed cannot raise the combined bottleneck beyond the crossing, while decreasing V makes the directional attack cheaper. A repair must also increase m , or alter the construction so that one reduction no longer applies.

Table 1: Status of the main claims.

Claim	Status
Projective direct core $m - 4$	Exact for every nonzero target once a seed root is supplied; Chevalley–Warning guarantees such a root in every seven-dimensional seed, and homogeneity quotients the residual by scaling.
Seed root is off oil	Fixed-key probability bound over public attack coins; no random-key assumption.
Common-isotropic extension	Deterministic and finite after the seed, using linear algebra and exhaustive search in $\mathbb{P}^6(\mathbb{F}_q)$.
Eligible projective-root probability	Exact lower bound in the independent-graph public-random degree-three model.
One regular direction determines an equivalent key	Exact for every expanded key satisfying the public rank and signing checks.
Returned signature or key is valid	Exact; every candidate is verified against the unchanged public map.
Finite cost rows	Reproducible symbolic-algorithm diagnostics, not measured circuits.
Full-parameter memory	Prohibitive for the direct representations analyzed here.
Fixed SHAKE/AES expansion	Key-by-key verification is exact; density statements transfer only in the stated ideal-primitive models, with explicit buffer and seed-collision terms.

Concrete conclusions

Using the QR-UOV specification’s broad Boolean-gate unit, explicit recursive Karatsuba circuits for the auxiliary fields, the corrected lifting precision, local separation, and complete success normalization, the best rows are

$$140.384, \quad 191.027, \quad 247.125$$

bits at Levels I, III, and V. Therefore all three parameter sets fall below their category targets in the stated model, by 2.616, 15.973, and 24.875 bits respectively. The Level-I conclusion is a candidate model-based break, not a practical computation: its finite margin depends on the explicit characteristic-polynomial degree transfer and fast online-product envelope. The reduced-parameter implementation nevertheless executes the entire attack pipeline and produces publicly accepted forgeries, so the construction is no longer supported only by disconnected component tests. The symbolic states are astronomical. We report time, memory, exact reductions, and distributional assumptions separately rather than presenting the rows as implementation benchmarks.

What is exact, and what is modeled

Organization

[Section 2](#) positions the two attacks. [Section 3](#) explains the official pull-back, the public-random models, and exact signing certificates. [Section 4](#) proves target-orthogonal slicing and the security scissors. [Sections 5](#) and [6](#) give directional graph recovery. [Sections 7](#) to [9](#) establish the characteristic-127 solver and cost refinements. [Section 19](#) combines the attacks and gives parameter consequences. Independent solver routes and seeded-expansion arguments are retained in the appendices.

2 Related work

Direct attacks on underdetermined MQ. Thomae–Wolf, Furue et al., and Hashimoto reduce underdetermined quadratic systems before applying an MQ solver [30, 31, 29]. Hashimoto’s

line is the benchmark used in the QR-UOV specification. The QR-UOV team’s January 2026 response reports classical Hashimoto estimates of 158, 217, and 285 bits for the recommended sets [28]. The affine residual dimensions 51, 75, 102 are therefore prior art. Our contribution is a self-contained homogeneous construction, a fixed-key seed-disjointness bound, a QR-UOV-specific nonsingular-root theorem, and the observation that full target normalization lets the solver quotient by radial scaling, reducing the square geometric cores to 50, 74, 101. We then cost those projective cores with a separate finite-characteristic solver. The recent “Just Guess” algorithm improves the quantum analysis of underdetermined MQ but does not improve the team’s reported classical QR-UOV estimates [13, 28].

Equivalent-key recovery for UOV. Pébereau showed that one suitable oil vector can suffice for polynomial-time UOV equivalent-key recovery [14]. We do not claim that endpoint as new. The QR-UOV-specific contribution is the public mechanism for finding the needed information: a direction exposes a planted root Gc , full derivative rank makes it isolated, and cross-direction graph identities recover all of G . Exact pull-back and signing checks then certify a key-only universal forgery.

Beullens’ intersection attack and later work recover hidden UOV geometry through structured linear algebra [9]. Rectangular-MinRank estimates were sharpened for QR-UOV and related schemes [10, 11]. Truncated-polynomial-ring methods improve some UOV parameters but do not report an improvement for the QR-UOV sets studied here [12]. Odd-characteristic symmetric-algebra attacks apply to QR-UOV but, at characteristic 127, do not improve the existing QR-UOV estimates [34].

Polynomial-system solving. The arbitrary-core engine is the nonsingular-solutions homotopy of Safey El Din and Schost [26], building on symbolic homotopy and geometric-resolution methods [27, 7]. We replace the start-system and separator choices responsible for its published characteristic restriction. The appendices retain independent checks using the near-quadratic solver of van der Hoeven–Lecerf and a locally closed Kronecker algorithm [6, 8].

Chevalley–Warning. The direct reduction uses the classical fact that, when the sum of the degrees is smaller than the number of variables, the number of common zeros over a finite field is divisible by the characteristic [32, 33]. Applied to three quadrics in seven variables, it guarantees a nonzero extension vector. Exhaustive projective search makes the theorem constructive, so no unpriced existence oracle is hidden in the preprocessing.

3 Expanded-key model and exact signing certificates

This section connects the algebra used by the attack to the official verification map. There are two coordinate systems:

- the official base-field system over $\mathbb{F}_{127}$, in which signatures are verified; and
- a smaller extension-field system over $K = \mathbb{F}_{127^3}$, in which the hidden oil space and the attack are easier to describe.

The specification gives an exact, invertible pull-back between them. We first recover and verify the hidden structure over K , then pull it back and check the resulting base-field factorization against the unchanged public key. One invertible oil system then certifies that the recovered decomposition can sign every target.

The Round-2 $q = 127$ parameter sets use quotient-ring degree $\ell = 3$. The specification has $v = \ell V$ base-field vinegar variables and $m = \ell O$ base-field oil variables and output coordinates.

After quotient-ring expansion, the attacker works with the same m outputs as quadratic forms in only $N = V + O$ variables over

$$K = \mathbb{F}_{q^\ell} = \mathbb{F}_{127^3}, \quad Q = |K| = 127^3 = 2,048,383, \quad \log_2 Q = 20.966054 \dots$$

Thus m denotes both the number of public output coordinates and the base-field oil dimension, whereas $O = m/\ell$ is the oil dimension over K .

Let $\Phi_f : K \rightarrow \mathbb{F}_q^{\ell \times \ell}$ be the regular representation from the specification, and let $W_f \in \mathbb{F}_q^{\ell \times \ell}$ be its invertible symmetrizer. Write ϕ for the blockwise inverse of Φ_f , and write $W_f^{(a)}$ for the block-diagonal matrix with a copies of W_f .

The attack uses the expanded extension-field tuple

$$P = (p_1, \dots, p_m), \quad p_i(x) = x^\top P_i x, \quad x \in K^N,$$

where the symmetric matrices P_i share a hidden totally isotropic O -space $\mathcal{O}$:

$$u^\top P_i w = 0 \quad (u, w \in \mathcal{O}, 1 \leq i \leq m). \quad (1)$$

The Round-2 sets analyzed in this paper are

Level	q	ℓ	v	m	$V = v/\ell$	$O = m/\ell$
I	127	3	156	54	52	18
III	127	3	228	78	76	26
V	127	3	306	105	102	35

Symbol	Meaning
$N = V + O$	Number of extension-field variables.
$e = m - V$	Number of public equations not used in the square V -equation core.
$D = 2^V$	Bézout degree of a smooth projective core.
$L_s = \mathbb{F}_{Q^s}$	Solver working field; the same field supplies slicing, recombination, and rational-root extraction.

All concrete theorems below concern these three sets, for which $m < q$. The directional and projective equivalent-key analyses use [Theorem 3.1](#) and allow arbitrary correlation between the secret graph and the symmetric public blocks. The direct-forgery root-counting theorem uses the stronger independent-graph specialization of [Theorem 3.2](#). The fastest projective route has cost $\tilde{O}(2^V 2^{(1+\varepsilon)(V-1)} 3 \log_2 127)$ for every fixed rational $\varepsilon > 0$, while the graph-direction route uses the explicit locally closed finite-field theorem of Giménez et al. [Section 24](#) gives separate ideal-XOF and ideal-cipher transfers for the two models and isolates the remaining fixed-primitive gap.

3.1 Public-random expansion model and relation to seeded expansion

Definition 3.1 (Public-random expansion with an arbitrarily correlated secret graph). First sample

$$(A_1, \dots, A_m) \leftarrow \text{Sym}_V(K)^m$$

from the product-uniform distribution. Conditional on the resulting tuple $A = (A_i)_i$, sample the secret graph $G \in K^{V \times O}$ according to an arbitrary conditional distribution. Conditional on (A, G) , sample

$$(E_1, \dots, E_m) \leftarrow (K^{V \times O})^m$$

from the product-uniform distribution, set each lower-right public block by the key-generation identity, and expose the resulting expanded public tuple. Values are sampled once and reused consistently. No independence between G and A is assumed.

Definition 3.2 (Independent-graph specialization). The *independent-graph public-random model* is [Theorem 3.1](#) with the additional requirement that the marginal law of G be independent of A . Equivalently, after centralization the product-uniform coefficient pairs $(A_i, B_i)_{1 \leq i \leq m}$ are independent of G . The marginal law of G may otherwise be arbitrary.

Proposition 3.3 (Exact central-block factorization under graph correlation). In the block notation of (2)–(4), define

$$B_i := E_i - A_i G.$$

Conditional on every fixed pair (A, G) , the tuple $(B_1, \dots, B_m)$ is product-uniform and its conditional law does not depend on (A, G) . Consequently, B is independent of (A, G) , and the pairs $(A_i, B_i)_{1 \leq i \leq m}$ are independent and uniform on $\text{Sym}_V(K) \times K^{V \times O}$. The graph G may nevertheless remain arbitrarily correlated with A . The lower-right public blocks C_i are determined by (4).

Proof. Fix $(A, G) = (a, g)$. For each i , the map

$$E_i \mapsto B_i = E_i - a_i g$$

is a translation of the additive space $K^{V \times O}$ and hence a bijection. It sends the product-uniform law of E to the product-uniform law of B , independently of (a, g) . Thus B is independent of (A, G) . Since A itself is product-uniform, $(A_i, B_i)_i$ has the claimed product law. The formula for C_i is the key-generation block identity (4). $\square$

The directional and projective bad-key calculations depend on the central pairs (A_i, B_i) , uniform public slices, and fresh attack coins. A secret coordinate change depending on G maps a uniform public slice to a uniform central slice for every fixed key, so those calculations tolerate arbitrary G – A correlation. The direct-forgery reduction itself is also key by key, but its average-case residual-root theorem conditions on an off-oil point and then uses independence of G from the central coefficient pairs; it is therefore stated under [Theorem 3.2](#). [Section 24](#) proves separate ideal-primitive transfers, including the public/secret seed-collision term needed for the independent-graph model. All returned keys and signatures are verified against the unchanged public map, independently of any distributional model.

3.2 Central and public graph coordinates

In secret central coordinates $K^V \oplus K^O$,

$$F_i = \begin{pmatrix} A_i & B_i \\ B_i^\top & 0 \end{pmatrix}, \quad F_i(v, o) = v^\top A_i v + 2v^\top B_i o, \quad (2)$$

with $A_i \in \text{Sym}_V(K)$ and $B_i \in K^{V \times O}$ independent and uniform. The planted oil space is $\mathcal{O}_0 = \{0\} \oplus K^O$.

In normalized public coordinates the hidden space is a graph

$$\mathcal{O} = \left\{ \begin{pmatrix} -Gc \\ c \end{pmatrix} : c \in K^O \right\}. \quad (3)$$

Writing

$$P_i = \begin{pmatrix} A_i & E_i \\ E_i^\top & C_i \end{pmatrix},$$

total isotropy is equivalent to

$$C_i = G^\top E_i + E_i^\top G - G^\top A_i G. \quad (4)$$

Proposition 3.4 (Verified graph gives an exact expanded UOV decomposition). Let $G' \in K^{V \times O}$ satisfy every identity (4), and put

$$B'_i = E_i - A_i G', \quad F'_i = \begin{pmatrix} A_i & B'_i \\ B_i^\top & 0 \end{pmatrix}, \quad S_{G'} = \begin{pmatrix} I_V & G' \\ 0 & I_O \end{pmatrix}.$$

Then

$$P_i = S_{G'}^\top F'_i S_{G'} \quad (1 \leq i \leq m).$$

Consequently, the specification's inverse pull-back gives an expanded base-field UOV decomposition of the unchanged verification map.

Proof. Since

$$S_{G'}^{-1} = \begin{pmatrix} I_V & -G' \\ 0 & I_O \end{pmatrix},$$

a direct block multiplication gives

$$S_{G'}^{-\top} P_i S_{G'}^{-1} = \begin{pmatrix} A_i & E_i - A_i G' \\ E_i^\top - G'^\top A_i & C_i - G'^\top E_i - E_i^\top G' + G'^\top A_i G' \end{pmatrix}.$$

The lower-right block is zero by (4); the remaining blocks are exactly F'_i . Rearranging proves the displayed factorization.

To make the interface with the specification explicit, let hats denote the corresponding base-field matrices. The forward pull-back of Section 4.1 of the specification has the form

$$P_i = \phi\left((W_f^{(N)})^{-1} \hat{P}_i\right), \quad F'_i = \phi\left((W_f^{(N)})^{-1} \hat{F}'_i\right), \quad S_{G'} = \phi(\hat{S}_{G'}).$$

Hence the inverse pull-back sets

$$\hat{P}_i = W_f^{(N)} \phi^{-1}(P_i), \quad \hat{F}'_i = W_f^{(N)} \phi^{-1}(F'_i), \quad \hat{S}_{G'} = \phi^{-1}(S_{G'}).$$

The pull-back identity in the specification converts the verified extension-field factorization into $\hat{P}_i = \hat{S}_{G'}^\top \hat{F}'_i \hat{S}_{G'}$ for the unchanged base-field verification map [1, Sec. 4.1]. These matrix identities publicly certify an exact expanded UOV decomposition. Signing uses this base-field pull-back, not the extension-field tuple with only $O = m/3$ oil variables. The separate test in [Theorem 3.5](#) certifies that the decomposition can invert every target. $\square$

For an expanded base-field decomposition $\hat{P}_i = \hat{S}^\top \hat{F}_i \hat{S}$, write the central variables as $(a, o) \in \mathbb{F}_q^v \oplus \mathbb{F}_q^m$. Because the oil-oil block of every $\hat{F}_i$ is zero, fixing a leaves a linear map in the oil variables. Denote its matrix by

$$\mathbb{L}_{\hat{F}}(a) \in \mathbb{F}_q^{m \times m}, \quad (\mathbb{L}_{\hat{F}}(a)o)_i := \hat{F}_i(a, o) - \hat{F}_i(a, 0). \quad (5)$$

Proposition 3.5 (Public signing witness). If $\det \mathbb{L}_{\hat{F}}(a) \neq 0$, then $(\hat{F}, \hat{S}, a)$ is a publicly checkable signing certificate. For every target $y \in \mathbb{F}_q^m$, one computes

$$o = \mathbb{L}_{\hat{F}}(a)^{-1}(y - \hat{F}(a, 0)), \quad x = \hat{S}^{-1}(a, o),$$

and obtains $\hat{P}(x) = y$.

Proof. Direct substitution gives $\hat{F}(a, o) = y$, followed by $\hat{P}(x) = \hat{F}(\hat{S}x) = y$. These signatures verify under the unchanged public map; matching the official signing distribution is neither needed nor claimed. $\square$

Corollary 3.6 (Key-only universal chosen-message forgery). Given a publicly verified signing certificate from [Theorem 3.5](#), an attacker can forge an accepted QR-UOV signature on every chosen message without querying a signing oracle. Concretely, for a message M , choose any allowed salt r , compute the official target

$$\mu = \text{SHAKE256}(\text{seed}_{\text{pk}} \parallel \text{BytesToBits}(M), 512), \quad y = \text{Hash}(\mu, r),$$

invert y using [Theorem 3.5](#), and output (r, x) . The official verifier accepts because $\hat{P}(x) = \text{Hash}(\mu, r)$.

Proof. The specification’s verifier recomputes μ and $\text{Hash}(\mu, r)$ and accepts exactly when the public map evaluated at the signature vector equals that target [1, Secs. 4.2–4.3]. [Theorem 3.5](#) inverts every target under the unchanged public map. No distributional claim about the forged signature is needed for verification, and the secret seed is never recovered. $\square$

Fix a nonzero $\xi \in \mathbb{F}_q^\ell$. For each quotient-ring vinegar block $1 \leq j \leq V$, let a_j place ξ in block j and zero elsewhere, and set

$$M_j := \mathbf{L}_{\hat{F}}(a_j) \in \mathbb{F}_q^{m \times m}.$$

For $0 \leq r \leq m$, write

$$\varrho_r := \Pr_{M \leftarrow \mathbb{F}_q^{m \times m}}[\text{rank } M = r] = q^{-m^2} \prod_{i=0}^{r-1} \frac{(q^m - q^i)^2}{q^r - q^i}. \quad (6)$$

In particular,

$$\pi_m := \varrho_m = \prod_{k=1}^m (1 - q^{-k}), \quad \sigma_m := 1 - \pi_m. \quad (7)$$

For the stronger fallback, define

$$\chi_m := q^{-(m-1)} + (1 - q^{-(m-1)})q^{-1}, \quad \zeta_{\text{pair}} := \varrho_{m-1}q^{-1}\chi_m + (1 - \varrho_m - \varrho_{m-1})(1 - \varrho_m). \quad (8)$$

Proposition 3.7 (Basis-first deterministic signing search). For the planted decomposition in the public-random expansion model, the matrices $M_1, \dots, M_V$ are independent and uniform in $\mathbb{F}_q^{m \times m}$. Consequently:

1. Testing the basis assignments $a_1, a_2, \dots$ in order uses fewer than

$$\pi_m^{-1} < 1.008$$

basis tests on average, stopping at the first success or after all V tests. It fails to find an invertible matrix exactly when all V basis matrices are singular, an event of probability

$$\eta_{\text{basis}} := \sigma_m^V, \quad (9)$$

whose base-two logarithms at Levels I, III, and V are

$$-362.823, \quad -530.280, \quad -711.692.$$

2. Much shorter fixed lists already suffice at the category scale. Testing the first

$$b_{\text{sign}} = 22, \quad 31, \quad 40$$

basis assignments at Levels I, III, and V fails with probabilities $\sigma_m^{b_{\text{sign}}}$, whose base-two logarithms are

$$-153.502, \quad -216.298, \quad -279.095. \quad (10)$$

3. Pair the basis matrices as $(M_1, M_2), (M_3, M_4), \dots$. Let $\mathcal{E}_{\text{sign}}$ be the event that every designated pair span contains only singular matrices. Then

$$\Pr[\mathcal{E}_{\text{sign}}] \leq \eta_{\text{sign}} := \zeta_{\text{pair}}^{\lfloor V/2 \rfloor}, \quad (11)$$

with base-two logarithms below

$$-544.820, \quad -796.276, \quad -1068.687.$$

Outside $\mathcal{E}_{\text{sign}}$, a deterministic fallback succeeds after at most

$$V + \lfloor V/2 \rfloor (q - 1) \quad (12)$$

oil-matrix tests: first test every M_j ; if necessary, for each pair (A, B) test $A + tB$ for all $t \in \mathbb{F}_q^\times$. The basis tests already cover the two remaining projective points A and B .

For any fixed exact candidate, define its oil-matrix determinant polynomial

$$\Delta(c_1, \dots, c_V) := \det \left(\sum_{j=1}^V c_j M_j \right).$$

Whenever Δ is nonzero, a uniform scalar-block vinegar assignment is invertible with probability at least $1 - m/q$. The corresponding expected numbers of assignments are at most

$$\frac{q}{q - m} = 1.740, \quad 2.592, \quad 5.773$$

at Levels I, III, and V. This last statement is key by key. For a recovered candidate not known to be planted, every successful determinant test is still an exact public signing certificate.

Proof. Write $b_{i,j,k} \in K = \mathbb{F}_{q^\ell}$ for entry (j, k) of the planted extension-field block B_i . Under the public-random experiment these elements are independent and uniform. The (i, k) block of the oil coefficient row produced by a_j is

$$2\xi^\top W_f \Phi_f(b_{i,j,k}) \in \mathbb{F}_q^{1 \times \ell}.$$

The factor 2 is invertible. The $\mathbb{F}_q$ -linear map

$$K \longrightarrow \mathbb{F}_q^{1 \times \ell}, \quad b \longmapsto \xi^\top W_f \Phi_f(b)$$

is injective: if $b \neq 0$, then $\Phi_f(b)$ is invertible, while $\xi^\top W_f \neq 0$. Equal dimensions make it an isomorphism. Thus each M_j is uniform, and distinct j use disjoint entries of the independent blocks B_i ; hence the M_j are independent.

A basis test succeeds with probability π_m . Truncating a geometric random variable at V gives expected test count

$$\sum_{j=0}^{V-1} \sigma_m^j = \frac{1 - \sigma_m^V}{\pi_m} < \pi_m^{-1}.$$

Independence gives σ_m^b for every fixed prefix of length b , yielding all displayed basis-list probabilities.

It remains to bound one pair. Let A, B be independent uniform $m \times m$ matrices, and let $E_{A,B}$ be the event that their span contains no invertible matrix. If A is invertible, this event is impossible. If $\text{rank } A = m - 1$, invertible left and right changes of basis reduce A to $\text{diag}(I_{m-1}, 0)$ without changing the distribution of B . Write

$$B = \begin{pmatrix} C & b \\ c^\top & d \end{pmatrix}.$$

On $E_{A,B}$, the polynomial $\det(A + tB)$ vanishes for every $t \in \mathbb{F}_q$. Its degree is at most $m < q$, so it is the zero polynomial. Its linear coefficient is d , and, after $d = 0$, its quadratic coefficient is $-c^\top b$. These conditions have probability at most $q^{-1}\chi_m$. If $\text{rank } A \leq m - 2$, then $E_{A,B}$ implies that B is singular, an event of probability $1 - \varrho_m$. Conditioning on the rank of A gives $\Pr[E_{A,B}] \leq \zeta_{\text{pair}}$. The designated pairs are independent, proving (11).

The basis matrices and $A + tB$ for $t \in \mathbb{F}_q^\times$ enumerate every projective point on each designated pair line, so the deterministic scan succeeds whenever one pair span contains an invertible matrix. Finally, Δ has total degree m . If it is nonzero, Schwartz–Zippel gives success probability at least $1 - m/q$ for a uniform coefficient vector. $\square$

4 Projective target-orthogonal direct forgery

The directional attack uses QR-UOV’s extension-field graph. This section gives a second route that works directly with the official base-field verification map

$$\hat{P} : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q^m, \quad n = v + m, \quad q = 127.$$

It does not recover a key. Given a nonzero target y , it constructs a public subspace on which three target equations disappear. Full output normalization then leaves $m - 4$ homogeneous zero-target equations in projective dimension $m - 4$; one final equation only recovers the affine scale. The square solver cores are

$$50, \quad 74, \quad 101$$

for Levels I, III, and V.

The preceding affine dimensions 51, 75, 102 coincide with the $\alpha = 3$ Hashimoto line already used in the QR-UOV specification’s direct-attack analysis [29, 1]. We do not claim those dimensions as new. The further one-variable reduction follows from two facts specific to the homogeneous verification problem: every target but one can be made zero, and the remaining zero equations are invariant under nonzero scalar multiplication. The other new ingredients are a fixed-key seed bound, a finite Chevalley–Warning construction, a second-moment lower bound for an eligible nonsingular projective root, and a characteristic-127 solver ledger independent of WXL semi-regularity estimates.

The algebraic reduction below is key by key. Its probability calculation uses the independent-graph public-random model of Theorem 3.2: the hidden graph G is independent of the product-uniform central coefficient pairs (A_i, B_i) . This independence is needed only for the direct attack’s average-case root analysis; the directional results continue to allow arbitrary G – A correlation. In the ideal-XOF and ideal-cipher models, official public and secret expansion have this independence whenever their independently sampled seeds differ. Section 24 adds the seed-collision and buffer terms explicitly. No such transfer is claimed for the fixed SHAKE or AES functions.

4.1 The pull-back still supplies full first-order randomness

The official degree-three pull-back produces a restricted family of base-field quadrics, not the set of all UOV quadrics over $\mathbb{F}_q$. We therefore record the exact local property needed below.

Let $E = K^{V+O}$, viewed as an $n = 3(V + O)$ dimensional vector space over $\mathbb{F}_q$, and let $\hat{O} \subset E$ be the hidden $m = 3O$ dimensional base-field oil space. In central coordinates, one public-random expanded form is obtained from

$$p(v, o) = v^\top A v + 2v^\top B o$$

with $A \in \text{Sym}_V(K)$ and $B \in K^{V \times O}$ uniform. The specification's pull-back evaluates a fixed nonzero $\mathbb{F}_q$ -linear functional of $p(v, o)$. Equivalently, after choosing a nonzero $\gamma \in K$, it has the form

$$Q(v, o) = \text{Tr}_{K/\mathbb{F}_q}(\gamma p(v, o)). \quad (13)$$

The precise value of γ depends on the chosen regular-representation basis and is irrelevant; nondegeneracy of the trace pairing is what matters.

Lemma 4.1 (Evaluation and one-jet surjectivity). Let $x \in E \setminus \widehat{\mathcal{O}}$, and let $U \subseteq E$ be an $\mathbb{F}_q$ -linear subspace containing x . For a uniform pull-back form Q as in (13):

1. $Q(x)$ is uniform in $\mathbb{F}_q$;
2. the map

$$Q \longmapsto (Q(x), dQ_x|_U)$$

is onto

$$\{(a, \lambda) \in \mathbb{F}_q \times U^* : \lambda(x) = 2a\};$$

consequently, conditioned on $Q(x) = a$, the derivative $dQ_x|_U$ is uniform in the displayed affine hyperplane;

3. for $x, z \in E \setminus \widehat{\mathcal{O}}$, the two evaluation functionals $Q \mapsto Q(x)$ and $Q \mapsto Q(z)$ are proportional over $\mathbb{F}_q$ only if $z = \lambda x$ for some $\lambda \in K$ with $\lambda^2 \in \mathbb{F}_q$. Since $[K : \mathbb{F}_q] = 3$ is odd, this forces $\lambda \in \mathbb{F}_q$.

Proof. Write $x = (v_x, o_x)$ in central K -coordinates. Since x is off oil, $v_x \neq 0$. Choose a K -linear functional $\varphi : E \rightarrow K$ with $\varphi|_{\widehat{\mathcal{O}}} = 0$ and $\varphi(x) = 1$. The map

$$h \longmapsto (u \mapsto \text{Tr}_{K/\mathbb{F}_q}(\gamma h(u)))$$

from K -linear maps $E \rightarrow K$ to $\mathbb{F}_q$ -linear maps $E \rightarrow \mathbb{F}_q$ is an isomorphism, by nondegeneracy of the trace pairing and equality of dimensions.

Fix a desired pair (c, λ) with $\lambda(x) = 2c$. Choose a K -linear h whose trace functional restricts to $\lambda/2$ on U , and define

$$b_h(u, w) = \varphi(u)h(w) + h(u)\varphi(w) - h(x)\varphi(u)\varphi(w).$$

This symmetric K -bilinear form vanishes on $\widehat{\mathcal{O}} \times \widehat{\mathcal{O}}$ and satisfies $b_h(x, u) = h(u)$. Hence

$$p_h(u) := b_h(u, u), \quad Q_h(u) := \text{Tr}_{K/\mathbb{F}_q}(\gamma p_h(u))$$

is a pull-back UOV form with

$$Q_h(x) = \text{Tr}(\gamma h(x)) = c, \quad d(Q_h)_x(u) = 2 \text{Tr}(\gamma h(u)) = \lambda(u) \quad (u \in U).$$

This proves the first two claims.

For the last claim, compare coefficients in central coordinates. If the two evaluation functionals are proportional by $\rho \in \mathbb{F}_q^*$, nondegeneracy of the trace pairing gives

$$v_z v_z^\top = \rho v_x v_x^\top, \quad v_z o_z^\top = \rho v_x o_x^\top.$$

The first rank-one identity gives $v_z = \lambda v_x$ and $\lambda^2 = \rho$ for some $\lambda \in K^*$. The second then gives $o_z = \lambda o_x$, so $z = \lambda x$ and $\lambda^2 \in \mathbb{F}_q$. Finally,

$$\gcd(2(q-1), q^3-1) = q-1$$

because $q^2 + q + 1$ is odd. Thus $\lambda^{q-1} = 1$ and $\lambda \in \mathbb{F}_q$. □

4.2 Normalize every target coordinate except one

If $y = 0$, then $x = 0$ is already a valid preimage, so only the nonzero case needs analysis. Let $y \in \mathbb{F}_q^m$ be nonzero. Choose a public matrix $T_y \in \text{GL}_m(\mathbb{F}_q)$ whose first $m - 1$ rows span $y^\perp$ and whose last row has inner product one with y . Then

$$T_y y = (0, \dots, 0, 1)^\top. \quad (14)$$

Write $Q_1, \dots, Q_m$ for the corresponding output recombinations of the public forms. Since T_y is fixed by y and invertible, product-uniform public forms remain product-uniform in the public-random expanded-key model.

Only Q_1, Q_2, Q_3 are used to construct the attack subspace. Choose a uniformly random seven-dimensional subspace $S_0 \subset E$, restrict these three forms to S_0 , and exhaustively test the points of $\mathbb{P}(S_0) \simeq \mathbb{P}^6(\mathbb{F}_q)$. Chevalley–Warning guarantees a nonzero common zero because three quadrics have total degree six in seven variables. Denote the first such point by w .

The hidden oil space is not known to the attacker, so the algorithm cannot test whether w is off oil. The random seed subspace makes this unnecessary: with overwhelming probability it contains no nonzero oil vector at all.

Proposition 4.2 (A random seed misses every fixed oil space). Fix an arbitrary m -dimensional subspace $\hat{\mathcal{O}} \subset \mathbb{F}_q^n$ and choose a uniform seven-dimensional subspace $S_0 \subset \mathbb{F}_q^n$. Then

$$\Pr[S_0 \cap \hat{\mathcal{O}} \neq \{0\}] \leq \epsilon_{\text{seed}} := \frac{q^m - 1}{q - 1} \frac{q^7 - 1}{q^n - 1}. \quad (15)$$

Consequently, except with probability at most ϵ_{seed} over public attack coins, every nonzero common zero found inside S_0 is off oil. For the three official profiles,

$$\log_2 \epsilon_{\text{seed}} \leq -1048.291, \quad -1551.477, \quad -2096.594. \quad (16)$$

The bound holds for every fixed hidden oil space; it does not average over key generation.

Proof. There are $(q^m - 1)/(q - 1)$ projective oil points. For any fixed projective point $[u]$, a uniform seven-dimensional subspace contains $[u]$ with probability $(q^7 - 1)/(q^n - 1)$. A union bound proves (15). The numerical values substitute the three official pairs (n, m) . $\square$

4.3 A deterministic Chevalley–Warning extension

For a quadratic form Q over a field of odd characteristic, write

$$B_Q(u, z) = \frac{Q(u + z) - Q(u) - Q(z)}{2}$$

for its polar bilinear form. A subspace is *totally isotropic* for Q when Q vanishes identically on it.

Theorem 4.3 (Common-isotropic extension from one seed point). Let $Q_1, \dots, Q_s$ be homogeneous quadrics on $\mathbb{F}_q^n$, and let $0 \neq w$ satisfy $Q_i(w) = 0$ for every i . If

$$n \geq (s + 1)r + s, \quad (17)$$

then a deterministic finite algorithm constructs an r -dimensional subspace U containing w on which every Q_i vanishes identically.

At each extension step the algorithm uses linear algebra and examines at most

$$|\mathbb{P}^{2s}(\mathbb{F}_q)| = \frac{q^{2s+1} - 1}{q - 1} \quad (18)$$

projective points.

Proof. Suppose a j -dimensional common totally isotropic subspace W containing w has already been constructed. Put

$$L_W = \bigcap_{i=1}^s W^{\perp_{B_{Q_i}}}.$$

The sj orthogonality conditions give

$$\dim(L_W/W) \geq n - (s+1)j.$$

For $j \leq r-1$, (17) makes this dimension at least $2s+1$. Choose any $(2s+1)$ -dimensional subspace S of the quotient. The s induced quadrics on S have total degree $2s < 2s+1$. The Chevalley–Warning theorem says that their number of common zeros is divisible by the characteristic [32, 33]. Since the zero vector is one zero, there is a nonzero zero. Enumerating (18) finds one deterministically.

Lift that zero to $z \in L_W$. Then $Q_i(z) = 0$ and $B_{Q_i}(z, W) = 0$, so $W \oplus \langle z \rangle$ is again common totally isotropic. Iterating reaches dimension r . $\square$

Corollary 4.4 (Target-orthogonal affine reduction). Let $P : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ be any homogeneous quadratic map, let $y \neq 0$, and choose $1 \leq s < m$. After an invertible output recombination sends y to the last coordinate, a nonzero common zero of any s of the first $m-1$ transformed quadrics can be extended to an $(m-s)$ -dimensional common totally isotropic subspace whenever

$$n \geq (s+1)m - s^2. \quad (19)$$

On that subspace, the affine problem has $m-s$ equations in $m-s$ variables.

Proof. Apply Theorem 4.3 with $r = m-s$. Its condition becomes $n \geq (s+1)(m-s) + s = (s+1)m - s^2$. The selected s forms vanish identically on the constructed subspace and have zero target, leaving the stated affine residual. $\square$

The affine formulation leaves one variable that homogeneity makes redundant. The next corollary removes it exactly.

Corollary 4.5 (Projective target-orthogonal reduction). Assume that q is odd. Under the hypotheses of Theorem 4.4, put $r = m-s$ and let U be the constructed r -dimensional subspace. Normalize the target to $(0, \dots, 0, 1)$ and write

$$H_1, \dots, H_{r-1}, H_0$$

for the restrictions to U of the remaining transformed forms, with H_0 corresponding to the nonzero target coordinate. Then:

1. affine solutions of

$$H_1(z) = \dots = H_{r-1}(z) = 0, \quad H_0(z) = 1 \quad (20)$$

modulo the pair $z \sim -z$ are in bijection with projective points $[z] \in \mathbb{P}(U)$ satisfying

$$H_1(z) = \dots = H_{r-1}(z) = 0 \quad (21)$$

and $H_0(z)$ a nonzero square in $\mathbb{F}_q$;

2. a solution of (20) has invertible affine Jacobian if and only if its projective point is nonsingular for (21); and
3. after choosing a uniform nonzero linear form $L \in U^*$, the chart $L(z) = 1$ turns (21) into $r-1$ quadrics in $r-1$ affine variables. Any fixed projective root lies in this chart with probability

$$1 - \frac{q^{r-1} - 1}{q^r - 1} > 1 - \frac{1}{q}.$$

Thus the square solver core has dimension $m - s - 1$ and geometric degree at most 2^{m-s-1} .

Proof. For a nonzero common zero z of $H_1, \dots, H_{r-1}$, replacing z by λz multiplies every quadratic value by λ^2 . Hence some representative satisfies $H_0(\lambda z) = 1$ exactly when $H_0(z)$ is a nonzero square, and then there are exactly two choices λ and $-\lambda$.

Euler's identity gives $dH_i(z)(z) = 2H_i(z) = 0$ for $1 \leq i < r$, whereas $dH_0(z)(z) = 2$. The first $r - 1$ Jacobian rows therefore have rank $r - 1$ exactly when their common kernel is the radial line $\langle z \rangle$; in that case the last row is not in their span and the affine Jacobian has rank r . Conversely, an invertible affine Jacobian forces the first $r - 1$ rows to have rank $r - 1$. Finally, among the $q^r - 1$ nonzero linear forms on U , exactly $q^{r-1} - 1$ vanish at a fixed projective point. Dehomogenizing in a chart that contains the point preserves nonsingularity and gives the claimed square affine system. $\square$

For the QR-UOV parameters, set $s = 3$ and $r = m - 3$. The subspace condition is

$$n \geq 4m - 9. \quad (22)$$

The three official profiles give

$$(n, 4m - 9) = (210, 207), \quad (306, 303), \quad (411, 411).$$

The resulting projective solver dimensions are

$$k = r - 1 = m - 4 \in \{50, 74, 101\}. \quad (23)$$

The seed root and the extension are both obtained by finite exhaustive searches, so this algebraic reduction is unconditional. One seed search and the $r - 1$ extension steps each evaluate three seven-variable quadrics on at most $|\mathbb{P}^6(\mathbb{F}_{127})|$ points. The companion audit additionally overcharges uniform seed generation, recomputation of every polar-orthogonal space by dense elimination, the random chart, and the complete success normalization. The resulting preprocessing bounds are

$$2^{63.545}, \quad 2^{64.077}, \quad 2^{64.501}$$

Boolean gates, negligible beside every terminal solver estimate below.

4.4 An eligible nonsingular projective root with constant probability

Let U be the subspace from [Theorem 4.3](#), and choose an $\mathbb{F}_q$ -basis to identify U with $\mathbb{F}_q^r$. Under the normalization (14), write

$$H_j = Q_{3+j}|_U \quad (1 \leq j \leq r - 1), \quad H_0 = Q_m|_U.$$

The affine residual is

$$H_1(z) = \dots = H_{r-1}(z) = 0, \quad H_0(z) = 1, \quad r = m - 3. \quad (24)$$

The projective solver uses only the first $r - 1$ equations; H_0 filters and scales their projective roots.

The next statement gives an explicit success probability rather than assuming that a random-looking square system has a root. Let

$$\pi_{r-1}(q) = \prod_{j=1}^{r-1} (1 - q^{-j}). \quad (25)$$

Proposition 4.6 (Affine nonsingular-root probability). Condition on the selected forms, the attack coins used to construct U , and an off-oil seed point. Put

$$d = \dim_{\mathbb{F}_q}(U \cap \hat{\mathcal{O}}), \quad \mu = 1 - q^{d-r}.$$

Then $d \leq r - 1$. Over the independent remaining public-random QR-UOV forms, the probability that (24) has an $\mathbb{F}_q$ -rational root with invertible $r \times r$ Jacobian is at least

$$\frac{\mu \pi_{r-1}(q)^2}{2 + \mu - (q - 1)q^{-r}}. \quad (26)$$

For $q = 127$ and each of $r = 51, 75, 102$, this is at least

$$p_* = 0.326336998767509 \dots \quad (27)$$

Proof. The off-oil seed point belongs to U , so U is not contained in $\hat{\mathcal{O}}$ and $d \leq r - 1$. Let $S = U \setminus \hat{\mathcal{O}}$ and let M count all roots of (24) in S . For a fixed $x \in S$, Theorem 4.1 gives

$$\Pr[x \text{ is a root}] = q^{-r}.$$

Hence $\mathbb{E}[M] = |S|q^{-r} = \mu$.

For two off-oil points, the evaluation functionals are independent unless the points are scalar multiples. If $z = \lambda x$ and both solve (24), homogeneity and the last equation give $\lambda^2 = 1$. By Theorem 4.1, the only dependent ordered pairs that can both solve the target are therefore $z = x$ and $z = -x$; the other $q - 3$ nontrivial base-field multiples have joint probability zero. Every nonproportional ordered pair has joint probability q^{-2r} . Consequently,

$$\mathbb{E}[M^2] = 2\mu + \mu^2 - (q - 1)\mu q^{-r}. \quad (28)$$

Let N count only roots with invertible Jacobian. At a fixed root x , the conditional one-jet statement makes the Jacobian rows independent and uniform subject only to Euler's relation $dQ_x(x) = 2Q(x)$. The target vector is nonzero, so one column is fixed and nonzero while the other $r - 1$ columns are uniform. Thus

$$\Pr[J(x) \text{ invertible} \mid x \text{ is a root}] = \pi_{r-1}(q),$$

and $\mathbb{E}[N] = \mu \pi_{r-1}(q)$. Since $N^2 \leq M^2$, the second-moment inequality gives

$$\Pr[N > 0] \geq \frac{\mathbb{E}[N]^2}{\mathbb{E}[M^2]},$$

which is (26). The right-hand side increases with μ , so use $\mu \geq 1 - q^{-1}$ and substitute the three values of r . $\square$

Corollary 4.7 (Eligible projective-root probability). Under the same conditioning, the probability that the projective system

$$H_1 = \dots = H_{r-1} = 0 \quad \text{in } \mathbb{P}^{r-1}(\mathbb{F}_q)$$

has a nonsingular point $[z]$ for which $H_0(z)$ is a nonzero square is at least the bound in (26), and hence at least p_* for every official profile.

Proof. By Theorem 4.5, every affine nonsingular root of (24) maps to an eligible nonsingular projective root; the two affine roots z and $-z$ map to the same projective point. Existence of an affine root therefore implies the displayed projective event. $\square$

The factor $1/p_* = 3.0643169600\dots$ costs 1.615566 bits in the work-to-success ratio. A random chart adds less than $\log_2(127/126) = 0.011405$ bits. Unlike a Poisson heuristic, the root bound follows from the QR-UOV pull-back distribution and a second-moment argument. The seed-disjointness statement is fixed-key; the residual-root statement averages over the independent-graph idealized expansion model. We do not claim that repeated invocations on every fixed seeded key are independent.

4.5 Verified projective direct forgery

Theorem 4.8 (Projective target-orthogonal QR-UOV forgery). Work in the independent-graph public-random model and the official degree-three pull-back. For any nonzero target y , the following public algorithm has one-sided error.

1. Normalize the target to $(0, \dots, 0, 1)$ as in (14).
2. Choose a uniform seven-dimensional seed subspace, enumerate its projective points, and take a nonzero common zero of the first three forms.
3. Use Theorem 4.3 to construct an $(m - 3)$ -dimensional common totally isotropic subspace for those forms.
4. Choose a uniform nonzero chart form L on that subspace. Dehomogenize the $m - 4$ remaining zero-target quadrics in the chart $L = 1$ and apply Theorem 7.6 to the resulting square system in

$$k = m - 4$$

variables.

5. For each represented $\mathbb{F}_q$ -rational root, recover its projective point, test whether the last quadratic value is a nonzero square, scale to make that value one, reverse the public transformations, and evaluate the unchanged public map.

Every returned vector is a valid signature preimage. Over the independent-graph public-random key and the public attack coins, one complete invocation reaches an off-oil residual with a represented eligible $\mathbb{F}_q$ -rational nonsingular projective root with probability at least

$$(1 - \epsilon_{\text{seed}})p_* \left(1 - \frac{1}{q}\right) \left(1 - \frac{2^{2k}}{|k'|}\right), \quad k = m - 4, \quad (29)$$

where $k'/\mathbb{F}_q$ is the solver working extension. Taking

$$[k' : \mathbb{F}_q] = 15, \quad 22, \quad 30$$

at Levels I, III, and V gives the finite costs reported in Section 19. The inverse of (29) is used as a conservative work-to-success normalization; it is not asserted to be a fixed-key Las Vegas restart factor.

Proof. The first $m - 1$ target coordinates are zero. By Theorem 4.2, the seed subspace is disjoint from the hidden oil space with probability at least $1 - \epsilon_{\text{seed}}$ for every fixed key. Chevalley–Warning then supplies an off-oil seed root, and Theorem 4.3 constructs the advertised subspace. Conditional on the selected first three forms, the graph, the attack coins, and this off-oil seed point, the remaining transformed forms are still independent uniform pull-back forms. Theorem 4.7 supplies an eligible nonsingular projective root with probability at least p_* . A random nonzero chart contains a fixed such root with probability greater than $1 - 1/q$. The finite-characteristic solver includes every root in the chart with the separator probability in (29). Square testing and scaling recover the associated affine solutions exactly. Re-evaluation of all m original public forms rejects every malformed or extraneous candidate, so the algorithm cannot output an invalid signature. $\square$

4.6 The fixed-output security scissors

The direct reduction becomes especially informative when combined with the directional equivalent-key attack. Let the quotient degree be ℓ , so that

$$n = \ell V + m.$$

Applying [Theorem 4.4](#) with a general value of s gives the required $(m - s)$ -dimensional common-isotropic subspace whenever

$$\ell V \geq s(m - s). \quad (30)$$

By [Theorem 4.5](#), the corresponding nonzero-target solver core has dimension $m - s - 1$.

Theorem 4.9 (Projective security scissors at fixed output count). Work over a field of odd characteristic. Fix m and quotient degree ℓ . For every nonzero target, within the two attack families analyzed in this paper an algebraic core of dimension at most $m - \ell - 1$ is available for every integer vinegar count V :

$$r_{\text{attack}}(V) \leq m - \ell - 1. \quad (31)$$

More precisely:

1. if $V \leq m - \ell - 1$, directional equivalent-key recovery uses a square core of dimension V ;
2. if $V \geq m - \ell$, target-orthogonal slicing with $s = \ell$, followed by projectivization, uses a square core of dimension $m - \ell - 1$.

These cases cover every integer V . For QR-UOV, $\ell = 3$, so the ceiling is $m - 4$.

Proof. In the first case, $V < m$, so the m directional equations admit a V -equation square recombination and the planted-root attack of [Section 5](#) has core $V \leq m - \ell - 1$. In the second case,

$$\ell V \geq \ell(m - \ell),$$

which is (30) with $s = \ell$. The affine direct residual has dimension $m - \ell$ and [Theorem 4.5](#) removes its radial coordinate, leaving core $m - \ell - 1$. Since consecutive integers $m - \ell - 1$ and $m - \ell$ meet the two inequalities, no case is omitted. $\square$

The theorem is an algebraic-core statement, not a claim that running time depends only on the number of variables. For the official degree-three sets, [Theorems 4.2](#) and [4.7](#) supply the success factors charged in the concrete ledger. Its design implication is exact: once V reaches $m - \ell$, increasing V while holding m fixed no longer enlarges the projective direct-forgery core. Conversely, decreasing V into the other case makes directional recovery no larger than the same ceiling. Moving both attack lines therefore requires changing m as well as V , or modifying the construction so that one reduction no longer applies.

4.7 Exact small-parameter execution

The companion program `QRUOV_direct_qr_toy.py` instantiates the degree-three model over $\mathbb{F}_{3^3}$, generates independent extension-field UOV forms, pulls each back by an explicit trace, hides the oil space with a random extension-field change of variables, and performs the public attack entirely over $\mathbb{F}_3$. It now executes the projective reduction itself: after building U , it enumerates the projective roots of the remaining zero-target forms and recovers the affine scale from the final equation. In the frozen 200-key run with

$$(V, O, n, m, r, k) = (3, 2, 15, 6, 3, 2),$$

199 keys yielded a verified forgery within twelve outer trials. Every returned vector passed both the base-field public map and an independent extension-field evaluator; the remaining key produced no candidate within the fixed trial budget. 185 of the 200 final seed subspaces were disjoint from the hidden oil space, while all 200 selected seed points were off oil. The distinction is expected at this tiny field size: seed-space disjointness is a sufficient production-scale guarantee, not a condition that the public algorithm tests. This is a correctness regression test, not a production benchmark.

Remark 4.10 (Relation to Hashimoto and the specification). The affine reduction to $m - 3$ variables is prior art. Hashimoto’s underdetermined-MQ method and the QR-UOV specification already analyze those residual sizes [29, 1]. The exact projective quotient to $m - 4$ variables is additional: it uses full target normalization, solves only the homogeneous zero-target equations modulo scaling, and lets the final nonzero-target equation reconstruct the affine representative. The other upgrades are the finite Chevalley–Warning construction, fixed-key seed bound, explicit degree-three nonsingular-root probability, and finite-characteristic solver cost.

5 Graph-direction and derivative-kernel recovery

The attack has two local endpoints. The graph-direction endpoint is the stronger structural statement: one affine root determines the complete hidden graph by public linear algebra. The derivative-kernel endpoint is retained for the faster projective route. Both endpoints are followed by the same exact graph, pull-back, factorization, and signing checks.

5.1 A public graph direction produces a planted root

Fix a direction $c \in K^O$. Substitute the public oil coordinate c and the negated vinegar coordinate $-x$ into the public form P_i :

$$q_{i,c}(x) := P_i(-x, c) = x^\top A_i x - 2x^\top E_i c + c^\top C_i c, \quad x \in K^V. \quad (32)$$

The system $q_{1,c} = \dots = q_{m,c} = 0$ is public and has only V variables. Define the public $m \times V$ matrix

$$\mathcal{D}_c(x) := \begin{pmatrix} (E_1 c - A_1 x)^\top \\ \vdots \\ (E_m c - A_m x)^\top \end{pmatrix}. \quad (33)$$

Recall $B_i = E_i - A_i G$, and write $\mathcal{B}(c)$ for the matrix whose i th row is $(B_i c)^\top$.

Proposition 5.1 (Graph-direction identity). For every $c \in K^O$, the point

$$x_c := Gc$$

is a common zero of the public system (32). More precisely, for every $z \in K^V$,

$$q_{i,c}(Gc + z) = z^\top A_i z - 2(B_i c)^\top z. \quad (34)$$

Consequently,

$$\mathcal{D}_c(x_c) = \mathcal{B}(c), \quad J_q(x_c) = -2\mathcal{B}(c).$$

Proof. Expand (32) at $x = Gc + z$. Its constant term is

$$c^\top (G^\top A_i G - G^\top E_i - E_i^\top G + C_i) c = 0$$

by (4). Its linear term is $2z^\top (A_i G - E_i) c = -2(B_i c)^\top z$, proving (34). The two matrix identities follow by differentiation and by $E_i - A_i G = B_i$. $\square$

Proposition 5.2 (One regular direction determines the entire graph). Let $c \in K^O$, let x be a common zero of all the equations $q_{i,c}$, and suppose $\text{rank } \mathcal{D}_c(x) = V$. For $d \in K^O$, define

$$b_{c,d}(x)_i := c^\top C_i d - x^\top E_i d \quad (1 \leq i \leq m). \quad (35)$$

Then the overdetermined public system

$$\mathcal{D}_c(x)y = b_{c,d}(x) \quad (36)$$

has at most one solution $y \in K^V$. If $x = Gc$, its unique solution is $y = Gd$. Thus solving (36) for the O coordinate vectors $d = e_j$ recovers G exactly. Requiring every identity (4) then gives a public polynomial-time certificate of a valid expanded UOV decomposition.

Proof. Full column rank gives uniqueness. For $x = Gc$, multiply (4) on the left by $c^\top$ and on the right by d . Rearrangement yields

$$(E_i c - A_i Gc)^\top Gd = c^\top C_i d - (Gc)^\top E_i d,$$

which is exactly row i of (36). Hence Gd is the unique solution. The final identities are precisely the hypotheses of Theorem 3.4. $\square$

Proposition 5.3 (Several partial-rank directions can be combined). Let $c_1, \dots, c_r \in K^O$ and suppose that roots $x_j = Gc_j$ of the corresponding public direction systems have been found. Stack their public completion matrices and right-hand sides:

$$\mathcal{D}_c(\mathbf{x}) := \begin{pmatrix} \mathcal{D}_{c_1}(x_1) \\ \vdots \\ \mathcal{D}_{c_r}(x_r) \end{pmatrix}, \quad b_{c,d}(\mathbf{x}) := \begin{pmatrix} b_{c_1,d}(x_1) \\ \vdots \\ b_{c_r,d}(x_r) \end{pmatrix}.$$

If the stacked matrix has column rank V , then for every $d \in K^O$ the public system

$$\mathcal{D}_c(\mathbf{x})y = b_{c,d}(\mathbf{x})$$

has the unique solution $y = Gd$. Thus no individual direction needs full rank: enough complementary derivative information across several recovered direction roots also determines the whole graph.

Proof. For each j , the proof of Theorem 5.2 shows that Gd satisfies the j th block of equations. It therefore satisfies the stacked system. Full column rank gives uniqueness. $\square$

Remark 5.4 (A compact attack certificate). A successful graph-direction branch can be certified by the tuple (c, x, G) together with one full-rank minor of $\mathcal{D}_c(x)$. A verifier checks the m equations $q_{i,c}(x) = 0$, the O linear systems (36), all m graph identities, the inverse pull-back, and the signing witness. No claim about the other roots or about completeness of the solver output enters this certificate.

5.2 How often a direction is regular, and how to choose a square core

Put

$$\rho_{m,V}(Q) := \prod_{a=0}^{V-1} (1 - Q^{a-m}), \quad \tau_{m,V}(Q) := 1 - \rho_{m,V}(Q). \quad (37)$$

Thus $\rho_{m,V}(Q)$ is the full-column-rank probability of a uniform $m \times V$ matrix over K .

The direction system has m equations but only V variables. The obvious deterministic reduction to a square system tries all $\binom{m}{V}$ row subsets. The next lemma replaces that list by an optimal-size family of public linear combinations.

Index the m equations by $r = 0, \dots, m-1$. For $0 \leq a < V$, write $(r)_a = r(r-1) \cdots (r-a+1)$, with $(r)_0 = 1$, and define

$$\mathbf{C}(t)_{a+1, r+1} := (r)_a t^{r-a} \quad (t \in K), \quad (38)$$

where the entry is zero when $r < a$. The first V columns of $\mathbf{C}(t)$ are triangular with diagonal $0!, 1!, \dots, (V-1)!$. They are therefore nonsingular because $V-1 < 127 = \text{char } K$.

Lemma 5.5 (A one-parameter rank condenser). Let $B \in K^{m \times V}$ have column rank V , where $m \leq 127$. Then

$$W_B(T) := \det(\mathbf{C}(T)B) \quad (39)$$

is a nonzero polynomial of degree at most $V(m-V)$.

Proof. For column j of B , form the polynomial $f_j(T) = \sum_{r=0}^{m-1} B_{r+1,j} T^r$. The columns of B are independent exactly when the polynomials $f_1, \dots, f_V$ are independent, and $\mathbf{C}(T)B$ is their ordinary Wronskian matrix.

Choose another basis $g_1, \dots, g_V$ of their span with distinct increasing degrees $0 \leq d_1 < \dots < d_V < m$. The leading coefficient of the Wronskian of the g_j is, up to the nonzero product of their leading coefficients,

$$\det((d_j)_a)_{0 \leq a < V, 1 \leq j \leq V} = \prod_{i < j} (d_j - d_i).$$

This is nonzero in K because all d_j are distinct and below 127. A change of polynomial basis only multiplies the Wronskian by a nonzero scalar, so W_B is nonzero. Finally,

$$\deg W_B \leq \sum_{j=1}^V d_j - \frac{V(V-1)}{2} \leq V(m-V),$$

where the last bound uses the V largest possible distinct degrees below m . $\square$

Proposition 5.6 (Key-by-key sampling and public-random density). For a fixed expanded key, suppose some $V \times V$ minor of $\mathcal{B}(c)$ is not the zero polynomial in c . A fresh uniform $c \leftarrow K^O$ then satisfies $\text{rank } \mathcal{B}(c) = V$ with probability at least

$$1 - \frac{V}{Q}. \quad (40)$$

Conditional on this event, a fresh uniform $t \leftarrow K$ makes $\mathbf{C}(t)\mathcal{B}(c)$ invertible with probability at least

$$1 - \frac{V(m-V)}{Q}. \quad (41)$$

In the public-random model, $\mathcal{B}(c)$ is a uniform $m \times V$ matrix for every fixed nonzero $c \in K^O$. In particular, one preselected coordinate direction fails with probability $\tau_{m,V}(Q)$, whose base-two logarithm is

$$-62.898161, \quad -62.898161, \quad -83.864216$$

at Levels I, III, and V. More strongly, the O coordinate-direction matrices $\mathcal{B}(e_1), \dots, \mathcal{B}(e_O)$ are independent uniform matrices. Therefore

$$\Pr[\text{rank } \mathcal{B}(e_j) < V \text{ for every } j] = \tau_{m,V}(Q)^O, \quad (42)$$

with base-two logarithms

$$-1132.166907, \quad -1635.352198, \quad -2935.247544.$$

Proof. Every maximal minor of $\mathcal{B}(c)$ is homogeneous of degree V in c , so Schwartz–Zippel gives (40). Conditional on full rank, [Theorem 5.5](#) and the univariate root bound give (41).

Under [Theorem 3.1](#), each B_i is an independent uniform linear map $K^O \rightarrow K^V$. For fixed nonzero c , the vector $B_i c$ is uniform in K^V , independently across i . For the coordinate directions, the vectors $B_i e_j$ use independent columns of the B_i , proving both the fixed-direction law and (42). The displayed values are exact evaluations of the rank formulas. $\square$

Corollary 5.7 (A short deterministic public list). Fix any public list of $V(m - V) + 1$ distinct elements of K . Whenever $\mathcal{B}(c)$ has column rank V , at least one t on the list makes the square system

$$\mathbf{C}(t) (q_{1,c}, \dots, q_{m,c})^\top = 0 \quad (43)$$

have the planted root as a nonsingular solution.

For one fixed coordinate direction, the list sizes at Levels I, III, and V are

$$105, \quad 153, \quad 307.$$

Scanning all coordinate directions uses respectively

$$1,890, \quad 3,978, \quad 10,745$$

square systems, and fails in the public-random model only with probability $\tau_{m,V}(Q)^O$. Thus all outer choices can be deterministic; only the polynomial-system solver needs random coins.

Proof. The determinant in (39) has degree at most $V(m - V)$ and cannot vanish at every point of a longer distinct list. At the planted root, the Jacobian of (43) is $-2\mathbf{C}(t)\mathcal{B}(c)$ by [Theorem 5.1](#). The numerical counts use $m - V \in \{2, 2, 3\}$. $\square$

Corollary 5.8 (Category-scale deterministic direction lists). Use the first

$$r_{\text{dir}} = 3, \quad 4, \quad 4$$

coordinate directions at Levels I, III, and V, and for each direction use the condenser list from [Theorem 5.7](#). The resulting fully public lists contain

$$315, \quad 612, \quad 1228$$

square systems. In the public-random model, the probability that none has the planted point as a nonsingular root is exactly $\tau_{m,V}(Q)^{r_{\text{dir}}}$, with base-two logarithms

$$-188.694, \quad -251.593, \quad -335.457. \quad (44)$$

These bounds are already below the Level-I, Level-III, and Level-V category exponents 143, 207, 272. Scanning all O coordinate directions gives the stronger probabilities in (42).

Proof. The coordinate-direction derivative matrices are independent by [Theorem 5.6](#). If one of the first r_{dir} matrices has full column rank, [Theorem 5.7](#) supplies a condenser value with nonsingular planted Jacobian. Multiplying the one-direction failure logarithm by r_{dir} gives (44); multiplying the per-direction list lengths 105, 153, 307 gives the displayed system counts. $\square$

Remark 5.9 (The list length is geometrically optimal). The Grassmannian of V -subspaces of an m -dimensional space has dimension $V(m - V)$. For any fixed full-rank linear map $R : K^m \rightarrow K^V$, the subspaces on which R is singular form a hyperplane section in Plücker coordinates. Over an algebraic closure, at most $V(m - V)$ such hyperplane sections always have a common point. Hence no family of at most $V(m - V)$ linear recombinations can work for every full-rank $m \times V$ matrix after all field extensions. The family in [Theorem 5.7](#) meets this natural lower bound exactly.

Randomized versus deterministic use. Sampling one field element t is convenient for the expected-cost theorem. The short list in [Theorem 5.7](#) removes that choice entirely and is far smaller than enumerating all V -row subsets.

5.3 Derivative-kernel recovery for the projective route

For a public common zero x , form

$$C_x = (P_1x \cdots P_mx) \in L^{(V+O) \times m}.$$

Lemma 5.10 (Derivative-kernel recovery). If $x \in \mathbb{P}(\mathcal{O}_L)$, then $\mathcal{O}_L \subseteq \ker C_x^\top$. If $\text{rank } C_x = V$, equality holds.

Proof. For $u \in \mathcal{O}_L$, (1) gives $u^\top P_i x = 0$ for every i . If $\text{rank } C_x = V$, its transpose kernel has dimension $(V + O) - V = O$. $\square$

At the oil point, a smooth restricted core has derivative rank V . Total isotropy bounds the ambient derivative rank by the same value, so [Theorem 5.10](#) recovers the complete oil space. Row reduction into the public graph chart, Frobenius descent to K , verification of (4), and [Theorem 3.4](#) then recover the expanded decomposition. Exact factorization certifies that decomposition, while [Theorem 3.7](#) supplies an invertible oil matrix whose inverse certifies target inversion.

Optional identification with the planted oil space. Neither endpoint needs uniqueness: the algorithm keeps every exactly verified candidate and tests all of them for a signing witness. [Section 25](#) separately shows that a second K -rational oil space is extraordinarily unlikely. Outside that additional event, every accepted oil space is the planted one.

6 Jacobian localization removes global regularity

Fix $c \in K^O$ and a public recombination $R \in K^{V \times m}$. Let

$$f = (f_1, \dots, f_V)^\top := R(q_{1,c}, \dots, q_{m,c})^\top$$

and put

$$h(x) := \det J_f(x). \tag{45}$$

The graph-direction condition is local: if $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, then [Theorem 5.1](#) gives

$$h(Gc) = (-2)^V \det(R\mathcal{B}(c)) \neq 0. \tag{46}$$

No assertion is needed about the system on the complementary hypersurface $h = 0$.

Lemma 6.1 (Localization gives a reduced regular sequence). Let k be a perfect field, let $f_1, \dots, f_V \in k[x_1, \dots, x_V]$, and let $h = \det J_f$. In the localization

$$S_h := k[x_1, \dots, x_V]_h,$$

the sequence $f_1, \dots, f_V$ is regular. For every $1 \leq s \leq V$, the ideal $(f_1, \dots, f_s)S_h$ is radical. Equivalently, the sequence is reduced and regular on the open set $h \neq 0$.

Proof. Let $\mathfrak{p}$ be a prime of S_h containing the first s equations. Because h is a unit in S_h , the full Jacobian matrix is invertible modulo $\mathfrak{p}$. Its first s rows are therefore linearly independent. The Jacobian criterion over a perfect field shows that the local zero set of $f_1, \dots, f_s$ is smooth of

codimension s at every such prime. Hence the localized quotient is reduced and every minimal prime has height s . The ring S_h is regular, hence Cohen–Macaulay; an ideal generated by s elements and of height s is a complete intersection. Thus the generators form a regular sequence, and smoothness gives radicality for every prefix. $\square$

Corollary 6.2 (Only the planted simple root is needed). If $\mathcal{B}(c)$ has full column rank and $R\mathcal{B}(c)$ is invertible, then Gc belongs to the locally closed set

$$\{f_1 = \dots = f_V = 0, h \neq 0\}.$$

The input tuple $(f_1, \dots, f_V, h)$ satisfies hypotheses (H1)–(H2) of the locally closed Kronecker algorithm of Giménez et al. These conclusions are independent of every root and irreducible component contained in $h = 0$.

Proof. Equation (46) puts the planted root in the open set. Theorem 6.1 is exactly the reduced-regular-sequence condition required after localization. $\square$

Remark 6.3 (Why this is a separate geometric route). The local theorem does not use the projective slice, the off-oil incidence, the cumulative-degree bound, the joint discriminant, smoothness of the full projective core, or radicality of components on $h = 0$. A failure in any of those arguments can affect the faster projective line without affecting graph-direction recovery.

7 Finite-characteristic nonsingular-root recovery

The directional reduction does not need a description of every component of the square core. It needs one isolated root at which the Jacobian is invertible. This section adapts the symbolic-homotopy algorithm of Safey El Din and Schost to that exact task in characteristic 127.

7.1 The solver target and its degree parameters

Let k be a perfect field and let

$$f = (f_1, \dots, f_V) \in k[X_1, \dots, X_V]^V$$

be a square system of affine quadrics. Write

$$Z_{\text{ns}}(f) := \{x \in \bar{k}^V : f(x) = 0, \det J_f(x) \neq 0\}. \quad (47)$$

Every point in $Z_{\text{ns}}(f)$ is isolated. Bézout’s theorem gives

$$C := 2^V, \quad C_0 := 2^V + V2^{V-1} = 2^{V-1}(V+2). \quad (48)$$

Here C bounds the number of isolated roots, while C_0 bounds the degree of the one-parameter homotopy curve. Rational reconstruction of a degree- C_0 curve uses coefficients through precision $2C_0$; the lifting ledger below therefore runs to that precision. The second formula is the coefficient sum of $(\vartheta_0 + 2\vartheta)^V$ after discarding ϑ_0^2 and ϑ^{V+1} , exactly as in the multihomogeneous degree definition of Safey El Din and Schost [26].

The published algorithm starts from a product system with known simple roots, follows its C branches symbolically, specializes at the target fiber, and removes singular outputs by a Jacobian-based CLEAN operation. Its large-characteristic hypothesis supports the particular finite families used for the start and the separator. We replace both families.

7.2 A simple product start over any sufficiently large finite field

Lemma 7.1 (Vandermonde product start). Let k' contain $2V$ distinct elements $a_1, \dots, a_{2V}$. For $a \in k'$, put

$$L_a(X) := X_1 + aX_2 + \dots + a^{V-1}X_V + a^V, \quad (49)$$

and, for $1 \leq i \leq V$, set

$$g_i(X) := L_{a_{2i-1}}(X)L_{a_{2i}}(X). \quad (50)$$

Then $g = (g_1, \dots, g_V)$ has exactly $C = 2^V$ affine roots over k' . Every root is simple. A zero-dimensional parametrization of all roots can be constructed in $O(V^2C)$ operations in k' .

Proof. Choose one node $b_i \in \{a_{2i-1}, a_{2i}\}$ from each pair. The equations $L_{b_1} = \dots = L_{b_V} = 0$ have coefficient matrix

$$\begin{pmatrix} 1 & b_1 & \dots & b_1^{V-1} \\ \vdots & \vdots & & \vdots \\ 1 & b_V & \dots & b_V^{V-1} \end{pmatrix},$$

which is Vandermonde and invertible. Hence the choice gives one solution.

At that solution, the monic polynomial

$$T^V + X_V T^{V-1} + \dots + X_2 T + X_1$$

has $b_1, \dots, b_V$ as its roots. It therefore vanishes at none of the V unchosen nodes. Consequently the solution satisfies exactly one factor in each product g_i , and different choices give different solutions. Conversely, any common zero of the products must choose at least one factor from each pair. It cannot satisfy both factors in one pair, because that would give at least $V + 1$ distinct roots of the displayed monic polynomial. Thus there are exactly 2^V roots.

At a root, row i of the Jacobian of g is the Vandermonde row belonging to the chosen node, multiplied by the nonzero value of the unchosen factor. The Jacobian is therefore an invertible Vandermonde matrix times an invertible diagonal matrix. Enumerating the 2^V choices and forming the corresponding degree- V polynomials gives the stated $O(V^2C)$ bound. $\square$

This construction uses no division by an integer and no condition on the characteristic beyond the existence of $2V$ distinct field elements. For the QR-UOV working extensions below, this condition is trivial.

7.3 A full random separator

A primitive linear form must separate finitely many branch values and avoid finitely many leading-coefficient vectors. The published proof draws from a short integer-indexed family; that is the source of its second characteristic bound. Sampling the entire dual space is simpler over a finite extension.

Lemma 7.2 (Separator over a finite extension). Let A be a domain containing the finite field k' , and let $z_1, \dots, z_s \in A^V$ be nonzero. For a uniform $\lambda = (\lambda_1, \dots, \lambda_V) \leftarrow (k')^V$,

$$\Pr[\lambda \cdot z_j = 0 \text{ for some } j] \leq \frac{s}{|k'|}. \quad (51)$$

In particular, if $s \leq C^2$, the separator succeeds with probability at least $1 - C^2/|k'|$.

Proof. Fix a nonzero vector z . Choose an index h with $z_h \neq 0$. After the other $V - 1$ coefficients of λ are fixed, at most one value of $\lambda_h \in k'$ can make $\lambda \cdot z = 0$: two such values would give a nonzero scalar in k' annihilating the nonzero domain element z_h . Thus one vector is killed with probability at most $1/|k'|$. A union bound proves the claim. $\square$

Safey El Din and Schost identify at most C^2 vectors that a well-separating form must avoid: differences between start roots, differences between relevant target specializations, and leading-coefficient vectors of branches [26, Lem. 14 and Rem. 15]. [Theorem 7.2](#) therefore replaces their integer family without changing the success criterion.

7.4 The lifting identities do not require large characteristic

For completeness, we isolate the two local identities used by precision doubling. They work over a commutative ring and do not divide by factorials or by the characteristic.

Lemma 7.3 (Newton–Hensel precision doubling). Let R be a commutative ring, let $I \subset R$ be an ideal, and let $F \in R[X_1, \dots, X_V]^V$. Suppose

$$F(x) \equiv 0 \pmod{I^m}$$

and that $J_F(x)$ is invertible modulo I^m . Put

$$x^+ := x - J_F(x)^{-1}F(x) \pmod{I^{2m}}. \quad (52)$$

Then $F(x^+) \equiv 0 \pmod{I^{2m}}$.

Proof. The correction $h = x^+ - x$ lies in I^m . Polynomial Taylor expansion gives

$$F(x+h) = F(x) + J_F(x)h + R_2(x, h),$$

where every monomial in R_2 contains at least two coordinates of h . Hence $R_2(x, h) \in I^{2m}$. The first two terms cancel by (52). No denominator is used. $\square$

Lemma 7.4 (First-order change of primitive parameter). Work modulo T^{2m} . Let $Q(T, U)$ be monic in U , let $X(T, U) \in (R[T, U]/(T^{2m}, Q))^V$, and let λ be linear. Suppose

$$e(T, U) := \lambda(X(T, U)) - U \in T^m R[T, U]/(T^{2m}, Q).$$

Define

$$X^* := X - (\partial_U X)e, \quad Q^* := Q - (\partial_U Q)e. \quad (53)$$

Then the change of parameter $U_{\text{old}} = U - e(T, U)$ represents the same lifted branches modulo T^{2m} , and

$$\lambda(X^*) \equiv U \pmod{(T^{2m}, Q^*)}. \quad (54)$$

Proof. Because $e \in T^m$, substitution of $U - e$ into any polynomial is, modulo T^{2m} , its first-order Taylor expansion. This gives exactly (53). Moreover

$$\lambda(X^*) - U = e - (\partial_U \lambda(X))e = -(\partial_U e)e.$$

Differentiation in U does not change T -adic order, so $\partial_U e \in T^m$ and the product lies in T^{2m} . Again, the argument uses no integer denominator. $\square$

7.5 Proof crosswalk for the finite-characteristic solver

Proposition 7.5 (Where the published characteristic bound is used). For a square system of quadrics over a finite perfect field, replace the start construction in Safey El Din–Schost Lemma 7 by [Theorem 7.1](#), and replace the separator family in their Lemma 14 by [Theorem 7.2](#). The remaining correctness proof of their NONSINGULARSOLUTIONS algorithm applies over the extension field without a further lower bound on the characteristic.

Proof. We follow the proof in the order in which its ingredients are used.

Start fiber. The original lower bound $\text{char } k \geq \sum_i d_i$ ensures that its integer-indexed affine forms are distinct and that the product start has the full multihomogeneous root count with invertible Jacobian. [Theorem 7.1](#) supplies the same three facts directly: degree two, exactly C explicit roots, and a simple start fiber.

Homotopy curve. The homotopy is

$$H(T, X) = (1 - T)g(X) + Tf(X).$$

The published construction saturates by the X -Jacobian determinant. Lazard’s lemma and the multihomogeneous degree bound then give a one-dimensional curve of degree at most C_0 . These arguments require the simple start just established, not an ordering of integers inside the field.

Target specialization. The target system can have fewer than C isolated nonsingular roots and may have singular or positive-dimensional components. The published proof handles this by generalized power series, explicitly because ordinary Puiseux series are not sufficient in arbitrary characteristic [\[26, Sec. 3.4\]](#). Its valuation and specialization lemmas therefore already cover the finite-field setting.

Precision doubling and primitive normalization. The global lifting step is Newton–Hensel lifting of a geometric resolution. Its local identities are [Theorems 7.3](#) and [7.4](#); interpolation and polynomial arithmetic use inverses guaranteed by the simple fiber and separating form. No factorial or characteristic-dependent scalar is introduced.

Separator and cleaning. The published well-separating condition is avoidance of at most C^2 nonzero vectors. [Theorem 7.2](#) supplies it with probability at least $1 - C^2/|k'|$. Finally, the published CLEAN call removes every point where the target Jacobian vanishes. As noted in its Remark 15, every output is therefore a subset of the nonsingular target roots; a bad separator may omit roots or cause failure, but cannot introduce an invalid point.

These replacements cover the two uses of the integer-indexed finite families. All other steps are field algebra, generalized-power-series specialization, or Jacobian cleaning over the perfect extension field. $\square$

Theorem 7.6 (Nonsingular roots in characteristic 127). Let k be a finite field and let $f = (f_1, \dots, f_V)$ be affine quadrics over k , represented by a straight-line program of size L . Let k'/k be a finite extension containing at least $2V$ elements. There is a randomized algorithm with the following properties.

1. Every returned point belongs to $Z_{\text{ns}}(f)$.
2. Every point of $Z_{\text{ns}}(f)$ is represented in the output with probability at least

$$1 - \frac{C^2}{|k'|}. \tag{55}$$

3. The arithmetic cost is

$$\tilde{O}\left(CC_0(L + 2V + V^2)V\right) \tag{56}$$

operations in k' .

In particular, $|k'| \geq 8C^2$ gives one-invocation inclusion probability at least $7/8$.

Proof. Run the symbolic-homotopy algorithm of Safey El Din and Schost with the start from [Theorem 7.1](#) and a uniform full separator from [Theorem 7.2](#). [Theorem 7.5](#) gives correctness over k' . The well-separating event has probability at least [\(55\)](#); on that event the published specialization

recovers every isolated nonsingular target branch. The final CLEAN step proves item 1 on every branch, including branches with a bad separator.

For V quadrics in one block, the multihomogeneous quantities in the published Proposition 5 are exactly C and C_0 from (48); the sum of degrees is $2V$. The new start costs $O(V^2C)$ and does not change the dominant lifting term. Substitution into that proposition gives (56). $\square$

Remark 7.7 (One-sided failure is enough for cryptanalysis). The original solver paper notes that it cannot cheaply certify completeness of a generic output. QR-UOV does not need such a certificate. We impose every unrecombined directional equation, require a K -rational root, complete the graph, and verify the unchanged public map. An omitted planted root causes a restart. An extraneous or malformed candidate is rejected.

7.6 End-to-end verified directional recovery

For a random $V \times m$ matrix R over K , put

$$f_{c,R} := R(q_{1,c}, \dots, q_{m,c})^\top.$$

If $\mathcal{B}(c)$ has rank V , then $R\mathcal{B}(c)$ is a uniform $V \times V$ matrix. Write

$$\mu_V(Q) := \prod_{j=1}^V (1 - Q^{-j}), \quad Q = |K| = 127^3. \quad (57)$$

Choose $K' = \mathbb{F}_{Q^r}$ with

$$r := \min\{s \geq 1 : Q^s \geq 8 \cdot 2^{2V}\}. \quad (58)$$

The values are

$$r = 6, \quad 8, \quad 10 \quad (59)$$

at Levels I, III, and V.

Theorem 7.8 (Verified directional equivalent-key recovery). Let a fixed expanded QR-UOV public key have a valid graph decomposition G . Suppose

1. some maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c ; and
2. the recovered planted decomposition has a nonzero signing determinant polynomial Δ from Theorem 3.7.

Then the following Las Vegas algorithm returns a publicly verified functional equivalent key.

1. Sample $c \leftarrow K^O$ and $R \leftarrow K^{V \times m}$.
2. Run Theorem 7.6 on $f_{c,R}$ over K' from (58).
3. Intersect the returned parametrization with all m original directional equations and with the K -rational locus.
4. For every retained root x , run the graph completion of Theorem 5.2; require all graph identities, the inverse pull-back, exact public-map factorization, and a signing witness.
5. Return the first accepted key; otherwise restart with fresh public coins.

One invocation includes the planted branch with probability at least

$$p_{\text{dir}} \geq \left(1 - \frac{V}{Q}\right) \mu_V(Q) \left(1 - \frac{2^{2V}}{Q^r}\right). \quad (60)$$

The expected numbers of full solver invocations are at most

$$1.0000262, \quad 1.0000561, \quad 1.0202201. \quad (61)$$

Every returned key is correct, regardless of whether failed solver invocations were complete.

Proof. By [Theorem 5.6](#), a random direction has full planted derivative with probability at least $1 - V/Q$. Conditional on that event, $R\mathcal{B}(c)$ is uniform square, so it is invertible with probability $\mu_V(Q)$. The planted point is then an isolated nonsingular root of $f_{c,R}$ by [Theorem 5.1](#). [Theorem 7.6](#) includes it with probability at least $1 - 2^{2V}/Q^r$.

The all-equation and rational-root filters retain the planted point. [Theorem 5.2](#) reconstructs G , and [Theorems 3.4](#) and [3.7](#) supply the exact factorization and signing certificate. Conversely, any candidate that fails an original equation, completion equation, graph identity, pull-back identity, factorization check, or oil-matrix test is rejected. This proves zero-error acceptance. Fresh restarts and (60) give the geometric tail and the numerical expectations. $\square$

Corollary 7.9 (Public-random key density). In the public-random expanded-key model, the probability that every coordinate direction has deficient planted derivative is $\tau_{m,V}(Q)^O$, with base-two logarithms

$$-1132.166907, \quad -1635.352198, \quad -2935.247544.$$

The probability that the deterministic pair-line signing search fails has base-two logarithm below

$$-544.820, \quad -796.276, \quad -1068.687.$$

Outside their union, a deterministic scan of coordinate directions and signing witnesses contains a successful outer branch; only the internal nonsingular-root solver requires random coins.

Proof. Combine [Theorems 3.7](#) and [5.6](#). The two exceptional events need not be independent; a union bound suffices. $\square$

8 Dense-quadratic batching and flattened lifting

8.1 Why the direct straight-line-program bound is not the last word

For dense quadrics, a direct straight-line program has size $L = O(V^3)$. Substituting it into (56) gives

$$\tilde{O}(V^5 4^V) \quad (62)$$

operations in K' . This is a valid generic specialization, but it repeatedly charges contractions that all use the same dense quadratic coefficient tensor.

Write

$$f_i(X) = X^\top A_i X + b_i^\top X + c_i. \quad (63)$$

At precision k , the lifted coordinates live in a coefficient algebra

$$\mathcal{A}_k := K'[T, U]/(T^k, Q_k(T, U)), \quad (64)$$

where Q_k is monic of U -degree at most C . The V coordinate vectors form a $V \times (Ck)$ coefficient matrix. Flattening the matrices A_i turns all Jacobian contractions into one rectangular matrix product over K' . Quadratic values follow from the Jacobian and X , while the Newton solve uses matrix algebra over $\mathcal{A}_k$.

Proposition 8.1 (Dense-quadratic lifting ledger). Let ω be an admissible square-matrix multiplication exponent, and let $\mu(C, k)$ be the cost of one multiplication in $\mathcal{A}_k$. The precision- k work of the global Newton lift can be organized using

$$\tilde{O}(V^3 Ck + (V^\omega + V^2)\mu(C, k)) \quad (65)$$

operations in K' . Summed over precision doubling through $2C_0$, this gives

$$\tilde{O}(V^3 C C_0 + (V^\omega + V^2)\mu(C, 2C_0)). \quad (66)$$

Proof. Stack the V matrices A_i into a $V^2 \times V$ scalar matrix. Multiplication by the $V \times (Ck)$ lifted-coordinate matrix forms every linear Jacobian contraction simultaneously in $O(V^3 Ck)$ scalar operations. This term is written explicitly rather than hidden inside the coefficient-algebra multiplication count.

The identity

$$f_i(X) = \frac{1}{2}X^\top(2A_iX + b_i) + \frac{1}{2}b_i^\top X + c_i$$

computes all values with $O(V^2)$ products in $\mathcal{A}_k$. Newton matrix inversion or solving costs $O(V^\omega)$ such products, and primitive-element renormalization lies within the same envelope. The terminal precision is $2C_0$; quasi-linear multiplication makes the last doubling stage dominate up to logarithmic factors. $\square$

8.2 One multiplication, not a product of two multiplication costs

The earlier stress analysis bounded multiplication in (64) by applying a quasi-linear polynomial-multiplication bound first in U and then in T . This gives a product $M(C)M(k)$. A bivariate array can instead be flattened once.

Lemma 8.2 (Collision-free Kronecker flattening). Let $R_k = K'[T]/(T^k)$ and let $Q_k(U) \in R_k[U]$ be monic of degree C . After precomputing the reciprocal of the reversed Q_k , one multiplication in $R_k[U]/(Q_k)$ can be performed using at most three ordinary polynomial multiplications whose input length is below $2Ck$, plus $O(Ck)$ additions. The reciprocal precomputation costs $O(M(2Ck) \log C)$ operations in K' .

Proof. For a polynomial

$$A(U, T) = \sum_{0 \leq i < C, 0 \leq j < k} a_{i,j} U^i T^j,$$

choose the radix $B = 2C - 1$ and encode

$$U^i T^j \mapsto Z^{i+Bj}. \quad (67)$$

In a product, the U -exponent is at most $2C - 2 = B - 1$. It therefore never carries into the T digit. The largest input exponent is $(C - 1) + B(k - 1) < Bk < 2Ck$, so one ordinary univariate multiplication recovers the complete bivariate product modulo T^k .

To reduce the U -degree modulo the monic polynomial Q_k , use the standard reciprocal division algorithm over the coefficient ring R_k . With the reciprocal precomputed, one product obtains the quotient estimate and one product forms the quotient times Q_k . Together with the unreduced product, this is three products of the same padded length. Newton inversion of the reversed monic polynomial gives the stated one-time reciprocal cost over any commutative coefficient ring. $\square$

Corollary 8.3 (Flattened lifting complexity). Let $M(n)$ denote a quasi-linear polynomial-multiplication cost. The batched lift can be implemented in

$$\tilde{O}(V^3 CC_0 + (V^\omega + V^2)M(CC_0)) = \tilde{O}(V^{\omega+1}4^V) \quad (68)$$

operations in K' .

For the explicit stress function

$$M(n) := n \log_2 n \log_2 \log_2 n, \quad (69)$$

the entire precision-doubling schedule for one coefficient-algebra product is bounded by

$$3M(8CC_0). \quad (70)$$

Proof. Apply [Theorem 8.2](#) at precisions $2C_0, C_0, C_0/2, \dots$. At the terminal stage, each encoded input has length below $4CC_0$. Monotonicity of the logarithmic factors gives

$$\sum_{j \geq 0} M\left(\frac{4CC_0}{2^j}\right) < 2M(4CC_0) < M(8CC_0).$$

There are three ordinary products per modular multiplication. Reciprocal precomputation contributes an additional logarithmic factor but is dominated by the explicit V^2 outer allowance. Finally, $CC_0 = 2^{2V-1}(V+2)$. $\square$

Remark 8.4 (What was improved and what was corrected). Flattening does not alter the exponential factor 4^V ; it replaces the product of two independent quasi-linear overheads by the overhead at the total coefficient-array length. The factor 8 in (70) is important: an earlier draft stopped at precision C_0 , whereas rational reconstruction requires $2C_0$. All finite values in [Section 19](#) use the corrected precision.

9 Sparse working-field models

Both solvers may choose any convenient representation of their finite working field. The directional solver starts over $K = \mathbb{F}_{127^3}$, so its auxiliary fields must contain K ; the direct solver starts over $\mathbb{F}_{127}$ and has no such divisibility requirement. Projectivization lowers the Level-I direct working degree from 16 to 15.

Lemma 9.1 (Explicit sparse auxiliary fields). The following polynomials are irreducible over $\mathbb{F}_{127}$:

$$\begin{aligned} \phi_{15}(X) &= X^{15} + X^{12} - X + 1, & \phi_{18}(X) &= X^{18} + X^4 + 1, \\ \phi_{22}(X) &= X^{22} + X + 1, & \phi_{24}(X) &= X^{24} + X^5 - X + 1, \\ \phi_{30}(X) &= X^{30} + X^7 + 1. \end{aligned}$$

The projective direct-forgery solver uses degrees 15, 22, 30 at Levels I, III, and V. The directional solver uses degrees 18, 24, 30.

Proof. The companion audit applies Rabin's irreducibility criterion [25]. For every displayed degree d , it verifies

$$X^{127^d} = X \pmod{\phi_d}$$

and, for each prime divisor e of d ,

$$\gcd(\phi_d, X^{127^{d/e}} - X) = 1.$$

All computations are exact over $\mathbb{F}_{127}$, and the machine-readable output records every gcd degree. The same audit certifies the additional sparse fields used in the retuning table of Table 8, including $X^{26} + X^3 - X + 1$. $\square$

Schoolbook convolution is unnecessary at these modest extension degrees. The next statement freezes an explicit recursive circuit rather than invoking an asymptotic multiplication theorem.

Proposition 9.2 (Audited Karatsuba extension-multiplication envelope). Let

$$g_{\text{mul}} = G_{\text{mul}}(\log_2 127), \quad g_{\text{add}} = 4 \log_2 127.$$

For length- d coefficient vectors, let (M_d, A_d) be chosen recursively as follows. The schoolbook candidate is

$$(d^2, (d-1)^2).$$

For $d \geq 2$, put $a = \lceil d/2 \rceil$ and $b = \lfloor d/2 \rfloor$. The Karatsuba candidate is

$$(2M_a + M_b, 2A_a + A_b + 4d - 4). \quad (71)$$

Choose whichever candidate has smaller value $Mg_{\text{mul}} + Ag_{\text{add}}$. If s_d is the number of nonleading terms of ϕ_d , one multiplication in $\mathbb{F}_{127^d}$ followed by sparse reduction is then charged by

$$G_{127^d}^{\text{sparse}} := M_d g_{\text{mul}} + (A_d + s_d(d-1))g_{\text{add}}. \quad (72)$$

For the official working fields, the selected plans are:

d	M_d	A_d	s_d
15	81	332	3
18	153	404	2
22	225	480	2
24	243	512	3
30	243	1112	2

Proof. The schoolbook counts are exact: d^2 coefficient products and $d^2 - (2d-1) = (d-1)^2$ additions. For the recursive candidate, write each operand as a low block of length a plus X^a times a high block of length b . The products of the two low blocks and of their sums have length a ; the high-block product has length b , giving $2M_a + M_b$ coefficient multiplications. Forming the sums, subtracting the low and high products from the middle product, and combining the three shifted outputs uses at most $4d-4$ additional coefficient additions, giving (71). Recursion therefore realizes the displayed pair (M_d, A_d) .

The unreduced product has only $d-1$ high coefficients. Eliminating each one modulo ϕ_d uses s_d additions or subtractions and no nontrivial constant multiplication because every nonleading coefficient is $+1$ or -1 . This proves (72). The companion audit executes the same recursive plan and compares 48 random products against schoolbook convolution before certifying the gate counts. $\square$

10 Local-separator descent and candidate Level-I break

Let $q = 127$, $a = m - 4$, $C = 2^a$, and $C_0 = 2^{a-1}(a+2)$ as in the parent attack. The supplied audit optimizes the extension degree over certified sparse irreducible polynomials.

Theorem 10.1 (Candidate local-separator bound). Assume the public-random model, homotopy degree bound, and Boolean-gate conventions of the preceding direct-attack construction and the split-before-lift refinement developed here. Replace global separation by the local conditions in Section 12, accept only simple linear target factors, and use the exact descent test in Section 13. Then the retry-aware diagnostic costs are those in Table 2. In particular, Level I costs

$$2^{140.383995063}$$

Boolean gates, leaving 2.616004937 bits below the category target.

Table 2: Audited stress estimates. “Global split” is the preceding split-before-lift bundle.

Level	Parent	Global split	Local split	Target	Margin
I	151.639	141.126	140.384	143	+2.616
III	203.507	191.630	191.027	207	+15.973
V	260.389	247.637	247.125	272	+24.875

The Level-I optimum in the main ledger uses $K' = \mathbb{F}_{127^8}$ with the Rabin-certified modulus

$$X^8 + X^4 - 1.$$

Degree 9, using $X^9 + X^2 + 1$, gives total 140.418965 and is retained as a nearby cross-check.

11 Characteristic polynomials do not need separation

Let $\Gamma \subset \mathbb{A}^{1+a}$ be the union of the saturated homotopy-curve components that dominate the t -line, with degree at most $D = C_0$. The Jacobian saturation removes the non-etale locus, so over $K'(t)$ its generic fiber is a finite product of separable field extensions,

$$B = K'(t)[X_1, \dots, X_a]/J, \quad \dim_{K'(t)} B = C.$$

Write x_j for the image of X_j and M_{x_j} for multiplication by x_j on B .

Introduce independent coefficients $\Lambda = (\Lambda_1, \dots, \Lambda_a)$ and the generic linear element

$$u_\Lambda = \sum_j \Lambda_j x_j.$$

Its characteristic polynomial is

$$\chi_\Lambda(t, U) = \det \left(UI - \sum_j \Lambda_j M_{x_j} \right). \quad (73)$$

Schost proves two facts that are exactly suited to the present use: specialization of Λ gives the characteristic polynomial of the specialized linear element, and the generic characteristic polynomial can be written

$$\chi_\Lambda = \frac{\Xi_\Lambda}{\Theta},$$

where the t -degrees of Ξ_Λ and Θ are at most $\deg \Gamma \leq D$ and Θ is independent of Λ [27, Lemmas 2–3 and Proposition 2]. This is the characteristic-polynomial, or generic u -resultant, form of the parametric degree theorem.

Schost phrases one corollary using a primitive element with coefficients in the ground field, together with a field-size hypothesis. The generic characteristic polynomial itself does not require

such an element. After extending to an algebraic closure of $K'(t)$, the finite étale algebra has C distinct geometric points. For every distinct pair, the difference

$$\Lambda \cdot (x_s - x_r)$$

is a nonzero polynomial in the independent indeterminates Λ . Hence u_Λ separates the geometric points over the rational-function field $K'(t)(\Lambda)$ and is primitive there, irrespective of the size of K' . Apply Schost's degree argument after this transcendental scalar extension. The determinant in (73) is formed from the original multiplication matrices, so its coefficients already lie in $K'(t)[\Lambda, U]$; scalar extension cannot lower a reduced t -degree and thereby hide a larger degree over K' . Thus no K' -rational global separator, and in particular no $|K'| \gg C^2$ assumption, is used.

Lemma 11.1 (Nonseparating eliminant). For every $\lambda \in K'^a$, including a colliding one,

$$q_\lambda(t, U) := \prod_{s=1}^C (U - \lambda(x_s(t))) = \chi_\Lambda(t, U)|_{\Lambda=\lambda}.$$

Every coefficient of q_λ has a common-denominator representation with numerator and denominator t -degrees at most D .

Proof. The first identity is the characteristic-polynomial product formula in the reduced generic fiber, with repeated factors allowed after specialization. The second assertion follows by substituting $\Lambda = \lambda$ in Ξ_Λ/Θ . Since Θ is independent of Λ , a separator collision cannot create a new parameter denominator or increase its degree. $\square$

The coordinate interpolation numerators are already encoded in the same generic characteristic polynomial.

Lemma 11.2 (Characteristic-polynomial polar). Define

$$v_j(t, U) = \sum_s x_{s,j}(t) \prod_{r \neq s} (U - \lambda(x_r(t))).$$

Then

$$v_j(t, U) = - \left. \frac{\partial \chi_\Lambda(t, U)}{\partial \Lambda_j} \right|_{\Lambda=\lambda}. \quad (74)$$

Consequently each v_j , and every fixed affine-linear combination of the v_j and q_U , has numerator and denominator t -degrees at most D , without assuming that λ separates the branches.

Proof. Over a splitting field, differentiate

$$\chi_\Lambda(t, U) = \prod_s \left(U - \sum_i \Lambda_i x_{s,i}(t) \right)$$

with respect to Λ_j and then specialize $\Lambda = \lambda$. This gives (74). Differentiating Ξ_Λ in the linear-form coefficients does not increase its t -degree, while the denominator Θ is unchanged. $\square$

This directly covers the sole linear sketch used later:

$$b_1(P) = \lambda^{q^{d-1}} \cdot P.$$

Its interpolation numerator is the corresponding fixed K' -linear combination of the v_j .

For the rational oil filter

$$\psi(X) = \frac{H_0(X)}{c + \lambda(X)},$$

the preceding split-before-lift note proves that the saturated graph curve has degree at most $2D$. Apply the same construction to its generic linear element

$$u_{\Lambda, \mu} = \Lambda \cdot X + \mu Y,$$

where $Y = \psi(X)$ is the graph coordinate. Specializing $(\Lambda, \mu) = (\lambda, 0)$ gives q_λ , and

$$-\left. \frac{\partial \chi_{\Lambda, \mu}}{\partial \mu} \right|_{(\lambda, 0)} = v_\psi.$$

The graph degree bound therefore gives numerator and denominator t -degrees at most $2D$ for the filter numerator. Consequently the existing precision

$$P = 4C_0 + 1$$

remains valid.

Remark 11.3. Separation is needed only when one wants the linear element to be a primitive coordinate for every point. The attack instead keeps the full characteristic polynomial and later accepts only a simple target factor. Characteristic polynomials and their coefficient polars remain defined at colliding linear forms.

12 Only local separation is needed

At target specialization, normalize q and each scalar numerator by the common t -valuation exactly as in Safey El Din–Schost Lemma 12 [26]. Call λ *valuation-generic* if every branch escaping to infinity with minimum coordinate valuation $\mu_s < 0$ satisfies

$$\text{ord } \lambda(x_s) = \mu_s.$$

This condition, not pairwise branch separation, is what the specialization proof uses to remove the infinite factors from the specialized eliminant.

Lemma 12.1 (Simple-factor recovery). Assume λ is valuation-generic. Let $\bar{q}(U)$ and $\bar{v}_b(U)$ be the normalized target specializations. If $\tau \in K'$ is a simple root of $\bar{q}$, then there is exactly one finite represented target point P with $\lambda(P) = \tau$, and

$$b(P) = \frac{\bar{v}_b(\tau)}{\bar{q}'(\tau)}.$$

No separation condition is needed away from that one fiber value.

Proof. Let S_∞ and S_0 index the escaping and finite branches. With

$$e = - \sum_{s \in S_\infty} \mu_s,$$

valuation genericity gives

$$(t^e q)(0, U) = \gamma \prod_{s \in S_0} (U - \lambda(P_s))$$

for a nonzero scalar γ . In the normalized numerator, the summand indexed by a finite branch s specializes to

$$\gamma b(P_s) \prod_{r \in S_0, r \neq s} (U - \lambda(P_r)).$$

A summand indexed by an escaping branch need not vanish term by term; rather, every such surviving summand is divisible by the full finite product $\prod_{r \in S_0} (U - \lambda(P_r))$. It therefore vanishes modulo $\bar{q}$, and in particular at every finite root of $\bar{q}$. Thus $\bar{v}_b$ modulo $\bar{q}$ is the ordinary interpolation numerator on the finite limits. A simple root τ belongs to exactly one finite branch, and evaluation followed by division by $\bar{q}'(\tau)$ gives the claimed value. $\square$

Condition on the existence of a canonical eligible nonsingular base-field target point $P_\star$; for example, choose the first under any fixed ordering. Nonsingularity makes it an isolated multiplicity-one point. Let I generic branches escape to infinity and let F have finite target limits, counted with branch multiplicity. Then $I + F = C$. There are at most I valuation-cancellation hyperplanes and at most $F - 1$ collision hyperplanes against $P_\star$. Each bad equality

$$\lambda(w) = 0 \quad \text{or} \quad \lambda(P - P_\star) = 0$$

cuts out at most $|K'|^{a-1}$ choices of λ in $(K')^a$. Thus a uniform $\lambda \in (K')^a$ fails valuation genericity or collides at $P_\star$ with probability at most

$$\frac{I + F - 1}{|K'|} = \frac{C - 1}{|K'|}. \quad (75)$$

The coefficient vector of the equation may lie only over an algebraic closure, so its solution set need not literally be a K' -defined hyperplane. The count is still valid: choose a nonzero coordinate of that vector and fix all other coefficients of λ ; the remaining equation has at most one solution in K' (and may have none).

For the filter denominator, first compute the C known start values $\lambda(\xi_s)$ and sample c uniformly from the complement of the forbidden set $\{-\lambda(\xi_s)\}_s$ in K' . Rejection sampling costs an expected factor at most $(1 - C/|K'|)^{-1}$ for a scan of the start values; this scan is negligible beside one length- P filter lift and is included in the miscellaneous envelope. Conditional on this choice, only the at most $F \leq C$ finite target values remain forbidden. Therefore

$$\Pr[c + \lambda(P) = 0 \text{ for a finite target branch}] \leq \frac{C}{|K'| - C}. \quad (76)$$

The old $C^2/|K'|$ event has disappeared.

13 Exact one-sketch Frobenius descent

Set

$$b_1(P) = \lambda^{q^{d-1}} \cdot P, \quad K' = \mathbb{F}_{q^d}.$$

The separator value $\tau = \lambda(P)$ is already available from the factor of $\bar{q}$, so no numerator for $\lambda \cdot P$ is needed.

Lemma 13.1 (Simple-root descent is exact). Let $\tau \in K'$ be a simple linear factor of the target eliminant, and let P be its unique represented point. Then

$$b_1(P)^q = \tau \iff P \in \mathbb{F}_q^a.$$

Proof. The forward identity for a base-field point is immediate. Conversely,

$$b_1(P)^q = \lambda \cdot P^q.$$

If it equals $\tau = \lambda \cdot P$, then P and P^q have the same separator value. The target equations and homotopy are defined over $\mathbb{F}_q$, so the represented finite target limits are stable under Frobenius. If $P^q \neq P$, the two distinct represented points produce multiplicity at least two at $U = \tau$, contradicting simplicity. Hence $P^q = P$. $\square$

This eliminates the probabilistic $C/|K'|$ false-positive term. It also eliminates the abort cap: every simple factor passing the Frobenius test and the nonzero-square oil filter is a genuine base-field target solution. The algorithm reconstructs coordinates for the first such factor and stops, so exactly one coordinate re-lift is charged per successful forgery.

14 The sharpened root bound is explicit

For completeness, this pass expands the probability calculation inherited from the split-before-lift note. Let M count all off-oil affine roots of the normalized residual system, and let N count those with invertible affine Jacobian. The parent manuscript proves, for an actual parameter $\mu \geq 1 - q^{-1}$,

$$\mathbb{E}[M] = \mu, \quad \mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}, \quad \mathbb{E}[N] = \mu\pi,$$

where $\pi = \prod_{j=1}^{r-1} (1 - q^{-j})$. The roots occur in pairs $z, -z$, and each nonsingular normalized pair yields one eligible projective point. Applying the second Bonferroni inequality to the projective-point events and bounding pairs of nonsingular roots by pairs of all roots gives

$$\begin{aligned} \Pr[\text{eligible nonsingular projective root}] &\geq \frac{\mathbb{E}[N]}{2} - \mathbb{E}\left[\binom{M/2}{2}\right] \\ &= \frac{\mu\pi}{2} - \frac{\mu^2}{8} + \frac{(q-1)\mu q^{-r}}{8}. \end{aligned}$$

The right-hand side is increasing for $\mu \in [1 - q^{-1}, 1]$, so substituting $\mu = 1 - q^{-1}$ yields

$$p_{\text{root}} = 0.3690869822 \dots$$

As a separate fallback, the ledger also substitutes the parent manuscript's already-proved lower bound $0.326336998767509 \dots$; the Level-I total then becomes 140.531461 , still 2.469 bits below target.

15 Finite ledger

At Level I the local variant has two scalar families, b_1 and ψ , instead of three. The terminal product tree therefore uses

$$\frac{a}{2}(1 + 2 \cdot 2) = 125$$

long-product units. Rational reconstruction contributes 99.885 units. Candidate coordinate evaluation is charged once over the complete Las Vegas execution, rather than per trial, and the 64-unit miscellaneous cushion is retained. This pass also charges the one-time formation of $u_s = \lambda \cdot x_s$ during that coordinate rerun.

Table 3: Level-I component ledger for extension degree $d = 8$.

Component	$\log_2$ gates
Initial base-field branch lifts	139.358
Valuation-safe filter series	132.120
Terminal scalar recombination	138.791
One on-demand coordinate re-lift	137.860
Extension-valued leaf formation	127.610
Coordinate-rerun separator leaves	124.898
Factorization and exact verification envelope	115.707
Preprocessing	63.365
Total	140.383995

The one-trial success lower bound is 0.353997, giving restart factor 2.824886. At $d = 8$, the local-separator failure bound in (75) is 0.016637, and the conditional target-denominator bound in (76) is 0.016918.

Table 4: Level-I sensitivity to the multiplier on $M(n) \log n$ in rational reconstruction, retaining $d = 8$.

Multiplier	8	10	16	32	64	128
Total	140.384	140.424	140.538	140.805	141.225	141.815

Table 5: Level-I cross-checks. Each row reoptimizes the extension degree.

Online-envelope multiplier	1	2	4	8	16
Best extension degree	8	8	9	9	9
Total	140.384	141.117	141.952	142.856	143.806

Thus the candidate remains below 2^{143} throughout the rational-reconstruction sensitivity range. It also remains below target after multiplying the entire claimed online-product envelope by eight, but not by sixteen; the finite margin therefore does not excuse the remaining obligation to provide a fully explicit schedule. The aggregate Level-I margin corresponds to an overall multiplicative cushion of $2^{2.616} = 6.13$ in the inherited gate model; this does not address memory traffic or a change of model.

16 Reduced-parameter end-to-end instantiation

The artifact now contains an executable instantiation of the complete attack rather than only independent checks of its ingredients. The reduced scheme follows the QR-UOV algebraic construction in the official specification [1]: central symmetric matrices over an extension field have a zero oil-oil block, the secret change of variables has the compact form

$$S = \begin{pmatrix} I & G \\ 0 & I \end{pmatrix},$$

and the extension-field matrices are pulled back to base-field quadratic forms through a nonzero linear functional, implemented here by the trace pairing. Seed expansion, byte serialization, and the exact domain-separated hash interface are deliberately simplified; the public quadratic relation attacked by the algorithm is unchanged.

The default demonstration uses

$$q = 13, \quad \ell = 3, \quad V = 3, \quad M = 2,$$

so the expanded UOV dimensions are

$$v = 9, \quad m = o = 6, \quad n = 15.$$

The residual projective core has $a = m - 4 = 2$, hence $C = 4$, $C_0 = 8$, and precision $P = 4C_0 + 1 = 33$. The attack function receives only the expanded public matrices, message, and salt. It executes target normalization, the anchored Chevalley–Warning construction, a common totally isotropic subspace for three forms, affine charting, all four coefficientwise homotopy lifts, local-separator recombination, rational reconstruction, valuation-safe specialization, exact Frobenius descent, one coordinate rerun, affine rescaling, and unchanged public verification.

For the canonical seed 20260731, the program first generated an honestly verifiable signature and then produced the forged signature

$$(0, 8, 2, 10, 0, 0, 0, 4, 10, 5, 11, 8, 0, 12, 1) \in \mathbb{F}_{13}^{15}$$

for the recorded message and salt. The public verifier accepted it. The successful run used the trace in Table 6.

Table 6: Canonical reduced-parameter attack trace. Wall time is machine-dependent and is not used in the security ledger.

Quantity	Value
Salt attempts / attack records	1/1
Target eliminant degree	4
Branch lifts	8 (4 initial, 4 coordinate rerun)
Checked lifted-branch residuals	4
Rational reconstructions	31
Coordinate reruns	1
Target roots examined / simple	2/1
Frobenius passes / exact verified candidates	1/1
Exhaustive $\mathbb{F}_{13}$ / $\mathbb{F}_{13^2}$ residual roots	2/2
Python wall time	0.415 seconds

The exhaustive residual-root enumeration is deliberately performed only *after* the local-separator algorithm has selected and reconstructed its candidate. It is not a solver subroutine. It checks that every extension-field root appears in the specialized eliminant, that every simple eliminant root has a singleton fiber, and that the accepted point is among the base-field roots.

The canonical transcript records the public key, target, isotropic subspace, chart, affine core, lifted-system parameters, separator, specialized characteristic polynomial and scalar polars, accepted root, recovered coordinates, signature, and operation counts. A separate public replay program consumes that transcript without the secret key, reconstructs the affine core, reruns the complete local-separator solver with the recorded separator, and reproduces the accepted forgery.

A fixed regression suite over the five consecutive seeds 20260000, . . . , 20260004 produced five publicly valid forgeries and five successful public replays. At the imposed finite retry limits, the largest salt count was two and the slowest individual run took 10.252 seconds. This is a regression test, not a statistical estimate of the asymptotic success probability. A supplementary run over $q = 29$ with the same dimensions also forged and replayed successfully.

This closes the narrow composability objection: the artifact now instantiates one finite end-to-end attack on reduced QR-UOV instances. It does *not* establish the Level-I gate count. The prototype uses naive truncated-series arithmetic at $a = 2$ and therefore neither implements nor benchmarks the fast dyadic multiplication schedule assumed by the full-parameter ledger. Full parameters remain infeasible to execute, and the public-random-model, seeded-transfer, characteristic-polynomial degree, and memory-model caveats remain.

17 Validation and remaining risk

The bundle includes:

1. `QRUOV_local_separator_audit.py`, which computes the optimized degree, finite probabilities, and all component costs;
2. `verify_local_separator_ledger.py`, a separately written implementation importing neither the primary audit nor its parent modules, matching all displayed totals to below $5 \cdot 10^{-10}$ bits;
3. the end-to-end reduced attack `implementation/qrnov_local_separator_forgery.py`, its public replay checker, five-instance regression suite, and complete public transcripts;

4. Rabin certificates for every newly used sparse modulus;
5. the original split-before-lift toys validating branch lifting, symmetric recombination, and Frobenius arithmetic, together with an exhaustive $\mathbb{F}_9/\mathbb{F}_3$ toy showing that a non-base point passes the one-sketch test exactly on separator collisions; and
6. `toy_infinite_branch_specialization.py`, which exhibits the subtle case in which an escaping-branch numerator term survives normalization but is a multiple of the finite eliminant and hence vanishes at every finite simple root.

The principal remaining proof obligations are external checks that the characteristic-polynomial degree theorem in [Theorems 11.1](#) and [11.2](#) applies to the exact saturated dominant homotopy algebra and to the rational-filter graph with the stated degree bounds. The determinant identities are exact and no defect is currently known, but this identification with the parent solver’s saturated dominant algebra is what the degree-8 retuning depends on. The reduced implementation supplies a literal finite schedule for the toy instance, but the dyadic fast online-product envelope used at Level I still needs a line-by-line full-scale algorithmic derivation; the multiplier table shows exactly how much error that part can absorb. The earlier caveats about the idealized public-random model, seeded transfer, and astronomical memory traffic are unchanged.

18 Assessment

This is a real strengthening rather than a cosmetic ledger improvement. The preceding candidate crossed Level I by splitting before lifting; the present argument removes the largest remaining auxiliary-field overkill. It replaces a global primitive-element event with a one-sided local isolation event and turns Frobenius filtering from probabilistic to exact. Under the inherited model, the cumulative improvement over the strict parent bundle is 11.254751 bits.

The result is now robust to far larger rational-reconstruction constants than the original 1.87-bit-margin version. The reduced implementation materially strengthens the evidence: the claimed transformations and filters have now been composed into an actual public-key forgery rather than tested only in isolation. This manuscript distinguishes this demonstrated toy-scale attack from the Level-I extrapolation and should await specialist verification that the characteristic-polynomial degree theorem applies to the exact saturated dominant algebra and its rational-filter graph.

19 Combined cost, memory, and parameter implications

19.1 Work unit and scope

The QR-UOV specification expresses classical attack costs as Boolean-gate exponents and charges multiplication of b -bit field elements by

$$G_{\text{mul}}(b) = 2b^2 + b \tag{77}$$

Boolean gates [\[1\]](#). We retain that broad unit. Every extension-field operation is overcharged as one multiplication using the explicit Karatsuba circuit of [\(72\)](#).

The principal symbolic-homotopy term uses the corrected reconstruction precision $2C_0$. For a square solver core of dimension a ,

$$C = 2^a, \quad C_0 = 2^{a-1}(a + 2),$$

and the flattened lifting schedule is charged by

$$3(a^3 + a^2)M(8CC_0) \tag{78}$$

Table 7: Classical gate diagnostics, including the local-separator split-before-lift refinement and the audited Karatsuba field circuit. A positive margin means that the attack row lies below the category target.

Attack	Level I		Level III		Level V	
	Cost	Margin	Cost	Margin	Cost	Margin
Original projective forgery	151.639	−8.639	203.507	3.493	260.389	11.611
Local-separator projective forgery	140.384	2.616	191.027	15.973	247.125	24.875
Directional equivalent-key recovery	154.826	−11.826	206.177	0.823	260.857	11.143
Best attack	140.384	2.616	191.027	15.973	247.125	24.875
Category target	143		207		272	

operations in the chosen working field. The projective direct row uses $a = m - 4$ and includes the fixed-key seed bound, nonsingular-root factor, random chart, and finite-field separator. The directional row uses $a = V$ and includes direction regularity and its separator. Polynomial-time preprocessing, all-equation filtering, factorization, graph completion, square testing and rescaling, and public verification are included through explicit lower-stage envelopes; they do not change the displayed decimals.

These are reproducible diagnostics for symbolic algorithms, not measured circuits. In particular, setting

$$M(t) = t \log_2 t \log_2 \log_2 t$$

assigns unit coefficient to an asymptotic multiplication bound. The exact reductions, Karatsuba circuits, sparse fields, and success probabilities do not depend on that convention; the final conversion to a practical implementation does.

19.2 The two attacks

The original direct rows use projective solver dimensions 50, 74, 101. The local-separator refinement keeps those algebraic cores but lowers the working-field degrees and removes global branch separation, the second Frobenius numerator, probabilistic false candidates, and repeated coordinate recovery. Its optimized field degrees are 8, 12, and 16 in the supplied ledger. The directional rows remain an independent equivalent-key route. The reduced implementation executes the local-separator route at toy scale; the full-parameter symbolic lift still dominates both time and memory.

For context, the QR-UOV team’s January 2026 response reports classical Hashimoto estimates of 158, 217, 285 bits for the same three public dimensions [28]. The new direct rows are lower by approximately 6.36, 13.49, 24.61 bits. This is not a claim that the implementations are identical: the team uses a semi-regular direct-MQ model, whereas our row uses the finite-characteristic nonsingular-root solver and its explicit success probability.

19.3 What follows at each level

Level I. The local-separator row is 140.384, or 2.616 bits below the 143-bit target. Thus the paper gives a candidate Level-I break in the stated Boolean-gate model. This is not a practical attack claim: the symbolic state is physically unrealizable, and the full-scale online-product schedule and characteristic-polynomial degree transfer remain explicit obligations. The reduced-parameter implementation establishes that the algebraic pipeline is a single executable attack, not that the Level-I extrapolation is practical.

Table 8: Smallest quotient-compatible (V, m) pairs found by the companion audit that put both displayed attack rows at the category target, or at the target plus a 16-bit attack-specific cushion.

Level	Current		Reach target		Reach target +16	
	V	m	V	m	V	m
I	52	54	52	54	57	60
III	76	78	78	81	87	90
V	102	105	108	111	117	120

Level III. The local-separator row is 191.027, leaving 15.973 bits below target. The older projective and independent equivalent-key rows are also below target. The combined evidence strongly supports changing the Level-III parameters.

Level V. The local-separator row is 247.125, leaving 24.875 bits below target. The older projective and independent equivalent-key rows are likewise below target. Level V should not be retained unchanged.

19.4 Why changing only the vinegar count is not a repair

The official parameter sets sit on the direct side of the projective scissors crossing:

Level	V	$m - 4$	Directional core	Projective direct core
I	52	50	52	50
III	76	74	76	74
V	102	101	102	101

Increasing V raises the directional core but leaves the projective direct core unchanged. Decreasing V eventually makes directional recovery no larger than $m - 4$. Consequently a fixed- m vinegar retuning cannot move the combined algebraic bottleneck.

The next table gives attack-specific thresholds under the same padded ledger. It is not a complete replacement proposal: every other attack, failure probability, key size, and implementation cost must be recomputed. The point is that both columns must move. Raising only V leaves the direct attack unchanged; raising only m leaves the directional attack unchanged.

The table enforces $m \equiv 0 \pmod{3}$ and the direct-side feasibility condition $V \geq m - 3$. Level I already exceeds its bare target in this model, but a 16-bit cushion requires moving both dimensions. At Levels III and V, even the target-only diagnostic requires increases in both V and m ; a 16-bit cushion requires substantially larger changes. A structural redesign may be preferable to paying both costs.

19.5 Memory remains the central practical limitation

The desired outputs are small: one direct attack needs a single vector in $\mathbb{F}_q^n$, and the equivalent-key attack ultimately needs a $V \times O$ graph. The direct symbolic representations are not small. One flattened coefficient array at terminal precision requires approximately

$$2^{49.41} \text{ EiB}, \quad 2^{98.51} \text{ EiB}, \quad 2^{153.40} \text{ EiB}$$

for the projective direct cores. The corresponding directional arrays require about $2^{53.73}$, $2^{102.68}$, and $2^{155.41}$ EiB. A full-parameter execution using these representations is physically unrealistic.

This limitation prevents a claim of a practical full-parameter implementation. No full-parameter forgery or equivalent-key recovery is reported, and the paper draws no conclusion from unimplemented attack variants. The exact reductions and the fixed-output scissors theorem are algebraic results; time, storage, and correctness are reported as separate resources.

20 Executable audits

The bundle contains the following executable checks and an end-to-end reduced-parameter attack.

Combined numerical audit. `QRUOV_scissors_audit.py` recomputes the Chevalley–Warning slacks, the projective $m-4$ cores, fixed-key seed–oil intersection bound, second-moment nonsingular-root bound, chart and separator probabilities, Karatsuba field circuits, direct and directional cost rows, memory envelopes, and attack-specific retuning thresholds. It also applies Rabin’s criterion to every sparse auxiliary-field polynomial used in the official and threshold rows. The script emits both human-readable and machine-readable output.

Directional algebra audit. `QRUOV_directional_audit.py` checks the corrected $2C_0$ precision, Vandermonde product starts, characteristic-127 lifting identities, flattened bivariate multiplication, sparse-field certificates, the recursive Karatsuba plans, and the complete graph-recovery pipeline on 200 random toy instances. Of those instances, 199 had a full-rank planted direction and all 199 yielded the exact graph; the remaining direction was correctly rejected for retry.

Exact degree-three direct-forgery toy. `QRUOV_direct_qr_toy.py` generates independent extension-field UOV forms over $\mathbb{F}_{3^3}$, hides their oil space, pulls each form back by a trace to a scalar base-field quadric, and performs full target normalization, random seed-subspace search, Chevalley–Warning extension, projective residual search, affine rescaling, and public verification. In the frozen 200-key run, 199 keys yielded a verified forgery within twelve trials. Every returned vector passed two independent evaluators; 185 final seed subspaces were disjoint from the hidden oil space, and every selected seed point was off oil. The experiment checks the attack logic; it is not a full-parameter benchmark.

End-to-end local-separator forgery. `implementation/qr_uov_local_separator_forgery.py` instantiates the complete reduced attack described in [Section 16](#). Its public replay checker reconstructs the residual system and reproduces the forgery without secret material. The frozen regression suite succeeds on five of five instances, and the supplementary $q = 29$ instance also succeeds. Exhaustive root enumeration is used only after acceptance as a validation oracle, never as the attack solver.

21 Conclusion

QR-UOV admits a projective direct-forgery reduction with square cores of dimensions 50, 74, 101 and a complementary directional attack that recovers an equivalent signing key. The local-separator refinement changes the concrete conclusion: global branch separation is unnecessary, characteristic-polynomial polars suffice for scalar recovery, and one exact Frobenius sketch identifies a base-field point on a simple target factor. Under the stated symbolic Boolean-gate model, the resulting rows lie below all three NIST category targets, including Level I by 2.616 bits.

The reduced-parameter implementation now composes the complete attack and produces publicly accepted forgeries from public inputs alone. Consequently, the result is no longer merely

a collection of independently tested micro-lemmas. What remains conditional is the full-parameter complexity projection: the exact characteristic-polynomial degree transfer to the saturated dominant algebra, the fast online-product schedule, seeded-instance transfer, and memory traffic all require further scrutiny. The paper therefore claims a finite, executable structural attack and a model-based candidate full-parameter break, not a practical full-parameter computation.

The parameter lesson is unchanged but stronger. Increasing only the vinegar count cannot repair the fixed-output scissors, while the new direct row already crosses Level I. Any QR-UOV revision should reconsider both dimensions and the underlying compressed structure rather than apply a vinegar-only retuning.

Appendix overview

The main body presents the direct forgery, directional equivalent-key recovery, characteristic-127 solver, and combined parameter consequences. The first appendices retain an independent smooth-projective route and two solver cross-checks. Later appendices cover seeded expansion, optional identification with the planted oil space, alternative low-space directions, and executable pseudocode. These details are separated so that readers can understand the attack and its parameter implications before entering the algebraic-geometry audit.

22 Independent projective smooth-core route

This section supplies the geometric input to [Theorem 23.13](#). Outside three explicit bad key loci, we show that some slice and output recombination gives a smooth 2^V -point projective core; we then bound the density of those loci. The next section chooses a finite working field and turns this existence statement into a verified randomized algorithm. Secret central coordinates are used only to expose the key distribution.

Let $K = \mathbb{F}_{127^3}$, $Q = |K|$, and

$$(V, O, m) \in \{(52, 18, 54), (76, 26, 78), (102, 35, 105)\}.$$

Put $e = m - V \in \{2, 2, 3\}$ and $N = V + O$. Work first in secret central coordinates

$$\mathcal{V} = K^V \oplus K^O, \quad \mathcal{O}_0 = \{0\} \oplus K^O.$$

The m central quadrics are

$$F_i(v, o) = v^\top A_i v + 2v^\top B_i o, \quad A_i \in \text{Sym}_V(K), \ B_i \in K^{V \times O}. \quad (79)$$

In the public-random expansion model, the pairs (A_i, B_i) are independent and uniform. Equivalently, each F_i is uniform in

$$\mathcal{U} := H^0\left(\mathbb{P}(\mathcal{V}), \mathcal{I}_{\mathbb{P}(\mathcal{O}_0)}(2)\right), \quad D_{\text{form}} := \dim \mathcal{U} = \frac{V(V+1)}{2} + VO.$$

The proof uses central coordinates only to expose the distribution. The secret change from public to central coordinates lies in $\text{GL}_N(K)$. Left multiplication by its inverse is a bijection on $L^{N \times (V+1)}$, so a uniform public slice remains uniform in central coordinates.

Smoothness, intersection multiplicity, and derivative-kernel recovery are coordinate invariant. In particular, the public and central tuples have identically vanishing joint discriminant polynomials simultaneously. Let

$$X_F := V(F_1, \dots, F_m) \subset \mathbb{P}(\mathcal{V}).$$

It contains the planted oil space $\mathbb{P}(\mathcal{O}_0) \simeq \mathbb{P}^{O-1}$.

Proposition 22.1 (Exact marginal law on a fixed transverse slice). Fix, independently of the sampled central forms, a K -rational projective V -plane Λ that meets $\mathbb{P}(\mathcal{O}_0)$ transversely in one point p . Under the marginal public-random law of the central pairs (A_i, B_i) , the restrictions $F_1|_\Lambda, \dots, F_m|_\Lambda$ are independent uniform elements of

$$H^0(\mathbb{P}^V, \mathcal{I}_p(2)),$$

the K -vector space of projective quadrics through p .

Proof. Choose homogeneous coordinates $[z : t_1 : \dots : t_V]$ on Λ with $p = [1 : 0 : \dots : 0]$. Transversality makes the vinegar projection of Λ/p invertible, so after changing the t -coordinates its central-coordinate parametrization is

$$v = t, \quad o = cz + Lt,$$

for some nonzero $c \in K^O$ and $L \in K^{O \times V}$. Hence

$$F_i|_\Lambda(z, t) = 2zt^\top B_i c + t^\top (A_i + B_i L + L^\top B_i^\top) t.$$

For every prescribed linear coefficient $\ell \in K^V$ and symmetric quadratic coefficient $H \in \text{Sym}_V(K)$, the matrices B_i satisfying $B_i c = \ell$ form an affine space of the same size, and then $A_i = H - B_i L - L^\top B_i^\top$ is forced. Thus $(A_i, B_i) \mapsto F_i|_\Lambda$ is a surjective linear map with equal-size fibers. The pairs (A_i, B_i) are marginally independent and uniform by [Theorem 3.3](#), proving the claim. $\square$

Remark 22.2 (Why the statement is marginal). The model permits the secret graph G to depend arbitrarily on A . Conditioning on G can therefore change the law of A , so no conditional-on- G uniformity is asserted. None is needed: the bad-key events in this section depend only on the marginal central tuple $(A_i, B_i)_i$, while a fresh public slice remains uniform after the fixed secret coordinate change for every individual key.

Proposition 22.3 (Generic isolation by the spare equations). Fix $p \in \mathbb{P}^V$ and let $m > V$. A Zariski-generic m -tuple in $H^0(\mathbb{P}^V, \mathcal{I}_p(2))^m$ has p as its only common projective zero. Generically its full Jacobian has rank V at p .

Proof. For each $x \neq p$, evaluation at x is a nonzero linear functional on $H^0(\mathbb{P}^V, \mathcal{I}_p(2))$. Requiring all m forms to vanish at x therefore imposes codimension m . The incidence with $x \in \mathbb{P}^V \setminus \{p\}$ has dimension at most the parameter-space dimension plus $V - m$, strictly smaller than the parameter-space dimension. By Chevalley's theorem, its image is constructible and its Zariski closure has dimension no larger than the incidence, so that closure is proper. Hence a generic tuple has no common zero away from p . At p , the first derivatives of a uniform quadric through p range over the full cotangent space. The rank-deficient $m \times V$ derivative matrices form a proper determinantal subvariety. $\square$

The two propositions explain why the overdetermined sliced system generically contains only the planted point. The main attack nevertheless solves a square core because [Theorem 22.3](#) is qualitative: it does not supply the finite-field density and solver interface needed for a quantified faster theorem.

Let L/K be the finite field chosen below. One invocation samples

$$M \leftarrow L^{N \times (V+1)}.$$

If $\text{rank } M < V + 1$, the trial fails. Otherwise the attack restricts the public quadrics to the projective V -plane $\Lambda_M = \mathbb{P}(\text{im } M)$:

$$q_i(t) = F_i(Mt), \quad t \in \mathbb{P}_L^V.$$

It next samples $R \leftarrow L^{V \times m}$. A rank defect again ends the trial; for full-rank R , the square core is

$$g_a(t) = \sum_{i=1}^m R_{a,i} q_i(t), \quad 1 \leq a \leq V.$$

Rank defects are counted as failed draws rather than conditioned away. This keeps the entries of (M, R) uniform, so the Schwartz–Zippel bound applies directly. By [Theorem 22.5](#), every rank defect already lies in $\{\Theta_F = 0\}$.

In secret affine coordinates, a transverse Λ_M is exactly a slice $o = c + Lv$ after translating its unique oil intersection. The projective parameterization is preferable in the proof because it has no inverses or denominators: every restricted coefficient is simply quadratic in the entries of M .

22.1 Why the full discriminant is enough

The finite-field Kronecker theorem used below assumes that every affine prefix is a reduced complete intersection. A smooth full projective core already implies these prefix conditions; no product of prefix discriminants is needed.

Lemma 22.4 (A smooth full core gives reduced regular prefixes). Let $g_1, \dots, g_V$ be homogeneous quadrics in $\mathbb{P}^V$ over a field of odd characteristic. If their common zero scheme is a smooth zero-dimensional complete intersection, then the ordered quadrics form a regular sequence and every homogeneous prefix ideal $(g_1, \dots, g_s)$ is geometrically radical. After any projective coordinate change and dehomogenization in an affine chart containing a final root, the affine sequence remains regular and every prefix ideal remains geometrically radical. These are precisely the solver’s hypotheses with localization polynomial equal to 1.

Proof. Work over an algebraic closure and let S be the homogeneous coordinate ring. The final ideal has height V and is generated by V elements. Since S is Cohen–Macaulay, $g_1, \dots, g_V$ is a regular sequence. Every prefix $I_s = (g_1, \dots, g_s)$ is therefore an unmixed complete intersection, and its coordinate ring is Cohen–Macaulay.

Suppose that some prefix were not radical. A Cohen–Macaulay ring satisfies Serre’s condition (S_1) , so a nonreduced prefix cannot be generically reduced: some irreducible component occurs in its complete-intersection cycle with multiplicity at least two. Each remaining quadric is a nonzerodivisor modulo the preceding prefix and hence contains no component. Until the final cut, every surviving component has positive projective dimension, so every remaining positive-degree hypersurface meets it. The associativity formula for intersection multiplicities preserves the component’s multiplicity factor through these proper cuts. The final zero-cycle would therefore contain a point with multiplicity at least two, contradicting smoothness of the final scheme. Thus every homogeneous prefix is geometrically radical.

Localization at a chart equation and dehomogenization preserve regular sequences and reducedness. The same conclusions hold in every affine chart containing a final root; roots at infinity are simply absent from the affine zero scheme. $\square$

Let $\Delta_{V,V}$ be the reduced discriminant for V quadrics in $\mathbb{P}^V$. It vanishes exactly when the common zero scheme is not a smooth complete intersection of codimension V . Because $\text{char } K = 127 \neq 2$, Benoist’s arbitrary-characteristic formula [5] gives the partial degree

$$d_V = (V + 1)2^{V-1} \tag{80}$$

in the coefficients of each equation. Thus it suffices to make the single polynomial

$$\Theta_F(M, R) := \Delta_{V,V} \left(\sum_i R_{1i} F_i(Mt), \dots, \sum_i R_{Vi} F_i(Mt) \right) \tag{81}$$

nonzero. Each coefficient of a restricted equation has degree at most two in M and one in the corresponding row of R . Since the discriminant has degree d_V in the coefficients of each of the V equations,

$$\deg_{M,R} \Theta_F \leq (2V + V)d_V = 3V(V + 1)2^{V-1}. \quad (82)$$

Lemma 22.5 (Nonvanishing certifies full parameter rank). If $\Theta_F(M, R) \neq 0$, then $\text{rank } M = V + 1$ and $\text{rank } R = V$.

Proof. If $\text{rank } M < V + 1$, then $\mathbb{P}(\ker M)$ is nonempty and every pull-back $F_i(Mt)$, together with all of its first derivatives, vanishes there; the core is singular. If $\text{rank } R < V$, the output equations are linearly dependent and cannot cut a smooth codimension- V complete intersection. In either case the full discriminant vanishes. $\square$

22.2 Geometric nonvanishing of the joint slice-and-output discriminant

For $x \in X_F$, let

$$\rho_F(x) := \text{rank}(dF_1(x), \dots, dF_m(x)),$$

where the differentials are taken on $T_x \mathbb{P}(\mathcal{V})$.

Definition 22.6 (Three key-level bad loci). Define:

$$\begin{aligned} \mathcal{B}_{\text{oil}} &:= \{ F : \rho_F(p) < V \text{ for every } p \in \mathbb{P}(\mathcal{O}_0)(\overline{K}) \}, \\ \mathcal{B}_{\text{off}} &:= \{ F : \exists x \in X_F(\overline{K}) \setminus \mathbb{P}(\mathcal{O}_0) \text{ with } \rho_F(x) < V \}, \\ \mathcal{B}_{\text{span}} &:= \{ F : \dim \text{span}(F_1, \dots, F_m) \leq V \}. \end{aligned}$$

The next three lemmas separate the proof into linear-section geometry, a second-order separability argument, and generic output mixing.

Lemma 22.7 (A transverse finite base slice through a regular oil point). Assume $F \notin \mathcal{B}_{\text{oil}} \cup \mathcal{B}_{\text{off}}$. Choose $p \in \mathbb{P}(\mathcal{O}_0)(\overline{K})$ with $\rho_F(p) = V$. Then a Zariski-open dense set of projective V -planes $\Lambda \subset \mathbb{P}(\mathcal{V}_{\overline{K}})$ through p has all of the following properties:

1. $\Lambda \cap \mathbb{P}(\mathcal{O}_0) = \{p\}$;
2. $X_F \cap \Lambda$ is finite;
3. at every $x \in X_F \cap \Lambda$, the restricted differential $T_x \Lambda \rightarrow \overline{K}^m$ has rank V .

Proof. Dimension of the base locus. First, $\dim X_F \leq O - 1$. Otherwise choose a closed off-oil point x on a component of dimension at least O . The Zariski tangent space is the common differential kernel and has dimension at least the local dimension, so

$$\rho_F(x) = (N - 1) - \dim T_x X_F \leq V - 1,$$

contrary to $F \notin \mathcal{B}_{\text{off}}$. Since $\mathbb{P}(\mathcal{O}_0) \subset X_F$, the maximal dimension is exactly $O - 1$.

Behavior at the planted point. At p , every differential annihilates $T_p \mathbb{P}(\mathcal{O}_0)$. Both this tangent space and $\ker dF(p)$ have dimension $O - 1$, so they are equal. Therefore a general V -plane through p is transverse at p and intersects the oil space only in p . We henceforth restrict to this nonempty open subset of the Grassmannian; no other oil point can occur in the slice.

Off-oil points not lying on a base line. Let $\mathcal{G}_p$ be the Grassmannian of projective V -planes through p , of dimension $V(O - 1)$. We now consider only $x \in X_F \setminus \mathbb{P}(\mathcal{O}_0)$. Put $K_x = \ker dF(x)$; the off-oil hypothesis gives $\dim K_x \leq O - 1$. For fixed x , planes through p and x form a family of dimension $(V - 1)(O - 1)$. If the tangent direction of the line $\overline{px}$ is not in K_x , the additional

condition $T_x\Lambda \cap K_x \neq 0$ is a proper Schubert condition and lowers this dimension by at least one. Allowing x to vary in dimension at most $O - 1$ gives total dimension at most $V(O - 1) - 1$.

Lines through the planted point. If the tangent direction of $\overline{px}$ lies in K_x , then every quadric F_i vanishes identically on that line: both endpoint values and the polar cross term vanish. Each such direction contributes a full projective line contained in X_F , so passing from points of X_F to line directions lowers dimension by at least one. The direction variety therefore has dimension at most $\dim X_F - 1 \leq O - 2$. Planes containing one such line again form a family of dimension at most

$$(O - 2) + (V - 1)(O - 1) = V(O - 1) - 1.$$

Conclusion. The off-oil rank-defect incidence is locally closed and has dimension at most $V(O - 1) - 1$ by the two cases above. Its closure in the projective product $X_F \times \mathcal{G}_p$ has the same dimension, and the projection of that closure is therefore a proper closed subset of $\mathcal{G}_p$. Outside it, every intersection point has restricted derivative rank V . At any such point the Zariski tangent space of $X_F \cap \Lambda$ is $T_x\Lambda \cap \ker dF(x) = 0$. Since local dimension never exceeds tangent-space dimension, no point can lie on a positive-dimensional component. Hence the intersection is finite. $\square$

Lemma 22.8 (A second-order spare equation forces separability). Let $f_1, \dots, f_m$ be quadrics on $\mathbb{P}^V$ with a common point p . Suppose their differential matrix at p has rank V . If there is a nonzero

$$h \in \text{span}(f_1, \dots, f_m)$$

whose first jet at p is zero, then the rational map

$$\phi_f : \mathbb{P}^V \dashrightarrow \mathbb{P}^{m-1}, \quad x \longmapsto [f_1(x) : \dots : f_m(x)]$$

has differential rank V on a nonempty open subset.

Proof. Choose projective coordinates with $p = [1 : 0 : \dots : 0]$ and affine coordinates $y = (y_1, \dots, y_V)$. After an equation-basis change, the first V forms have expansions

$$f_j(y) = y_j + q_j(y),$$

where each q_j is quadratic. Since h is a quadric with zero value and zero differential at p , it is a nonzero homogeneous quadratic $q(y)$ on this chart.

Consider the $(V + 1) \times (V + 1)$ first-jet minor whose rows are $f_1, \dots, f_V, h$ and whose columns are value followed by the V first derivatives. Up to a nonzero sign, its lowest-degree homogeneous term is

$$q(y) - \nabla q(y) \cdot y = -q(y),$$

using Euler's identity $\nabla q(y) \cdot y = 2q(y)$. It is therefore nonzero. Hence the full first-jet rank is $V + 1$ generically, which is equivalent to differential rank V for ϕ_f away from the base locus. $\square$

Lemma 22.9 (Finite regular base locus plus separability). Let $f_1, \dots, f_m$ be quadrics on $\mathbb{P}^V$ over an algebraically closed field. Assume:

1. their common base scheme $B = V(f_1, \dots, f_m)$ is finite and nonempty;
2. the $m \times V$ projective differential matrix has rank V at every point of B ;
3. ϕ_f has differential rank V on a nonempty open subset.

Then a nonempty Zariski-open set of matrices $R \in k^{V \times m}$ makes

$$\left(\sum_i R_{1i} f_i, \dots, \sum_i R_{Vi} f_i \right)$$

a smooth zero-dimensional complete intersection of degree 2^V .

Proof. Let $\mathcal{R} = k^{V \times m}$. For $x \notin B$, write

$$a(x) = (f_1(x), \dots, f_m(x)) \neq 0$$

and let $J(x)$ be the differential matrix. A matrix R has x as a root exactly when $Ra(x) = 0$, a codimension- V linear condition on $\mathcal{R}$.

On the open set where the first-jet matrix $[a(x) \mid J(x)]$ has rank $V + 1$, the map

$$a(x)^\perp \longrightarrow T_x^* \mathbb{P}^V, \quad r \longmapsto rJ(x)$$

is surjective. Consequently the additional condition $\det(RJ(x)) = 0$ has codimension one inside $Ra(x) = 0$. After allowing x to vary in a V -dimensional space, this singular-root incidence still has codimension at least one in $\mathcal{R}$.

The first-jet rank-defect locus away from B has dimension at most $V - 1$ by assumption 3. There the root condition $Ra(x) = 0$ alone has codimension V , so its total incidence also has dimension at most $\dim \mathcal{R} - 1$.

Finally, at each $x \in B$, the root condition is automatic, but the rank- V derivative assumption makes $\det(RJ(x)) = 0$ a genuine hypersurface in $\mathcal{R}$. Since B is finite, these basepoint hypersurfaces do not fill $\mathcal{R}$.

Globally, the singular-root incidence is the closed subset of $\mathbb{P}^V \times \mathcal{R}$ cut out by $Ra(x) = 0$ and the maximal minors of $RJ(x)$. Its projection is closed because $\mathbb{P}^V$ is proper, and the preceding stratified dimension bounds show that this projection is proper. Choose R outside it. The resulting projective zero scheme is nonempty because it contains B . It cannot have positive dimension: at every root the Jacobian has rank V , so the Zariski tangent space is zero-dimensional. Therefore the zero scheme is smooth and zero-dimensional. Since it is cut out by V quadrics in $\mathbb{P}^V$, it is a complete intersection of degree 2^V . $\square$

Theorem 22.10 (Existence of a smooth projective core). If

$$F \notin \mathcal{B}_{\text{oil}} \cup \mathcal{B}_{\text{off}} \cup \mathcal{B}_{\text{span}},$$

then the joint polynomial $\Theta_F(M, R)$ in (81) is nonzero.

Proof. Choose a regular geometric oil point $p \in \mathbb{P}(\mathcal{O}_0)(\overline{K})$. By Theorem 22.7, a dense open subset of the irreducible Grassmannian of V -planes through p has finite base intersection and full restricted derivative rank at every basepoint.

After scalar extension, let

$$T_{F, \overline{K}} : \overline{K}^m \longrightarrow \mathcal{U}_{\overline{K}}, \quad (\lambda_i)_i \longmapsto \sum_i \lambda_i F_i.$$

The coefficient kernel of the derivative map at p has dimension $m - V = e$ over $\overline{K}$. Scalar extension preserves the rank of the coefficient map, so

$$\dim_{\overline{K}} \ker T_{F, \overline{K}} = m - \dim_K \text{span}_K(F_1, \dots, F_m).$$

This relation space lies inside the derivative kernel. Since $F \notin \mathcal{B}_{\text{span}}$, its dimension is strictly smaller than e . We may therefore choose a derivative-kernel vector outside $\ker T_{F, \overline{K}}$; it produces a nonzero global quadric $H \in \text{span}_{\overline{K}}(F_i)$ with zero first jet at p .

The condition $H|_{\Lambda} \not\equiv 0$ is open and nonempty: if H vanished on every V -plane through p , it would vanish on their union, which is all of $\mathbb{P}(\mathcal{V})$. Intersecting this open set with the one supplied by [Theorem 22.7](#) gives a plane Λ satisfying both properties. On this plane, [Theorem 22.8](#) makes the restricted linear system generically separable, and [Theorem 22.9](#) supplies an output matrix R whose core is a smooth complete intersection of degree 2^V . Hence Θ_F is not the zero polynomial. $\square$

22.3 Finite-field density of the three bad loci

22.3.1 First-jet surjectivity away from oil

Lemma 22.11 (The central-form space spans every off-oil first jet). For every $x \in \mathbb{P}(\mathcal{V}) \setminus \mathbb{P}(\mathcal{O}_0)$, the first-jet evaluation map

$$\mathcal{U} \longrightarrow K \oplus T_x^* \mathbb{P}(\mathcal{V})$$

is surjective after scalar extension to $\overline{K}$.

Proof. Choose a vinegar coordinate v_j that is nonzero at x and normalize to the chart $v_j = 1$. The quadrics

$$v_j^2, \quad v_j v_k \ (k \neq j), \quad v_j o_a \ (1 \leq a \leq O)$$

belong to $\mathcal{U}$ and restrict to the affine functions 1, the remaining vinegar coordinates, and the oil coordinates. Their first jets form a basis of the value-and-cotangent space. $\square$

22.3.2 The off-oil incidence

The codimension count has a simple structure. At a fixed off-oil point, the values and first derivatives of each central form can be prescribed independently by [Theorem 22.11](#). Making the point a common zero costs m conditions, and forcing derivative rank at most $V - 1$ adds the usual determinantal codimension. Projecting away the point can recover at most its $N - 1$ coordinates. The next lemma makes this count uniform and controls the projection degree.

Let $\mathcal{A} = \mathcal{U}^m$ be the affine key space. For an affine variety, write cdeg for the sum of the degrees of its irreducible components. For each vinegar chart $C_j \simeq \mathbb{A}^{N-1}$, let $Z_j \subset \mathcal{A} \times C_j$ be the incidence of pairs (F, x) such that all $F_i(x)$ vanish and the $m \times (N - 1)$ derivative matrix has rank at most $V - 1$.

Lemma 22.12 (Codimension and cumulative degree of the off-oil incidence). Each Z_j is pure of codimension

$$m + c_{\det}, \quad c_{\det} = (m - V + 1)O = (e + 1)O,$$

and satisfies

$$\text{cdeg}(Z_j) \leq 3^m (2V)^{c_{\det}}.$$

If Y_j is the Zariski closure of its projection to key space, then

$$\text{codim}_{\mathcal{A}} Y_j \geq c_{\text{off}} := m + c_{\det} - (N - 1) = e(O + 1) + 1, \quad \text{cdeg}(Y_j) \leq \text{cdeg}(Z_j).$$

Proof. Structure and codimension. On the chart $v_j = 1$, the monomials v_j^2 , $v_j v_k$ for $k \neq j$, and $v_j o_a$ restrict to 1, the remaining vinegar coordinates, and the oil coordinates. They give a polynomial splitting of value-and-first-jet evaluation. After a triangular polynomial change

of coefficient coordinates, Z_j is the product of an affine space, the equations setting m value coordinates to zero, and the determinantal variety of $m \times (N - 1)$ matrices of rank at most $V - 1$. Hence Z_j is irreducible and pure of codimension

$$m + (m - V + 1)((N - 1) - (V - 1)) = m + (m - V + 1)O.$$

Cumulative degree. In the original coefficient-and-point coordinates, every value equation has total degree at most three. Every derivative entry has degree at most two, so each $V \times V$ minor has degree at most $2V$. At a generic rank- $(V - 1)$ point, the determinantal variety is smooth of codimension $c_{\det}$, and the differentials of its maximal minors span the conormal space. Choose $c_{\det}$ generic linear combinations of those minors whose differentials form a basis there. Together with the m value equations, they cut a complete intersection of codimension $m + c_{\det}$ that contains the irreducible variety Z_j as a component. Cumulative Bézout therefore gives

$$\text{cdeg}(Z_j) \leq 3^m (2V)^{c_{\det}}$$

by the cumulative-degree bounds of Heintz and Lachaud–Rolland [23, 24].

Projection to key space. Projection can recover at most the $N - 1$ chart coordinates, which gives

$$\text{codim}_{\mathcal{A}} Y_j \geq m + c_{\det} - (N - 1) = e(O + 1) + 1.$$

The coordinate projection $\mathcal{A} \times C_j \rightarrow \mathcal{A}$ is linear. The standard degree-under-linear-projection inequality gives

$$\deg \pi(\overline{Z_j}) \leq \deg Z_j$$

for an irreducible affine variety [23, 24]. Since both Z_j and its image closure Y_j are irreducible, this is exactly $\text{cdeg}(Y_j) \leq \text{cdeg}(Z_j)$. $\square$

Proposition 22.13 (Explicit off-oil density). Under the ideal-expansion distribution,

$$\Pr[F \in \mathcal{B}_{\text{off}}] \leq \eta_{\text{off}} := V 3^m (2V)^{(e+1)O} Q^{-\{e(O+1)+1\}}. \quad (83)$$

Proof. Every off-oil point lies in one of the V vinegar charts, so $\mathcal{B}_{\text{off}} \subseteq \bigcup_j Y_j$. The affine finite-field point bound

$$|Y_j(\mathbb{F}_Q)| \leq \text{cdeg}(Y_j) Q^{\dim Y_j}$$

from Lachaud–Rolland [24], followed by the union bound and [Theorem 22.12](#), gives the formula after division by $|\mathcal{A}(\mathbb{F}_Q)|$. $\square$

22.3.3 The oil and low-span loci

Let

$$\tau_{m,V} = 1 - \prod_{j=0}^{V-1} (1 - Q^{j-m})$$

be the rank-deficiency probability of a uniform $m \times V$ matrix. If every geometric oil direction is rank deficient, then in particular all O standard oil-coordinate directions are rank deficient. Their derivative matrices are independent in the public-random expansion model. Hence

$$\Pr[F \in \mathcal{B}_{\text{oil}}] \leq \eta_{\text{oil}} := \tau_{m,V}^O. \quad (84)$$

Let $e = m - V$ and let $T_F : K^m \rightarrow \mathcal{U}$ be the coefficient map from the proof of [Theorem 22.10](#). The event $F \in \mathcal{B}_{\text{span}}$ is exactly $\dim \ker T_F \geq e$. For each fixed e -space $W \subset K^m$, the condition

$W \subseteq \ker T_F$ imposes eD_{form} independent linear conditions on the uniform map T_F . A union bound over the Grassmannian therefore gives

$$\Pr[F \in \mathcal{B}_{\text{span}}] \leq \eta_{\text{span}} := \begin{bmatrix} m \\ e \end{bmatrix}_Q Q^{-eD_{\text{form}}}. \quad (85)$$

Thus the geometric proof requires only $\dim \text{span}(F_1, \dots, F_m) > V$; full linear independence is unnecessary.

Combining [Theorem 22.10](#) with (83), (84), and (85) gives the key theorem.

Theorem 22.14 (Almost-all-key joint regularizability). In the public-random expansion model,

$$\Pr[\Theta_F \equiv 0] \leq \eta_{\text{off}} + \eta_{\text{oil}} + \eta_{\text{span}}. \quad (86)$$

For the Round-2 parameter sets:

Level	c_{off}	$\log_2 \eta_{\text{off}}$	$\log_2 \eta_{\text{oil}}$	$\log_2 \Pr[\Theta_F \equiv 0]$
I	39	-364.564	-1132.167	< -364.563
III	55	-457.920	-1635.352	< -457.919
V	109	-1038.067	-2935.248	< -1038.066

The low-span term is below 2^{-94850} already at Level I and is invisible at the displayed precision.

Proof. By [Theorem 22.10](#), the event $\Theta_F \equiv 0$ is contained in $\mathcal{B}_{\text{oil}} \cup \mathcal{B}_{\text{off}} \cup \mathcal{B}_{\text{span}}$. Apply (83), (84), (85), and the union bound. The displayed values are direct evaluations of those formulas, rounded outward at the shown precision. $\square$

23 Finite-field solver interfaces and verified restart

We first specialize a finite-field theorem for locally closed sets to the graph-direction system. This gives a recovery route whose correctness needs only the local Jacobian condition of [Section 6](#). We then return to the faster projective route and its near-quadratic solver. In both cases, every accepted key is checked against the unchanged public map.

23.1 Locally closed finite-field solving

The following proposition records the exact external interface used below. It is a specialization of Theorem 12 of Giménez et al. [8].

Proposition 23.1 (Locally closed finite-field solver). Let $F_1, \dots, F_r, H \in \mathbb{F}_q[X_1, \dots, X_n]$ have degree at most d . Suppose that, in the localization $\mathbb{F}_q[X]_H$,

1. $F_1, \dots, F_r$ form a regular sequence; and
2. $(F_1, \dots, F_s)$ is radical for every $1 \leq s \leq r$.

Let V_s be the Zariski closure of $\{F_1 = \dots = F_s = 0, H \neq 0\}$, put $\delta_s = \deg V_s$, and let $\delta = \max_s \delta_s$. If the input has division-free straight-line-program length L , $0 < \epsilon < 1/4$, and

$$q > 2\epsilon^{-1}n^2rd\delta^3, \quad (87)$$

then a probabilistic algorithm computes a Kronecker representation of a lifting fiber of V_r with probability at least $1 - 2\epsilon$ and bit complexity

$$\tilde{O}\left((n^4 + r(L + n^2 + r^4))d\delta^2 \log_2 q\right). \quad (88)$$

When $r = n$, the lifting fiber is the entire zero-dimensional locally closed set. The solver is Monte Carlo: its paper does not provide an efficient certificate that a returned representation is complete.

Proof. The two displayed algebraic conditions are hypotheses (H1)–(H2) of the cited theorem. Equation (87), success probability $1 - 2\epsilon$, and (88) are its finite-field statement. The source explicitly notes that its algorithms do not appear to be Las Vegas because output correctness cannot in general be verified at comparable cost. For $r = n$, no free projection coordinates remain, so a lifting fiber is the zero-dimensional set itself. The source treats $\delta > d$ as without loss of generality; if $\delta \leq d$, replacing it by the safe upper bound $d + 1$ only enlarges the field and cost bounds. $\square$

Corollary 23.2 (Graph-local specialization). Assume $\text{rank } \mathcal{B}(c) = V$ and $R\mathcal{B}(c)$ is invertible. Apply Theorem 23.1 to the V affine quadrics $f = Rq_c$ and the localization polynomial $h = \det J_f$. Then

$$n = r = V, \quad d = V, \quad \delta \leq D := 2^V.$$

The input has straight-line-program length

$$L_{\text{loc}} = O(L_{\text{core}} + V^4), \quad L_{\text{core}} := M_V + 2V(M_V + V + 1), \quad M_V := \frac{V(V+1)}{2}. \quad (89)$$

Over any extension $L = \mathbb{F}_{Q^s}$ satisfying

$$Q^s > 2\epsilon^{-1}V^4 8^V, \quad (90)$$

the solver contains the planted root Gc with probability at least $1 - 2\epsilon$ and has bit complexity

$$\tilde{O}\left((V^4 + V(L_{\text{loc}} + V^2 + V^4))V4^V \log_2(Q^s)\right). \quad (91)$$

Proof. Theorem 6.2 gives the localized regularity and radicality hypotheses and places Gc in the open set. Bézout bounds every intermediate degree by $2^s \leq 2^V$. The maximum input degree is V because h has degree V . The dense quadratic core has the shared-monomial circuit in (89); Baur–Strassen differentiation followed by a division-free Samuelson–Berkowitz determinant circuit adds $O(V^4)$ operations. Substitution into (87)–(88) proves the result. $\square$

Corollary 23.3 (Smooth-projective Giménez cross-check). On a smooth projective core, choose a hyperplane at infinity and dehomogenize in the complementary affine chart. Conditional on the hyperplane avoiding all $D = 2^V$ projective roots, apply Theorem 23.1 with $H = 1$. Then

$$n = r = V, \quad d = 2, \quad \delta \leq D,$$

and, over $\mathbb{F}_{Q^s}$ with

$$Q^s > 4\epsilon^{-1}V^3 8^V, \quad (92)$$

the direct displayed complexity is

$$\tilde{O}\left(2(V^4 + V(L_{\text{core}} + V^2 + V^4))4^V \log_2(Q^s)\right). \quad (93)$$

The conditional solver success probability is at least $1 - 2\epsilon$. If the hyperplane is sampled uniformly over $\mathbb{F}_{Q^s}$, the unconditional success probability of the chart-plus-solver step is at least

$$1 - 2\epsilon - \frac{D}{Q^s}. \quad (94)$$

Proof. Smoothness of the full projective zero scheme implies regularity and geometric radicality of every prefix by Theorem 22.4. A hyperplane avoiding the finite zero set exists, and dehomogenization in its complementary affine chart preserves those properties. Substitute $d = 2$ and $\delta \leq 2^V$ into Theorem 23.1. A uniformly random hyperplane contains any fixed projective point with probability at most $1/Q^s$; a union bound over the D roots gives the additional D/Q^s term in (94). $\square$

Why Monte Carlo output is sufficient here. A successful call to [Theorem 23.1](#) contains the planted nonsingular root. A failed call may omit it or return unusable data. The attack parses every returned representation, retains only actual roots of all original equations, completes and verifies the graph, checks the inverse pull-back and public-map factorization, and finally checks an invertible oil system. If no candidate passes, it retries with fresh coins. Thus the external solver's lack of a completeness certificate can delay termination but cannot create a false accepted key.

23.2 Fast projective route: field size, failure probability, and cost

Put

$$D := 2^V, \quad L_s := \mathbb{F}_{Q^s}, \quad q_s := |L_s|.$$

The finite-field theorem of van der Hoeven and Lecerf has an exponent parameter and a field-size requirement. They must be chosen together. We use $\varepsilon > 0$ for the desired final exponent and set the theorem's internal parameter to $\eta = \varepsilon/3$. For V quadrics, its field-size threshold is

$$B_V(\eta) := \max\left\{\max\left\{2, (1 + \eta)(8/\eta + 1)V2^{\eta/8}\right\}(2^V - 1), 1008^V\right\}. \quad (95)$$

Define the smallest admissible extension degree and a one-degree safety choice by

$$s_\varepsilon^0 := \min\{s \geq 1 : Q^s > B_V(\varepsilon/3)\}, \quad s_\varepsilon^+ := s_\varepsilon^0 + 1. \quad (96)$$

The first choice minimizes field size. The second divides all finite-invocation failure bounds by the large factor $Q = 2,048,383$.

For any $s \geq s_\varepsilon^0$, sample

$$M \leftarrow L_s^{(V+O) \times (V+1)}, \quad R \leftarrow L_s^{V \times m}$$

and form the restricted square core. For every fixed key with $\Theta_F \neq 0$, Schwartz–Zippel gives

$$\beta_{V,s} := \Pr[\Theta_F(M, R) = 0] \leq \frac{E_V^{\text{joint}}}{Q^s}, \quad E_V^{\text{joint}} := 3V(V+1)2^{V-1}. \quad (97)$$

The proof of the finite solver theorem bounds its three internal bad choices by

$$\alpha_{V,s} := \frac{5D^2 + 26D^3}{Q^s}. \quad (98)$$

These are bounds for one finite invocation. A denominator failure, degree mismatch, or failed final check returns fail; the algorithm never runs an unproved internal restart loop on an irregular outer draw.

For the primary choice $\varepsilon = 0.3$, the field threshold is governed by 1008^V . The minimum and safety choices are:

Level	$s_{0.3}^0$	success at $s_{0.3}^0$	$s_{0.3}^+$	success at $s_{0.3}^+$
I	8	≥ 0.992337625602	9	≥ 0.999999996259
III	12	≥ 0.999997944684	13	≥ 0.999999999998
V	15	≥ 0.927725586161	16	≥ 0.999999964716

Here success means that the outer core is smooth and the solver's internal choices are good, conditional on $\Theta_F \neq 0$; it is at least $(1 - \beta_{V,s})(1 - \alpha_{V,s})$. At the safety fields, the base-two logarithms of $(\alpha_{V,s}, \beta_{V,s})$ are

$$(-27.994, -124.681), \quad (-39.858, -183.459), \quad (-24.756, -219.513).$$

Thus the minimum fields are adequate for expected-time verified restart, while the safety fields give a clean one-invocation theorem. The dependence on ε in (96) is essential: no fixed list of extension degrees works for every arbitrarily small fixed $\varepsilon > 0$.

23.3 Published near-quadratic Kronecker solver

Proposition 23.4 (Fast smooth-core specialization). Let $g_1, \dots, g_V$ be homogeneous quadrics over a finite field $\mathbb{F}_{p^k}$. Assume that they form a regular sequence and every projective prefix is geometrically radical. Put $D_i = 2^i$ and $D = D_V$. Fix a rational $\varepsilon > 0$, set $\eta = \varepsilon/3$, and assume

$$p^k > B_V(\eta).$$

One finite invocation of Theorem 5.4 of van der Hoeven and Lecerf uses

$$\tilde{O}\left(D 2^{(1+\varepsilon)(V-1)} k \log_2 p\right) \quad (99)$$

bit operations and fails because of its random coordinate, intersection, or evaluation choices with probability at most

$$\frac{5D^2 + 26D^3}{p^k}. \quad (100)$$

Outside those bad choices it returns a complete Kronecker parametrization. The checks in Theorem 23.6, including the requirement that the separator have degree D , certify completeness for this task. Repeating checked finite invocations on an input satisfying the hypotheses gives the Las Vegas expected-time form of Corollary 5.5.

For $K = \mathbb{F}_{127^3}$ and any $s \geq s_\varepsilon^0$, the explicit finite-invocation cost over L_s is obtained from (99) with $k = 3s$ and $p = 127$. Since $s = O(V) = O(\log D)$ for the choices above, this field-degree factor is absorbed by soft- O notation. Including field construction, casting, public slicing, output recombination, solving, and verification, the full expected projective route has scale

$$T_{\varepsilon,V} := \tilde{O}\left(2^V 2^{(1+\varepsilon)(V-1)} 3 \log_2 127\right). \quad (101)$$

Proof. The first statement is Theorem 5.4 specialized to the degree sequence $(2, \dots, 2)$, with internal parameter $\eta = \varepsilon/3$. Its complete field condition is (95); its complexity exponent $1 + 3\eta$ becomes $1 + \varepsilon$. The three bad-choice bounds in the proof sum to (100).

A smooth zero-dimensional projective core satisfies the required regularity and prefix-radicality assumptions by Theorem 22.4. Corollary 5.5 of the source constructs a suitable extension, casts the input, verifies the returned parametrization, and obtains the same expected soft- O scale. Sampling M and R , restricting the dense quadrics, and forming the output combinations costs only polynomially many operations in the working field and is lower order than the degree-dependent term. $\square$

Corollary 23.5 (One checked projective invocation). Fix $\varepsilon > 0$, choose any $s \geq s_\varepsilon^0$, and fix a key with $\Theta_F \neq 0$. Run one outer draw over L_s , one finite invocation from Theorem 23.4, and all checks from Theorem 23.6; return fail if any check rejects. The invocation has soft- O cost $T_{\varepsilon,V}$ and succeeds with probability at least

$$(1 - \beta_{V,s})(1 - \alpha_{V,s}). \quad (102)$$

For $\varepsilon = 0.3$, the minimum-field and safety-field probabilities are those in the preceding table.

Proof. The outer discriminant fails with probability at most $\beta_{V,s}$. Conditional on a smooth core, Theorem 23.4 fails internally with probability at most $\alpha_{V,s}$. The two groups of coins are fresh. Exact checking changes malformed or incomplete output into fail, never into a false accepted result. $\square$

23.4 A classical arithmetic cross-check

Let

$$M_2 := \binom{V+2}{2}, \quad L_{\text{quad}} := M_2 + V(2M_2 - 1). \quad (103)$$

The first term computes every degree-two monomial in the $V+1$ homogeneous variables once. Each of the V dense quadrics then uses M_2 scalar multiplications and $M_2 - 1$ additions. Thus $L_{\text{quad}} = \Theta(V^3)$ is an explicit division-free circuit for the full square core.

Giusti–Lecerf–Salvy give the older arithmetic bound

$$L_{\text{quad}} \tilde{O}(D^2) \quad (104)$$

field operations under the same regularity conditions [7, 6]. We retain (104) only as an epsilon-free growth-rate cross-check. We do *not* transfer the explicit finite-field failure constants of Theorem 5.4 to this older bound: the literature cited here does not state that combined theorem. Accordingly, the SLP line below has no proved one-invocation failure probability and is not used for the universal-forgery corollary.

Proposition 23.6 (Task-specific resolution checks). Let $f = (f_1, \dots, f_V)$ be affine quadrics over a finite field L , and put $D = 2^V$. Suppose a proposed geometric resolution consists of a monic polynomial $m(T)$, coordinate polynomials $v_1(T), \dots, v_V(T)$, and a linear form u . Reject it unless every v_j has degree below $\deg m$ and

1. $1 \leq \deg m \leq D$ and m is squarefree;
2. $u(v_1(T), \dots, v_V(T)) \equiv T \pmod{m}$;
3. $f_i(v_1(T), \dots, v_V(T)) \equiv 0 \pmod{m}$ for every i ;
4. $\gcd(m(T), \det J_f(v_1(T), \dots, v_V(T))) = 1$.

If the solver returns standard numerators w_j with denominator m' , first verify $\gcd(m, m') = 1$ and set $v_j = w_j(m')^{-1} \pmod{m}$. Passing these checks certifies that the represented points are distinct simple solutions of the square core. When this proposition is used with a smooth zero-dimensional complete intersection of V quadrics, requiring in addition $\deg m = D$ certifies completeness by Bézout. The checks cost $\tilde{O}(\text{poly}(V)D \log |L|)$ Boolean gates.

Proof. Squarefreeness gives distinct roots τ of m over $\bar{L}$. The primitive-form identity makes the represented points $P_\tau = (v_1(\tau), \dots, v_V(\tau))$ distinct. The equation checks put them in $V(f)$, and the Jacobian $\gcd$ makes each one simple. Evaluate the quadrics and Jacobian in $L[T]/(m)$; compute the determinant there with a division-free algorithm because this quotient need not be a field. Polynomial arithmetic, gcds, and the $V \times V$ determinant have the stated cost. $\square$

Why the extra checks remain useful. The published Las Vegas solver already verifies its complete parametrization. The task-specific checks make the cryptanalytic interface self-contained and protect against malformed implementations, truncated outputs, and conversion errors. On every branch, final acceptance additionally requires all unrecombined equations, exact public-map factorization, and an invertible oil system. No malformed solver transcript can create a false accepted key.

23.5 Root selection and oil-space recovery

Lemma 23.7 (Recovery from the projective oil point). Let L/K be a finite extension. Suppose $\Theta_F(M, R) \neq 0$, and let $[t^*] \in \mathbb{P}^V(L)$ be the unique point for which $x^* = Mt^*$ lies in $\mathbb{P}(\mathcal{O})_L$. Then

$$\text{rank}(P_1 x^* \cdots P_m x^*) = V, \quad \ker(P_1 x^* \cdots P_m x^*)^\top = \mathcal{O}_L.$$

Row reduction recovers the public graph of the oil space. For the planted point that graph is defined over K and, after the specification's inverse pull-back, yields the planted expanded UOV decomposition.

Proof. The full-rank matrix M defines a projective V -plane over L . Its vector-space intersection with the O -dimensional oil space is nonzero by dimension count. If that intersection had dimension at least two, all restricted quadrics would vanish on a positive-dimensional projective linear space, contradicting smooth zero-dimensionality. Thus it is one L -rational point.

Smoothness of the square core gives derivative rank at least V there. Total isotropy puts the O -dimensional oil space inside the ambient common derivative kernel, so the rank is at most $(V + O) - O = V$. Equality holds, and [Theorem 5.10](#) identifies the kernel with $\mathcal{O}_L$. The remaining statements follow from K -descent and [Theorem 3.4](#). $\square$

Lemma 23.8 (Separator descent). Let a reduced zero-dimensional scheme over $L = \mathbb{F}_q$ have a Kronecker representation whose primitive linear form u separates its geometric points. A geometric point P is L -rational if and only if $u(P) \in L$.

Proof. The forward implication is immediate. Conversely, if $u(P) \in L$, then

$$u(P^{(q)}) = u(P)^q = u(P).$$

Because u is defined over L and separates the geometric points, Frobenius must fix P . $\square$

Proposition 23.9 (Filter all original equations before factoring). Let a consistency-checked resolution over L_s have squarefree separator polynomial $m_u(T)$ and coordinate polynomials $v(T)$. Let $\tilde{q}_1, \dots, \tilde{q}_m$ be the dehomogenized original restricted quadrics in the same affine chart, and define

$$a(T) := \gcd(m_u(T), \tilde{q}_1(v(T)), \dots, \tilde{q}_m(v(T))). \quad (105)$$

Then the roots of a are exactly the separator values of represented points satisfying all m original equations. Moreover,

$$a_{L_s}(T) := \gcd(a(T), T^{q_s} - T) \quad (106)$$

has exactly the separator values of those common points that are L_s -rational. Computing a and a_{L_s} costs $\tilde{O}(D \text{ poly}(V, m, \log |L_s|))$ Boolean gates. Only a_{L_s} , whose degree may be much smaller than D , must be factored.

Proof. Work over an algebraic closure of L_s . Because m_u is squarefree, it is the product of the distinct factors $T - \tau$ indexed by the represented points. For each root τ ,

$$\tilde{q}_i(v(T)) \bmod m_u \big|_{T=\tau} = \tilde{q}_i(v(\tau)).$$

Thus $T - \tau$ divides (105) exactly when every original restricted equation vanishes at the represented point. The second claim follows from [Theorem 23.8](#) and the identity $z^{q_s} = z$ for $z \in L_s$. Dense quadratic evaluation in $L_s[T]/(m_u)$, modular exponentiation, and polynomial gcd computation at degree at most D give the stated cost. Since $\log |L_s| = O(V)$ for the selected fields, it is lower order than the degree-squared solver bound. $\square$

Check the resolution, prefilter, and extract rational roots. Convert the solver output to canonical coordinate polynomials and run the consistency checks of [Theorem 23.6](#). On failure, discard the invocation. On a successful Las Vegas solver call, the returned separator polynomial $m_u(T) \in L_s[T]$ represents the full affine zero scheme and contains the planted oil point. The additional checks reject malformed or incomplete implementation outputs.

Apply [Theorem 23.9](#): evaluate all m original restricted equations in $L_s[T]/(m_u)$, compute $a(T)$, and then compute $a_{L_s}(T)$. Factor only a_{L_s} over L_s . A Las Vegas finite-field factorization algorithm terminates with probability one and has expected subquadratic dependence on the input degree, uniformly for inputs of degree at most D ; its expected contribution is therefore below the solver’s near-quadratic-in- D scale [21]. Reconstruct the corresponding affine points, undo the coordinate changes, and recheck all m original equations as a final exact verification.

Certify the decomposition. For each survivor x , compute the common derivative kernel. Require an O -dimensional graph defined over K , total isotropy, and every identity (4). Apply the specification’s inverse pull-back and verify the base-field factorization against the original public map. An output is called an *exact candidate* only after all these checks pass.

Certify signing or restart. Deduplicate exact candidates by their K -rational graph. For each remaining candidate, first test the V basis assignments from [Theorem 3.7](#). If they all fail, scan the remaining nonzero combinations on the designated pair lines. Return the first candidate with an invertible oil matrix, together with that matrix’s inverse. If no candidate passes, discard the trial and sample a fresh slice. On a correct solver branch the candidate list contains the planted decomposition, so outside the event in (11) this scan succeeds deterministically. At most

$$D(V + \lfloor V/2 \rfloor (q - 1))$$

oil matrices are tested per trial, below the solver’s near-quadratic-in- D scale. In the public-random model, the basis-first part succeeds after fewer than 1.008 tests on average.

Let $A_{\epsilon, V, s}$ denote the event that the full trial accepts a publicly verified signing decomposition and inverse oil matrix. On a key outside $\mathcal{E}_{\text{sign}}$, a draw with $\Theta_F(M, R) \neq 0$ yields the planted exact candidate whenever the finite solver invocation succeeds, after which the deterministic signing scan succeeds. The outer and internal coins are fresh, so

$$\Pr[A_{\epsilon, V, s}] \geq (1 - \beta_{V, s})(1 - \alpha_{V, s}). \quad (107)$$

This one-sided implication is the only interface needed between the solver guarantee and the verified attack. On irregular inputs or bad internal choices, a failed check returns fail; it never creates a false accepted key.

Lemma 23.10 (Verified-restart accounting). Suppose each fresh trial, conditional on any history of earlier failures, accepts with probability at least $p > 0$. Then the number N of trials satisfies

$$\Pr[N > t] \leq (1 - p)^t, \quad \mathbb{E}[N] \leq p^{-1}.$$

If the current trial uses fresh coins and has expected work at most T , uniformly over prior histories, then the expected total work is at most T/p .

Proof. The tail bound follows inductively from the conditional success lower bound, and summing the tail probabilities gives $\mathbb{E}[N] \leq 1/p$. For the work bound, the event that trial j is reached depends only on earlier trials. Fresh current-trial coins therefore give

$$\mathbb{E}[\mathbf{1}_{\{N \geq j\}} W_j] \leq T \Pr[N \geq j].$$

Summing over j and applying the trial bound proves the claim. $\square$

23.6 Main recovery theorem

Use $T_{\epsilon, V}$ from (101). For $0 < \epsilon < 1/4$ and an extension degree s satisfying (90), write

$$T_{\text{loc}}(\epsilon, s) := \tilde{O}\left((V^4 + V(L_{\text{loc}} + V^2 + V^4))V4^V \log_2(Q^s)\right).$$

Theorem 23.11 (Key-by-key recovery from one graph direction). Let an expanded QR-UOV key over K have a valid planted decomposition. Suppose that at least one maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c . Fix $0 < \epsilon < 1/4$ and choose s so that (90) holds. Put

$$p_{\text{dir}} := \left(1 - \frac{V}{Q}\right) \left(1 - \frac{V(m-V)}{Q}\right) (1 - 2\epsilon).$$

Then the following verified attacks are correct for this fixed key.

1. A randomized graph-direction trial finds an exactly verified expanded UOV decomposition with probability at least p_{dir} . Repeating fresh trials terminates almost surely, uses at most p_{dir}^{-1} trials on average, and has expected work

$$O(T_{\text{loc}}(\epsilon, s)/p_{\text{dir}}). \quad (108)$$

Every accepted output is correct.

2. If the determinant polynomial Δ of the planted decomposition is nonzero, testing one fresh uniform scalar-block vinegar assignment in each trial returns a signing certificate with expected trial count at most

$$(p_{\text{dir}}(1 - m/q))^{-1}. \quad (109)$$

3. If one planted basis matrix is invertible, the deterministic basis-first signing scan succeeds on every successful recovery trial. Under the designated-pair condition of Theorem 3.7, the full basis-and-line scan does so after at most $V + \lfloor V/2 \rfloor (q - 1)$ oil-matrix tests per candidate. In either case the expected trial and work bounds are those of item 1.
4. If a coordinate direction e_j satisfies $\text{rank } \mathcal{B}(e_j) = V$, all outer attack choices can be made deterministic: scan the $O[V(m - V) + 1]$ direction/condenser pairs of Theorem 5.7, and repeat the finitely many probabilistic solver calls in rounds. This still terminates almost surely and adds only a polynomial factor. The shorter lists of Theorem 5.8 give category-scale failure bounds in the public-random model.

The theorem uses only a local nonsingularity condition at the planted root. It does not require the projective discriminant, the off-oil incidence argument, a key-generation distribution, or uniqueness of the recovered decomposition.

Proof. **A good trial contains the planted root.** By Theorem 5.6, a random c makes $\mathcal{B}(c)$ full rank with probability at least $1 - V/Q$. Conditional on that event, a random t makes $\mathcal{C}(t)\mathcal{B}(c)$ invertible with probability at least $1 - V(m - V)/Q$. The planted root then lies in the localized smooth set by Theorem 6.2. By Theorem 23.2, the solver returns a representation containing it with probability at least $1 - 2\epsilon$.

One planted root gives the whole hidden graph. For every returned root x , solve the public systems (36). At $x = Gc$, Theorem 5.2 returns every column of G . The attack then checks all original direction equations, rank conditions, graph identities, inverse pull-back identities, and public-map factorizations. Any root that does not yield a valid graph is discarded.

Restart and signing. A fresh trial therefore reaches the planted expanded decomposition with probability at least p_{dir} . Apply Theorem 23.10. The randomized and deterministic signing claims follow from Theorem 3.7. Filtering and completing at most D roots costs

$\tilde{O}(D \text{ poly}(V, O, m, \log |L|))$, below (91). Exact verification prevents false acceptance even when the internal Monte Carlo solver fails.

For item 4, [Theorem 5.7](#) guarantees that a regular coordinate direction has a condenser value with nonsingular planted Jacobian. The deterministic scan reaches it. Repeating all finitely many solver calls in rounds gives a fixed positive success probability per round and hence almost-sure termination. $\square$

Corollary 23.12 (Public-random density and compact deterministic certificates). In the public-random expansion model, the probability that all O coordinate directions are rank deficient is exactly $\tau_{m,V}(Q)^O$, whose base-two logarithms are

$$-1132.166907, \quad -1635.352198, \quad -2935.247544.$$

Hence, except with probability at most

$$\eta_{\text{local}} := \tau_{m,V}(Q)^O + \eta_{\text{sign}}, \quad (110)$$

the full deterministic direction list and deterministic signing fallback recover a universal signing certificate.

A much shorter fixed attack already has category-scale coverage. Use the first 3, 4, 4 coordinate directions and their condenser lists, together with the first 22, 31, 40 basis signing assignments, at Levels I, III, and V. These attacks solve only 315, 612, 1228 square systems and test at most 22, 31, 40 oil matrices per exact candidate. Their failure probabilities are at most

$$\tau_{m,V}(Q)^{r_{\text{dir}}} + \sigma_m^{b_{\text{sign}}},$$

with base-two logarithms below

$$-153.502, \quad -216.298, \quad -279.095. \quad (111)$$

No independence between the recovery and signing failures is needed for this union bound.

Proof. The all-direction statement is (42), followed by [Theorem 3.7](#) and the union bound. For the compact statement, combine [Theorem 5.8](#) with the fixed-prefix signing bound (10). On the complement, the deterministic outer list contains a square system with the planted point nonsingular, and the fixed signing list contains an invertible planted oil matrix. Round-robin solver repetition and exact verification then give the claimed universal certificate. $\square$

Theorem 23.13 (Fast projective key-by-key recovery). Let an expanded QR-UOV key over K have a valid planted expanded decomposition, and assume that the public joint slice-and-output discriminant $\Theta_F(M, R)$ is not the zero polynomial. Fix a rational $\varepsilon > 0$, choose any $s \geq s_\varepsilon^0$, and put

$$p_{\varepsilon,V,s} := (1 - \beta_{V,s})(1 - \alpha_{V,s}).$$

Then:

1. A decomposition-only verified-restart algorithm terminates with probability one and returns an exactly verified expanded UOV decomposition. Every accepted output is correct, the expected number of outer trials is at most $p_{\varepsilon,V,s}^{-1}$, and the expected work is

$$O\left(\frac{T_{\varepsilon,V}}{p_{\varepsilon,V,s}}\right). \quad (112)$$

For $\varepsilon = 0.3$, the minimum fields are $s = 8, 12, 15$. Their expected-trial bounds are below

$$1.00773, \quad 1.000003, \quad 1.07791.$$

The safety fields $s = 9, 13, 16$ give the near-one single-invocation probabilities in [Theorem 23.5](#).

2. Suppose the planted determinant polynomial Δ from [Theorem 3.7](#) is nonzero. If each outer trial tests one fresh uniform scalar-block vinegar assignment for every exact candidate, verified restart returns an equivalent signing key and inverse oil matrix. The expected number of outer trials is at most

$$\frac{1}{p_{\varepsilon,V,s}(1 - m/q)},$$

and the expected work is

$$O\left(\frac{T_{\varepsilon,V}}{p_{\varepsilon,V,s}(1 - m/q)}\right). \quad (113)$$

At most D oil matrices are tested per trial.

3. If one planted basis matrix M_j is invertible, the basis-first deterministic scan returns an equivalent signing key on every successful core invocation and tests at most DV oil matrices per outer trial. Under the designated-pair condition of [Theorem 3.7](#), the full deterministic fallback tests at most

$$D(V + \lfloor V/2 \rfloor (q - 1))$$

oil matrices per trial. In either case the expected trial count is at most $p_{\varepsilon,V,s}^{-1}$ and the expected work remains $O(T_{\varepsilon,V}/p_{\varepsilon,V,s})$.

These are statements about each fixed expanded key satisfying the displayed conditions. They do not depend on how the key was generated. Recovering the original seed or serialized private key is unnecessary.

Proof. A regular draw yields a complete core resolution. The discriminant hypothesis and (97) imply that a fresh (M, R) gives a smooth full core with probability at least $1 - \beta_{V,s}$. [Theorem 22.4](#) supplies the solver hypotheses. Conditional on a smooth core, the finite invocation in [Theorem 23.4](#) returns and verifies the complete geometric resolution with probability at least $1 - \alpha_{V,s}$. Every invocation has a finite operation schedule and returns fail if a denominator, degree, or verification check rejects. An irregular outer draw therefore cannot trap the attack inside an unjustified restart loop.

The planted decomposition appears among the exact candidates. On a regular draw, the quotient-algebra prefilter of [Theorem 23.9](#), rational-root extraction, and [Theorem 23.7](#) retain the planted oil point and recover the planted expanded decomposition. Exact all-equation checks, inverse pull-back, and public-map factorization reject every invalid candidate.

Signing certification. If $\Delta \neq 0$, [Theorem 3.7](#) gives success probability at least $1 - m/q$ for a fresh uniform assignment on the planted candidate. If a basis matrix or designated pair witness exists, the corresponding deterministic scan succeeds on every regular draw.

Restart and auxiliary work. Apply [Theorem 23.10](#). Finite-field factorization has subquadratic expected dependence on degree, and all filtering, derivative-kernel, descent, pull-back, verification, and signing work costs $\tilde{O}(D \text{poly}(V, O, m, q, \log |L_s|))$. For every fixed $\varepsilon > 0$ this is lower order than $T_{\varepsilon,V}$. Exact verification prevents false acceptance on every branch. $\square$

Corollary 23.14 (Public-random density for the fast projective route). In the public-random expansion model, the probability that either $\Theta_F \equiv 0$ or the planted determinant polynomial Δ is identically zero is below

$$2^{-364.563}, \quad 2^{-457.919}, \quad 2^{-1038.066}$$

at Levels I, III, and V. Every other key satisfies the randomized equivalent-key conclusion of [Theorem 23.13](#). In fact, outside the same bounded union of bad events, one designated pair span contains an invertible matrix, so the deterministic refinement applies.

Proof. The event $\Theta_F \equiv 0$ is bounded by [Theorem 22.14](#). If Δ is the zero polynomial, every designated pair span is singular, so this event is contained in $\mathcal{E}_{\text{sign}}$ and is bounded by [Theorem 3.7](#). The union has probability at most

$$\eta_{\text{off}} + \eta_{\text{oil}} + \eta_{\text{span}} + \eta_{\text{sign}},$$

which gives the displayed numerical bounds. On the complement of this particular union, the designated-pair condition itself holds, so item 3 of [Theorem 23.13](#) applies. $\square$

Corollary 23.15 (One-invocation key-only universal forger). In the public-random expansion model, set $\varepsilon = 0.3$, use the safety field $s = s_{0.3}^+$, run the checked projective invocation of [Theorem 23.5](#), and then run the basis-first scan and projective-line fallback of [Theorem 3.7](#). The adversary uses no signing oracle, never outputs an invalid signature, and can forge every chosen message whenever it succeeds. Its success probability over the key and attack coins is at least

$$1 - \delta_{\text{key}} - \alpha_{V, s_{0.3}^+} - \beta_{V, s_{0.3}^+}, \quad (114)$$

where δ_{key} is the key-level exceptional probability in [Theorem 23.14](#). In particular, the success probabilities at Levels I, III, and V are respectively at least

$$0.999999996259, \quad 0.999999999998, \quad 0.9999999964716.$$

Proof. Outside the key-level exceptional set, the projective discriminant and a deterministic pair-line witness both exist. A successful checked solver pass returns the complete core, so the planted point survives exact filtering and yields the planted expanded decomposition. The deterministic signing scan then returns a certificate that inverts every public target. Apply [Theorem 3.6](#) and union-bound the key, outer-discriminant, and internal-solver failures. Exact verification ensures that failure affects only whether a forgery is returned, never whether a returned forgery is valid. $\square$

24 Official seeded expansion: ideal primitives and fixed functions

The key-by-key theorem needs no expansion model. This appendix asks how often official seeded key generation satisfies the sufficient algebraic conditions for deterministic equivalent-key recovery. The official expansion uses public counter $2i - 1$ for the symmetric block A_i , public counter $2i$ for the rectangular block E_i , and secret counter 1 for the graph G [[1](#), Secs. 4.7–4.10]. For every recommended set, the secret buffer length τ_2 is at most the public A_i buffer length τ_1 . Hence, if the independently sampled public and secret seeds happen to coincide, the secret counter-1 raw stream is a prefix of the public A_1 stream. The resulting correlation between G and A_1 is expressly allowed by [Theorem 3.1](#); it is not a bad event.

For Levels I, III, and V, let $\lambda \in \{128, 192, 256\}$ be the seed length. The specification uses m public buffers of length τ_1 bytes and m public buffers of length τ_2 bytes. Each public rejection sampler exhausts its prescribed buffer with probability at most $2^{-\lambda}$. Put

$$\varepsilon_{\text{pub}} := 2m2^{-\lambda}. \quad (115)$$

No secret-buffer exhaustion term is needed: [Theorem 3.1](#) permits an arbitrary conditional law for G .

24.1 SHAKE in the ideal-XOF model

Proposition 24.1 (Exact public-random transfer in the ideal-XOF model). Model SHAKE as a random function from each domain-separated input to one consistent infinite output string. The

total-variation distance between the official expanded-key distribution and the public-random model of [Theorem 3.1](#) is at most

$$\delta_{\text{XOF}} := \varepsilon_{\text{pub}} = 2m2^{-\lambda}.$$

This statement includes keys with equal public and secret seeds.

Proof. The $2m$ public seed/counter inputs are pairwise distinct, so their ideal-XOF strings are independent and uniform. If the seeds differ, the secret string is independent of all public strings. If they coincide, the secret counter-1 string is the required prefix of the public A_1 string; after rejection sampling, this induces an arbitrary conditional distribution of G given A_1 , exactly as permitted by [Theorem 3.1](#). In either case, the even-counter E_i strings are independent uniform strings conditioned on (A, G) . Abort only if one of the $2m$ public buffers contains too few accepted values. The union bound (115) pays for that event. Conditioned on no public exhaustion, the accepted A_i values have their product-uniform marginal and the accepted E_i values are product-uniform conditioned on (A, G) , so the distribution is exactly [Theorem 3.1](#). $\square$

24.2 AES in the ideal-cipher model

Let

$$T := m \left(\left\lceil \frac{\tau_1}{16} \right\rceil + \left\lceil \frac{\tau_2}{16} \right\rceil \right)$$

be the number of 128-bit public cipher blocks, and define

$$\delta_{\text{perm}}(T) := 1 - \prod_{j=0}^{T-1} \left(1 - \frac{j}{2^{128}} \right), \quad \delta_{\text{IC}} := \delta_{\text{perm}}(T) + \varepsilon_{\text{pub}}. \quad (116)$$

Proposition 24.2 (Exact public-random transfer in the ideal-cipher model). Model AES as an ideal cipher with an independent random permutation for every key. The total-variation distance between the official expanded-key distribution and the public-random model of [Theorem 3.1](#) is at most δ_{IC} . This statement also includes equal public and secret AES keys.

Proof. For every fixed public key, the specification's T public counter blocks are distinct. Their ideal-cipher outputs are therefore a uniform ordered sample without replacement from 2^{128} blocks. The exact total-variation distance from T independent uniform blocks is the collision probability of the independent sample, namely $\delta_{\text{perm}}(T)$.

Couple the without-replacement public block vector to an independent-block vector at this distance. If the public and secret keys differ, generate G from the independent secret-key permutation in both experiments. If the keys coincide, generate G from the prescribed prefix of the coupled public A_1 block vector in each experiment. In both cases this is a common randomized postprocessing kernel, so total variation cannot increase. Under the independent-block vector, A has the product-uniform marginal and E is product-uniform conditioned on (A, G) . Truncation and rejection sampling are further postprocessing. Finally, public-buffer exhaustion contributes at most (115). $\square$

24.3 Independent-graph transfer for direct forgery

The direct residual-root theorem uses [Theorem 3.2](#), so equal public and secret seeds or AES keys must now be paid for rather than absorbed as an allowed graph-coefficient correlation. Define

$$\delta_{\text{XOF}}^{\text{ind}} := \delta_{\text{XOF}} + 2^{-\lambda}, \quad \delta_{\text{IC}}^{\text{ind}} := \delta_{\text{IC}} + 2^{-\lambda}. \quad (117)$$

Table 9: Official public-buffer parameters and exact ideal-primitive transfer bounds. The last column gives base-two logarithms. No seed-collision term is present.

Level	λ	m	τ_1	τ_2	T	$\log_2 \delta_{\text{XOF}} / \log_2 \delta_{\text{IC}}$
I	128	54	4267	2916	24300	-121.245 / -99.863
III	192	78	9020	6123	73866	-184.715 / -96.655
V	256	105	16144	11018	178290	-248.286 / -94.112

Proposition 24.3 (Independent-graph transfer in the ideal-primitive models). The total-variation distance between official expanded-key generation and [Theorem 3.2](#) is at most $\delta_{\text{XOF}}^{\text{ind}}$ when SHAKE is modeled as an ideal XOF, and at most $\delta_{\text{IC}}^{\text{ind}}$ when AES is modeled as an ideal cipher.

Proof. Condition first on the independently sampled public and secret seeds (or AES keys) being distinct. This event fails with probability exactly $2^{-\lambda}$. On the conditioning event, the ideal primitive supplies an independent secret stream for G and independent public streams for A and E . The public buffer and, in the ideal-cipher model, without-replacement couplings are exactly those of [Theorems 24.1](#) and [24.2](#). Their losses are δ_{XOF} and δ_{IC} , respectively. A union bound with the seed-collision event gives (117). $\square$

For Levels I, III, and V, the corresponding base-two logarithms are

$$\log_2 \delta_{\text{XOF}}^{\text{ind}} = -121.232, -184.705, -248.279, \quad \log_2 \delta_{\text{IC}}^{\text{ind}} = -99.863, -96.655, -94.112,$$

rounded upward in the last displayed digit.

Corollary 24.4 (Official seeded direct-forgery invocation in ideal-primitive models). Let p_{dir} denote the right-hand side of the direct-forgery success bound in (29). Under official seeded key generation, the success probability of the checked direct-forgery invocation is at least $p_{\text{dir}} - \delta_{\text{XOF}}^{\text{ind}}$ in the ideal-XOF model and at least $p_{\text{dir}} - \delta_{\text{IC}}^{\text{ind}}$ in the ideal-cipher model. Every returned signature remains valid against the unchanged verifier.

Proof. Total variation changes the probability of the ideal-model success event by at most the corresponding distance in [Theorem 24.3](#). Public verification is distribution-free and rejects every invalid candidate. $\square$

24.4 From key-distribution distance to verified recovery

Let

$$\eta_{\text{fast}} := \eta_{\text{off}} + \eta_{\text{oil}} + \eta_{\text{span}} + \eta_{\text{sign}}, \quad \eta_{\text{local}} := \tau_{m,V}(Q)^O + \eta_{\text{sign}}$$

be the public-random exceptional-key probabilities for the fast projective and graph-direction equivalent-key theorems. Total variation is applied only to these key-level events, not to a bounded attack execution.

Corollary 24.5 (Official seeded recovery in ideal-primitive models). Under the official seeded key generation:

1. with SHAKE modeled as an ideal XOF, the probability that a key fails the graph-direction equivalent-key hypotheses is at most $\eta_{\text{local}} + \delta_{\text{XOF}}$, while the corresponding fast-projective bound is $\eta_{\text{fast}} + \delta_{\text{XOF}}$;
2. with AES modeled as an ideal cipher, the corresponding bounds are $\eta_{\text{local}} + \delta_{\text{IC}}$ and $\eta_{\text{fast}} + \delta_{\text{IC}}$.

Every other key for the relevant route satisfies the corresponding key-by-key recovery theorem: accepted outputs are always correct, and verified restart terminates almost surely with the expected-work bounds of [Theorems 23.11](#) and [23.13](#). At Levels I, III, and V, both ideal-key exceptions are negligible compared with the transfer terms, so the base-two logarithms round respectively to

$$(-121.245, -184.715, -248.286) \quad \text{and} \quad (-99.863, -96.655, -94.112)$$

for SHAKE and AES.

Proof. By [Theorems 24.1](#) and [24.2](#), the probability of every key event changes by at most the corresponding total-variation distance. Apply this separately to the graph-direction event of [Theorem 23.12](#) and the fast-projective event of [Theorem 23.14](#). On either complement, the corresponding theorem is key by key and independent of how the expanded key was sampled. Exact verification is distribution-free. The ideal exceptional probabilities are far below the transfer terms at the displayed precision. $\square$

24.5 Why this does not prove a fixed-primitive theorem

Proposition 24.6 (Support barrier for deterministic seeded expansion). Let $E : \{0, 1\}^\lambda \rightarrow \Omega$ be any deterministic public expansion map, let S be uniform on $\{0, 1\}^\lambda$, and let U be uniform on the finite set Ω . Then

$$\Delta_{\text{TV}}(E(S), U) \geq 1 - \frac{2^\lambda}{|\Omega|}.$$

In particular, if $\Omega = \{0, 1\}^B$, the distance is at least $1 - 2^{\lambda-B}$.

Proof. The distribution $E(S)$ is supported on at most 2^λ points. Let R be its support. Then $\Pr[E(S) \in R] = 1$, while $\Pr[U \in R] = |R|/|\Omega| \leq 2^\lambda/|\Omega|$. The variational characterization of total variation gives the claim. $\square$

Thus the fixed public SHAKE/AES stream vector cannot be information-theoretically close to independent uniform streams. Nor is there a generic public-seed PRG reduction: the public seed is revealed, expansion is deterministic, and a distinguisher can recompute the official stream and compare it with the transcript. The ideal-XOF and ideal-cipher key-density conclusions above are exact statements inside those models; they do not instantiate themselves for fixed primitives.

The key-by-key recovery theorem nevertheless applies to every fixed-primitive key satisfying the key-by-key conditions. A high-probability standard-model statement must therefore analyze the relevant algebraic bad events under the actual fixed expansion map, or introduce an explicitly event-specific hardness assumption. The support barrier does not rule out such work; it rules out a generic statistical or transcript-pseudorandomness transfer that replaces the revealed deterministic expansion by independent uniform streams.

25 Optional identification with the planted oil space

Neither recovery theorem requires uniqueness: it keeps every verified candidate and searches all of them for a signing witness. This appendix proves the stronger generic identification statement that, for almost every key, no second K -rational common isotropic O -space exists. Fix the planted oil space $\mathcal{O}_0$. Let $\mathcal{O}' \neq \mathcal{O}_0$ be another O -space and put

$$s = O - \dim(\mathcal{O}' \cap \mathcal{O}_0), \quad 1 \leq s \leq O.$$

Lemma 25.1 (Relative Grassmannian stratum). The locus of such $\mathcal{O}'$ with fixed s has dimension $s(V + O - s)$.

Proof. Choose the $(O - s)$ -dimensional intersection inside $\mathcal{O}_0$, the s -dimensional image of $\mathcal{O}'$ in $K^{V+O}/\mathcal{O}_0 \simeq K^V$, and the graph map back to the s -dimensional quotient of $\mathcal{O}_0$. The dimensions are $(O - s)s$, $s(V - s)$, and s^2 , respectively. $\square$

Lemma 25.2 (Independent extra isotropy conditions). For fixed $\mathcal{O}'$ in this stratum, requiring one central form that already vanishes on $\mathcal{O}_0 \times \mathcal{O}_0$ also to vanish on $\mathcal{O}' \times \mathcal{O}'$ imposes exactly

$$c_s = \binom{O+1}{2} - \binom{O-s+1}{2} = \frac{s(2O-s+1)}{2}$$

additional independent linear conditions.

Proof. Restriction to $\mathcal{O}'$ already vanishes on $\text{Sym}^2(\mathcal{O}' \cap \mathcal{O}_0)$. Conversely, after choosing complements, every symmetric form on $\mathcal{O}'$ that vanishes on this subspace extends to an ambient symmetric form vanishing on $\mathcal{O}_0 \times \mathcal{O}_0$. The restriction map therefore has the displayed rank. $\square$

Theorem 25.3 (Uniqueness codimension and rational first moment). For m independent central forms, the key locus admitting an alternative common isotropic O -space of type s has codimension at least

$$\delta_s = m \frac{s(2O-s+1)}{2} - s(V + O - s).$$

The minimum over s equals 903, 1927, 3539 for Levels I, III, and V. With Gaussian binomial coefficients denoted by $\begin{bmatrix} a \\ b \end{bmatrix}_Q$,

$$\Pr[\text{a second } K\text{-rational oil space}] \leq \sum_{s=1}^O \begin{bmatrix} O \\ s \end{bmatrix}_Q \begin{bmatrix} V \\ s \end{bmatrix}_Q Q^{s^2 - mc_s}.$$

Writing the displayed first-moment bound as η_{uniq} , we have

$$\log_2 \eta_{\text{uniq}} < -18932.346, \quad -40401.586, \quad -74198.865$$

at Levels I, III, and V, respectively.

Proof. Over the stratum of dimension $s(V + O - s)$, each of the m form fibers loses c_s dimensions by [Theorem 25.2](#). The incidence variety therefore has codimension at least $mc_s - s(V + O - s)$ after projection to key space. The function δ_s is concave in s , so its minimum on $1 \leq s \leq O$ occurs at an endpoint; direct evaluation gives the displayed values and shows that $s = 1$ minimizes in all three parameter sets.

For the finite-field statement, the number of K -rational O -spaces in the type- s stratum is

$$\begin{bmatrix} O \\ s \end{bmatrix}_Q \begin{bmatrix} V \\ s \end{bmatrix}_Q Q^{s^2}.$$

For each fixed $\mathcal{O}'$, the m independent forms satisfy the c_s extra conditions with probability Q^{-mc_s} . The union bound gives the formula; exact evaluation gives the stated logarithms. $\square$

Corollary 25.4 (Generic identification with the planted decomposition). In the public-random model, outside the exceptional event of [Theorem 23.12](#) and the additional event of probability η_{uniq} in [Theorem 25.3](#), every exact graph-direction candidate has the planted oil space. The same conclusion holds for the fast projective route outside the exceptional event of [Theorem 23.14](#) and the uniqueness event. In either case, the returned decomposition is the planted expanded decomposition up to the verified coordinate representation. The secret seed is still not recovered or needed.

Proof. Every exact candidate defines a common isotropic O -space. Uniqueness forces that space to be $\mathcal{O}_0$, and its graph determines the expanded decomposition used by [Theorem 3.4](#). $\square$

26 Alternative elimination and low-space directions

The main text uses complete univariate resolutions because they are the only route for which we have both a suitable theorem and full failure accounting. This appendix records two exact ways to extract a root more compactly once a quotient algebra exists, and then explains why several natural attempts do not yet improve the proved exponent.

26.1 Extracting an isolated root from an existing quotient algebra

Let $A = L[y_1, \dots, y_V]/I$ be the reduced affine coordinate algebra of a square core, of degree $d \leq D$. Write Z for its geometric root set, and let $q_1, \dots, q_m$ denote the original restricted equations in A .

Proposition 26.1 (Norm-gradient root formula). Suppose $p \in L^V$ is the unique point of Z satisfying all m original equations. Choose $\rho \leftarrow L^m$, put $h = \sum_i \rho_i q_i \in A$, and define

$$\mathcal{N}_h(z_0, \dots, z_V) := \text{Norm}_{A/L} \left(h + z_0 + \sum_{j=1}^V z_j y_j \right). \quad (118)$$

With probability at least $1 - (d-1)/|L|$, h vanishes at p and at no other geometric core point. On that event,

$$\partial_{z_0} \mathcal{N}_h(0) \neq 0, \quad p_j = \frac{\partial_{z_j} \mathcal{N}_h(0)}{\partial_{z_0} \mathcal{N}_h(0)} \quad (1 \leq j \leq V). \quad (119)$$

Equivalently, if M_h and M_{y_j} are the multiplication operators in A , then $\ker M_h$ is one-dimensional and every nonzero $w \in \ker M_h$ satisfies $M_{y_j} w = p_j w$.

Proof. After scalar extension to an algebraic closure, reducedness gives

$$A \otimes_L \bar{L} \cong \prod_{x \in Z} \bar{L}.$$

At every $x \neq p$, the vector $(q_1(x), \dots, q_m(x))$ is nonzero. The condition $h(x) = 0$ cuts out a proper L -linear subspace of L^m , so a uniform ρ satisfies it with probability at most $1/|L|$. A union bound over at most $d-1$ points proves the selector bound. On the good event, the norm factors as

$$\mathcal{N}_h(z) = \prod_{x \in Z} \left(h(x) + z_0 + \sum_{j=1}^V z_j x_j \right).$$

Differentiating at the origin gives

$$\partial_{z_0} \mathcal{N}_h(0) = \prod_{x \neq p} h(x), \quad \partial_{z_j} \mathcal{N}_h(0) = p_j \prod_{x \neq p} h(x),$$

which proves (119). In the same product decomposition, M_h is diagonal with a unique zero eigenvalue and the coordinate operators act on its kernel by the scalars p_j . $\square$

Remark 26.2 (Several surviving roots). For an arbitrary selector h , let $S = \{x \in Z : h(x) = 0\}$. The lowest nonzero homogeneous part of (118) at the origin is

$$\left(\prod_{x \notin S} h(x) \right) \prod_{x \in S} \left(z_0 + \sum_{j=1}^V z_j x_j \right).$$

Thus the first nonzero norm jet is the Chow product of all selected roots. The linear formula (119) is exactly the case $|S| = 1$.

Proposition 26.3 (Resultant-gradient root formula). Assume $\text{char } L \neq 2$. Let $F_0, \dots, F_V \in L[t_0, \dots, t_V]$ be homogeneous quadrics with a unique geometric common point $p = [p_0 : \dots : p_V]$. Assume that their value Jacobian on the tangent space $T_p \mathbb{P}^V$ has rank V . Let $\mathcal{R}$ be their multivariate resultant, written in the monomial coefficients $c_{i,\alpha}$ of F_i , $|\alpha| = 2$. Then the coefficient tuple is a smooth point of the resultant hypersurface. If $\lambda = (\lambda_0, \dots, \lambda_V)$ spans the left kernel of the tangent Jacobian, there is a scalar $\kappa \neq 0$ such that

$$\frac{\partial \mathcal{R}}{\partial c_{i,\alpha}} = \kappa \lambda_i p^\alpha \quad (0 \leq i \leq V, |\alpha| = 2). \quad (120)$$

In particular, for any i, k with $\lambda_i p_k \neq 0$,

$$\frac{p_j}{p_k} = \frac{\partial \mathcal{R} / \partial c_{i,e_j+e_k}}{\partial \mathcal{R} / \partial c_{i,2e_k}} \quad (0 \leq j \leq V). \quad (121)$$

The coefficient convention in (121) treats $t_j t_k$ as one monomial, with no symmetric-matrix factor of two.

Proof. Consider the incidence variety of coefficient tuples and projective points satisfying $F_i(p) = 0$ for every i . Its first-order equations at the given pair are

$$\delta F_i(p) + dF_i(p)[\delta p] = 0 \quad (0 \leq i \leq V).$$

The tangent Jacobian has rank V , so a tangent displacement δp exists exactly when

$$\sum_{i=0}^V \lambda_i \delta F_i(p) = 0.$$

Uniqueness and transversality make the incidence projection locally one-to-one onto a smooth branch of the resultant hypersurface. Its tangent hyperplane is therefore the kernel of the displayed nonzero functional. The differential of the irreducible resultant is proportional to that functional, giving (120); the ratio formula follows by evaluating the two quadratic monomials at p . $\square$

Remark 26.4 (Compression achieved, evaluator still missing). The norm gradient outputs one root using only $V + 1$ field elements, and the resultant gradient uses one nonzero coefficient block. If quotient multiplication is available as a black box, a one-dimensional kernel can be found with $O(d)$ stored field elements and $O(d)$ multiplication-operator products by standard black-box linear algebra [22]. Alternatively, Baur–Strassen converts any arithmetic circuit for the norm or resultant into a circuit for all first derivatives with constant-factor overhead [15]. These statements compress the *root-extraction interface*; they do not construct the degree- d quotient algebra, multiplication operators, norm, or resultant in $o(d)$ space. With current constructions, the time remains near-quadratic in d (with the fixed ε loss of Theorem 23.4) and at least one degree- d state vector remains. Moreover, Theorem 22.3 gives no production-parameter density bound for uniqueness of the full-system root. The formulas are therefore exact algorithmic targets, not a practical attack.

26.2 Why the spare equations do not yet lower the finite exponent

The full restricted system has $e = m - V \in \{2, 2, 3\}$ equations beyond the square core, and generically those equations isolate the planted point. We tested whether they also lower the solving exponent. The following deliberately optimistic semi-regular screen uses the Hilbert series

$$\frac{(1 - t^2)^{V+e}}{(1 - t)^{V+1}}$$

for $V + e$ homogeneous quadrics in $V + 1$ projective variables. Because the actual system contains a planted point, this is a diagnostic rather than a theorem. At the last positive coefficient, the monomial counts are:

Level	V	degree	$\log_2 N$	optimistic $\log_2 N^2$	current $\log_2 D^2$
I	52	27	69.787	139.574	104
III	76	39	102.585	205.170	152
V	102	48	131.814	263.629	204

Even charging only N^2 sparse black-box work is worse than the baseline D^2 geometric scale and also worse than the primary $\varepsilon = 0.3$ fast-line indicators after field and circuit conversion; dense linear algebra is worse still. This does not rule out a coefficient-specific method, but it rules out the most direct overdetermined-Macaulay improvement under standard linear algebra.

Proposition 26.5 (Underslicing loses more in hit probability than it saves in degree). Let $0 \leq d < V$, and choose a uniform projective d -plane over $K = \mathbb{F}_Q$. Its probability of meeting the planted projective oil space is

$$\pi_d = 1 - \prod_{i=0}^d \frac{1 - Q^{i-V}}{1 - Q^{i-(V+O)}}. \quad (122)$$

If the same degree-squared square-core solver is run independently on every sampled plane, its expected leading degree factor is $4^d/\pi_d$. For $d = V - r$ and the QR-UOV base field $Q = 127^3$, removing $r = 1, 2, 3$ slice dimensions increases the base-two exponent relative to the $d = V$ slice by

$$18.966053, \quad 37.932107, \quad 56.898161$$

bits, respectively, before any other work is charged. Choosing the slices over the larger solver field only worsens this tradeoff.

Proof. Write $k = d + 1$. A vector k -subspace disjoint from the fixed O -dimensional oil space is the graph of a linear map from a k -subspace of the V -dimensional quotient to the oil space. Hence there are

$$Q^{O_k} \begin{bmatrix} V \\ k \end{bmatrix}_Q$$

disjoint subspaces, out of $\begin{bmatrix} V+O \\ k \end{bmatrix}_Q$ total. Taking the ratio and expanding the Gaussian binomial coefficients gives (122). Independent repetition needs expected $1/\pi_d$ planes. Substitution into $4^d/\pi_d$ gives the displayed values, which are recomputed by the companion numerical audit. Asymptotically each removed dimension costs $\log_2 Q - 2 = 18.966054\dots$ bits: it saves two degree-squared bits but loses roughly one factor of Q in intersection probability. $\square$

The same calculation covers the equivalent hybrid strategy of guessing one base-field coordinate of the oil point before solving: every guessed coordinate costs roughly $\log_2 Q$ bits and saves only two bits in the degree-squared solver. Thus the minimal guaranteed slice is also the best random underslice under the current solver model.

The other natural routes also stop short of an exponent improvement. Explicit resultant matrices remain exponentially larger than the D -dimensional quotient representation in the dense quadratic regime [16]. Chow-form algorithms are polynomial in geometric degree but do not provide a subquadratic dependence specialized to these parameters [17]. Faster modular composition lowers an operation used after a univariate algebra exists, not the geometric-intersection stage that constructs that algebra [18].

We also checked two recent refinements of Macaulay/Gröbner solving. Degree smoothing replaces a full Macaulay matrix by a sufficient submatrix lying between two adjacent integer

degrees [19]. Our optimistic screen already charges the complete monomial space at the *lower* adjacent degree and is still 35.574, 53.170, and 59.629 bits above the D^2 geometric scale; smoothing within that interval cannot reverse the comparison. The recent F4-tracer analysis for generic square degree- δ systems has leading arithmetic scale $\tilde{O}(\delta^{\omega_V+1}e^{cV})$ and relies on trace compatibility and genericity conjectures [20]. For quadrics, even its factor 2^{ω_V} exceeds $D^2 = 2^{2V}$ for every currently proved $\omega > 2$. Thus these improvements are valuable in their intended regimes but do not lower the exponent here. The norm and resultant gradients identify a smaller *output* target; no proved route found in this audit computes that target below the published near-quadratic-in- D line used here.

27 Reproducibility and executable attack description

27.1 Companion executable audits

The bundle contains four standard-library Python programs.

- `QRUOV_scissors_audit.py` recomputes the target-orthogonal dimension inequalities, projective $m - 4$ cores, fixed-key seed-oil intersection bounds, second moments, eligible-root probabilities, random-chart and separator factors, both finite-size cost rows, memory envelopes, Karatsuba plans, sparse-field certificates, and attack-specific retuning diagnostics. It emits text and JSON and freezes every displayed number.
- `QRUOV_directional_audit.py` checks the corrected $2C_0$ reconstruction precision, Vandermonde product starts, characteristic-127 lifting identities, flattened multiplication, sparse-field certificates, recursive Karatsuba convolution, and complete graph recovery on 200 toy keys.
- `QRUOV_direct_qr_toy.py` generates exact degree-three extension-field UOV instances, applies the trace pull-back, and executes full target normalization, seed search, Chevalley–Warning extension, projective residual search, affine rescaling, and public verification end to end.
- `QRUOV_cw_helper.py` supplies the finite-field and Chevalley–Warning helper routines used by the direct experiment.

These programs audit formulas and attack logic. They do not instantiate the astronomically large full-parameter symbolic state.

27.2 Executable attack description

Projective target-orthogonal direct-forgery route.

1. Derive the official nonzero verification target y from the message and salt. Choose a public invertible output transformation sending y to $(0, \dots, 0, 1)$, and apply the same transformation to the public quadrics.
2. Sample a uniform seven-dimensional seed subspace $S_0 \subset \mathbb{F}_q^n$. Enumerate $\mathbb{P}(S_0)$ until finding a nonzero common zero of the first three transformed quadrics; Chevalley–Warning guarantees one.
3. Starting from its span, repeatedly intersect the three polar-orthogonal spaces and solve the induced three quadrics on a seven-dimensional quotient slice. Extend until obtaining a common totally isotropic subspace U of dimension $m - 3$.
4. Choose a random nonzero chart form on U . Restrict the next $m - 4$ zero-target quadrics to the chart and run the checked finite-characteristic nonsingular-root solver on the resulting $m - 4$ variable square system.

5. At every represented rational projective root, evaluate the final transformed quadratic. If its value is a nonzero square, recover either affine scale making that value one, reverse the transformations, and re-evaluate all original public equations.
6. Return the first vector passing the unchanged QR-UOV verifier. If the chart, solver invocation, or residual search fails, change the salt and attack coins and repeat. Exact verification gives one-sided error.

Primary finite-characteristic direction route.

1. Expand the public key to matrices P_i over $K = \mathbb{F}_{127^3}$. Choose a nonzero public direction $c \in K^O$ and form $q_{i,c}(x) = P_i(-x, c)$.
2. Choose a random $V \times m$ output-recombination matrix R and form the square system $f = Rq_c$. Work in $K' = \mathbb{F}_{127^d}$ using the sparse degree-18, 24, or 30 model and the audited recursive Karatsuba circuit for the relevant parameter level.
3. Run the finite-characteristic nonsingular-root algorithm of [Theorem 7.6](#), using the Vandermonde product start and a full random separator. A bad invocation may omit roots or fail; it cannot create an accepted false key.
4. Re-evaluate all m original directional equations at every represented K -rational root. At each survivor, solve the O public linear systems of [Theorem 5.2](#). Keep only graphs passing every graph identity, total-isotropy check, inverse pull-back, and exact public-map factorization.
5. Run the basis-first signing scan and, if needed, the pair-line fallback. Return the first exact candidate with an invertible oil matrix. If no candidate passes, sample fresh solver coins and then a fresh outer direction.

Coordinate directions and Wronskian condensers provide a deterministic outer fallback; the primary finite-size ledger uses the simpler random full-rank recombination analyzed in [Theorem 7.8](#).

Fast projective route.

1. Set $\varepsilon = 0.3$. Use the minimum fields $s = 8, 12, 15$ for expected-restart analysis or the safety fields $s = 9, 13, 16$ for the one-invocation theorem. Sample a full-rank slice matrix M and an output recombination R over L_s ; restrict the public quadrics and form the square core.
2. Run one finite checked invocation of the van der Hoeven–Lecerf solver. Return fail on an exceptional division, degree mismatch, or consistency failure. Do not hide an unbounded internal restart inside one outer draw.
3. Canonicalize and verify the resolution. Evaluate all original restricted equations in its quotient algebra, prefilter with [Theorem 23.9](#), extract L_s -rational roots, and recheck all equations at each reconstructed point.
4. At every survivor, compute the common derivative kernel. Require dimension O , descent to a graph over K , total isotropy, all graph identities, the official inverse pull-back, and exact factorization of the original public map.
5. Run the basis-first signing scan and, if needed, the pair-line fallback. Return the first verified signing certificate; otherwise begin a fresh outer trial.

Both wrappers have one-sided error: a solver failure may cause a retry, but exact public verification prevents an invalid equivalent key or invalid signature from being returned. The output is a verified expanded signing decomposition and an inverse oil matrix, not the original seed or an official serialized private key.

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